PLC_1 [CPU 1215C AC/DC/Rly]

PLC_1			
Project information			
Name	PLC_1	**Author**	beirn
Comment		**Slot**	1
Rack	0		
Catalog information			
Short designation	CPU 1215C AC/DC/Rly	**Description**	Work memory 200 KB; 120/240VDC power supply with DI14 x 24VDC SINK/SOURCE, DQ10 x relay and AI2 and AQ2 on board; 6 high-speed counters and 4 pulse outputs on-board; signal board expands on-board I/O; up to 3 communication modules for serial communication; up to 8 signal modules for I/O expansion; PROFINET IO controller, 2 ports, I-device, transport protocol TCP/IP, secure Open User Communication, S7 communication, Web server, OPC UA: Server DA
Article number	6ES7 215-1BG40-0XB0	**Firmware version**	V4.6
	False		

Connection resources

	Station resources - Reserved - Maximum	Station resources - Reserved - Configured	Station resources - Dynamic - Configured	Module resources - PLC_1 [CPU 1215C AC/DC/Rly] - Configured
Maximum number of resources:		34	34	68
	Maximum	Configured	Configured	Configured
PG communication:	4	-	-	-
HMI communication:	12	0	0	0
S7 communication:	8	0	0	0
Open user communication:	8	0	0	0
Web communication:	2	-	-	-
OPC UA client/server communication:	0	-	-	-
Other communication:	-	-	0	0
Total resources used:		0	0	0
Available resources:		34	34	68

Overview of addresses\Overview of addresses\Overview of addresses			
Inputs	True	**Outputs**	True
Address gaps	False	**Slot**	True

Type	Addr. from	Addr. to	Module	PIP	Device name	Device number	Size	Master / IO system	Rack	Slot
I	0	1	DI 14/DQ 10_1	Automatic update	PLC_1 [CPU 1215C AC/DC/Rly]	-	2 Bytes	-	0	1 1
O	0	1	DI 14/DQ 10_1	Automatic update	PLC_1 [CPU 1215C AC/DC/Rly]	-	2 Bytes	-	0	1 1
I	64	67	AI 2/AQ 2_1	Automatic update	PLC_1 [CPU 1215C AC/DC/Rly]	-	4 Bytes	-	0	1 2
O	64	67	AI 2/AQ 2_1	Automatic update	PLC_1 [CPU 1215C AC/DC/Rly]	-	4 Bytes	-	0	1 2
I	1000	1003	HSC_1	Automatic update	PLC_1 [CPU 1215C AC/DC/Rly]	-	4 Bytes	-	0	1 16
I	1004	1007	HSC_2	Automatic update	PLC_1 [CPU 1215C AC/DC/Rly]	-	4 Bytes	-	0	1 17
I	1008	1011	HSC_3	Automatic update	PLC_1 [CPU 1215C AC/DC/Rly]	-	4 Bytes	-	0	1 18
I	1012	1015	HSC_4	Automatic update	PLC_1 [CPU 1215C AC/DC/Rly]	-	4 Bytes	-	0	1 19
I	1016	1019	HSC_5	Automatic update	PLC_1 [CPU 1215C AC/DC/Rly]	-	4 Bytes	-	0	1 20
I	1020	1023	HSC_6	Automatic update	PLC_1 [CPU 1215C AC/DC/Rly]	-	4 Bytes	-	0	1 21
O	1000	1001	Pulse_1	Automatic update	PLC_1 [CPU 1215C AC/DC/Rly]	-	2 Bytes	-	0	1 32
O	1002	1003	Pulse_2	Automatic update	PLC_1 [CPU 1215C AC/DC/Rly]	-	2 Bytes	-	0	1 33
O	1004	1005	Pulse_3	Automatic update	PLC_1 [CPU 1215C AC/DC/Rly]	-	2 Bytes	-	0	1 34
O	1006	1007	Pulse_4	Automatic update	PLC_1 [CPU 1215C AC/DC/Rly]	-	2 Bytes	-	0	1 35

PLC_1 [CPU 1215C AC/DC/Rly] / Program blocks

Main [OB1]

Main Properties					
General					
Name	Main	**Number**	1	**Type**	OB
Language	LAD	**Numbering**	Automatic		
Information					
Title	"Main Program Sweep (Cycle)"	**Author**		**Comment**	
Family		**Version**	0.1	**User-defined ID**	

Name	Data type	Default value
Input		
Initial_Call	Bool	
Remanence	Bool	
Temp		
Constant		

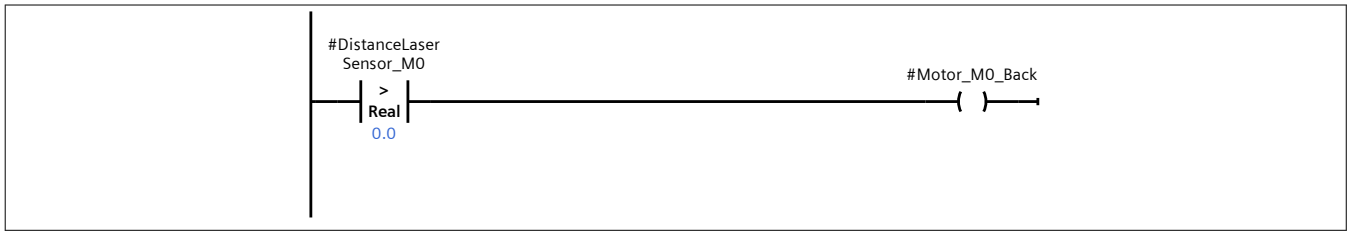
PLC_1 [CPU 1215C AC/DC/Rly] / Program blocks

Homing [FB1]

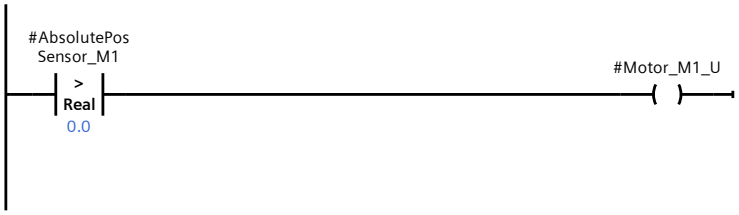
Homing Properties					
General					
Name	Homing	**Number**	1	**Type**	FB
Language	LAD	**Numbering**	Automatic		
Information					
Title		**Author**		**Comment**	
Family		**Version**	0.1	**User-defined ID**	

Name	Data type	Default value	Retain
Input			
DistanceLaserSensor_M0	Real	0.0	Non-retain
AbsolutePosSensor_M1	Real	0.0	Non-retain
InclinatieSensor_M0	Real	0.0	Non-retain
AbsoluteHoekEncoder_M2	Real	0.0	Non-retain
RadarSensor_M2	Real	0.0	Non-retain
Output			
M0_Motor_Backwards	Bool	false	Non-retain
M1_Motor_Up	Bool	false	Non-retain
M2_Motor_Right	Bool	false	Non-retain
M2_Motor_Left	Bool	false	Non-retain
InOut			
Static			
Motor_M0_Back	Bool	false	Non-retain
Motor_M1_U	Bool	false	Non-retain
Motor_M2_MoveToRight	Bool	false	Non-retain
Motor_M2_MoveToLeft	Bool	false	Non-retain
Temp			
Constant			

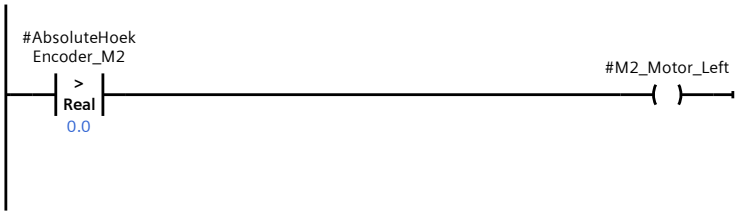
Network 1:



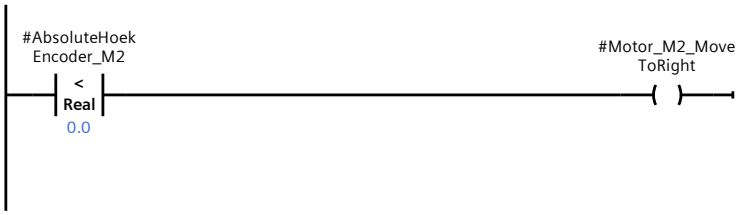
Network 2:



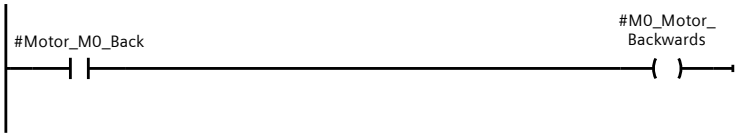
Network 3:



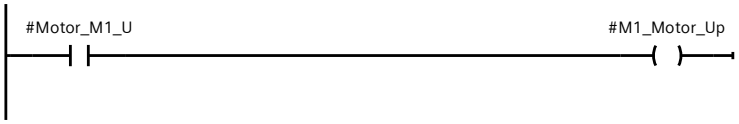
Network 4:



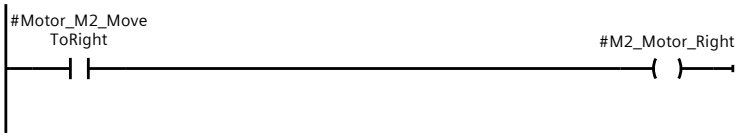
Network 5:



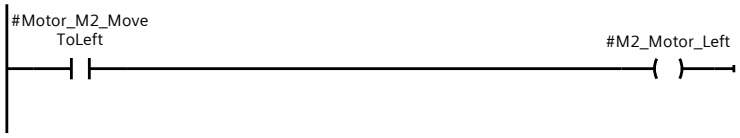
Network 6:



Network 7:



Network 8:


#Motor_M2_Move
ToLeft
#M2_Motor_Left

PLC_1 [CPU 1215C AC/DC/Rly] / Program blocks

Homing_DB [DB1]

Homing_DB Properties					
General					
Name	Homing_DB	**Number**	1	**Type**	DB
Language	DB	**Numbering**	Automatic		
Information					
Title		**Author**		**Comment**	
Family		**Version**	0.1	**User-defined ID**	

Name	Data type	Start value	Retain
Input			
DistanceLaserSensor_M0	Real	0.0	False
AbsolutePosSensor_M1	Real	0.0	False
InclinatieSensor_M0	Real	0.0	False
AbsoluteHoekEncoder_M2	Real	0.0	False
RadarSensor_M2	Real	0.0	False
Output			
M0_Motor_Backwards	Bool	false	False
M1_Motor_Up	Bool	false	False
M2_Motor_Right	Bool	false	False
M2_Motor_Left	Bool	false	False
InOut			
Static			
Motor_M0_Back	Bool	false	False
Motor_M1_U	Bool	false	False
Motor_M2_MoveToRight	Bool	false	False
Motor_M2_MoveToLeft	Bool	false	False

PLC_1 [CPU 1215C AC/DC/Rly] / Program blocks

DB_GlobalParams [DB2]

DB_GlobalParams Properties					
General					
Name	DB_GlobalParams	**Number**	2	**Type**	DB
Language	DB	**Numbering**	Automatic		
Information					
Title		**Author**		**Comment**	
Family		**Version**	0.1	**User-defined ID**	

Name	Data type	Start value	Retain
Static			
doel_afstand_M0_m	Real	3.0	False
doel_hoogte_M1_mm	Real	1100.0	False
doel_hoek_M2_deg	Real	0.0	False
start_hoogte_M1_mm	Real	1100.0	False
start_hoek_M2_deg	Real	0.0	False
M0_max_snelheid_m_s	Real	0.5	False
M0_rem_afstand_m	Real	0.3	False
M0_positie_tolerantie_m	Real	0.02	False
M1_Kp	Real	2.5	False
M1_Ti_s	Real	0.8	False
M1_Td_s	Real	0.05	False
M1_Tt_s	Real	0.4	False
M1_out_min_mm	Real	0.0	False
M1_out_max_mm	Real	2200.0	False
M1_deadband_mm	Real	0.5	False
M1_tolerantie_mm	Real	1.0	False
M2_Kp	Real	3.0	False
M2_Ti_s	Real	1.0	False
M2_Td_s	Real	0.08	False
M2_Tt_s	Real	0.5	False
M2_out_min_deg	Real	-90.0	False
M2_out_max_deg	Real	90.0	False
M2_deadband_deg	Real	0.1	False
M2_tolerantie_deg	Real	0.2	False
h_M1_nominaal_mm	Real	1100.0	False
d_pivot_M0_m	Real	0.5	False
L_M2_arm_m	Real	0.46	False
alfa_inclinatie	Real	0.85	False
dt_s	Real	0.01	False
auto_start	Bool	false	False
e_stop	Bool	false	False
reset_MSE	Bool	false	False
M1_spoed_mm_omw	Real	20.0	False
M1_overbreng	Real	3.0	False

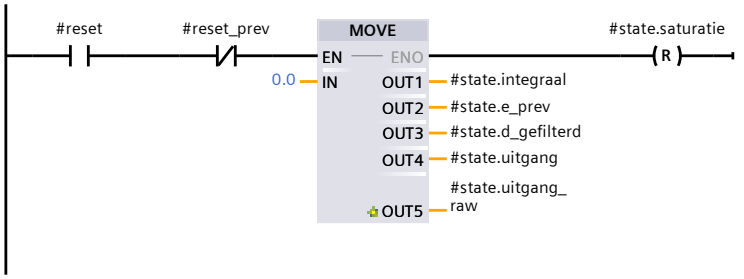
PLC_1 [CPU 1215C AC/DC/Rly] / Program blocks

FB_PID_M0 [FB3]

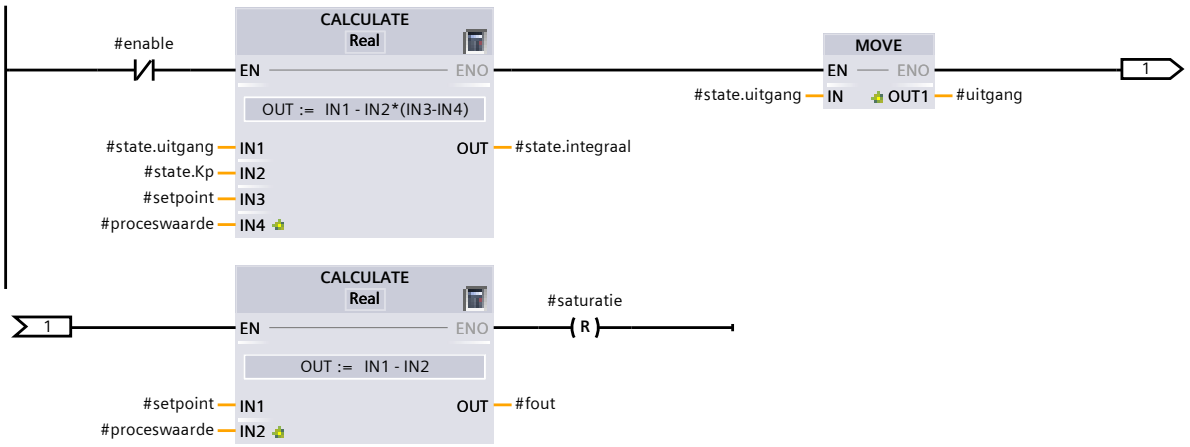
FB_PID_M0 Properties					
General					
Name	FB_PID_M0	**Number**	3	**Type**	FB
Language	LAD	**Numbering**	Automatic		
Information					
Title		**Author**		**Comment**	
Family		**Version**	0.1	**User-defined ID**	

Name	Data type	Default value	Retain
Input			
setpoint	Real	0.0	Non-retain
proceswaarde	Real	0.0	Non-retain
enable	Bool	false	Non-retain
reset	Bool	false	Non-retain
Output			
uitgang	Real	0.0	Non-retain
fout	Real	0.0	Non-retain
saturatie	Bool	false	Non-retain
InOut			
state	"UDT_PIDState"		
Static			
e_huidig	Real	0.0	Non-retain
P_term	Real	0.0	Non-retain
I_term	Real	0.0	Non-retain
D_term	Real	0.0	Non-retain
u_raw	Real	0.0	Non-retain
u_sat	Real	0.0	Non-retain
track_err	Real	0.0	Non-retain
alfa_D	Real	0.0	Non-retain
Kd	Real	0.0	Non-retain
in_deadband	Bool	false	Non-retain
reset_prev	Bool	false	Non-retain
abs(e_huidig)	Real	0.0	Non-retain
abs(state.deadb)	Real	0.0	Non-retain
Temp			
Constant			

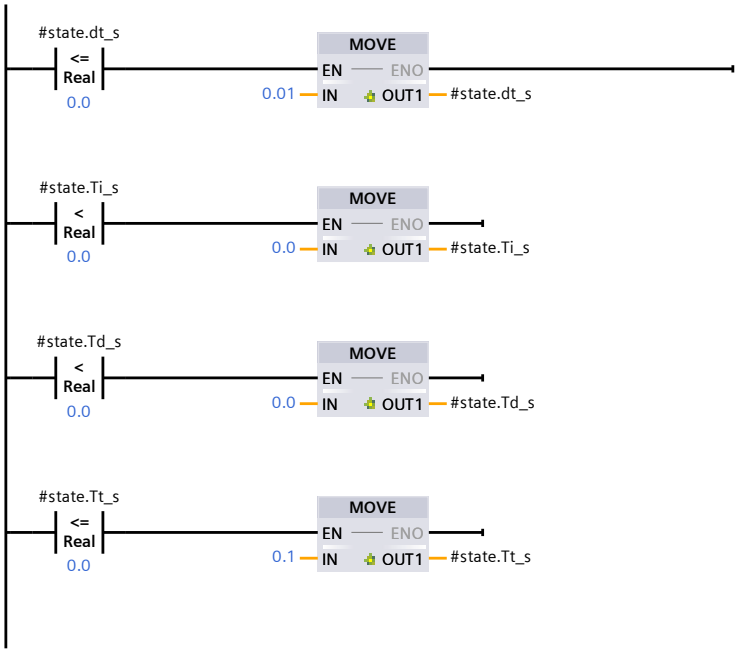
Network 1:



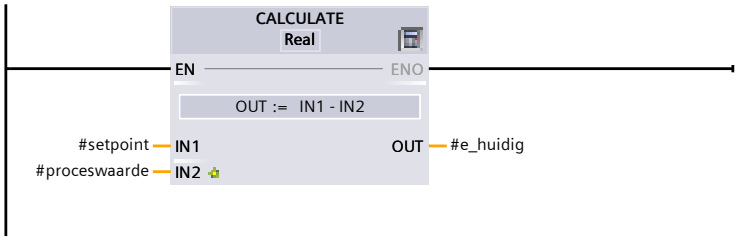
Network 2:



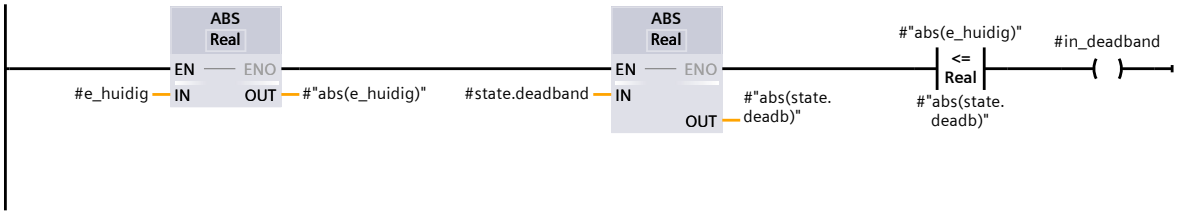
Network 3:



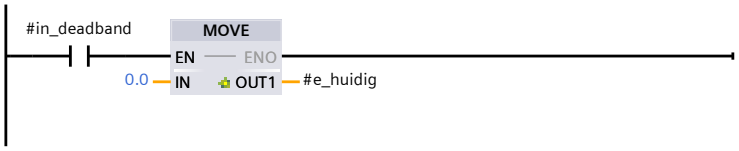
Network 4:



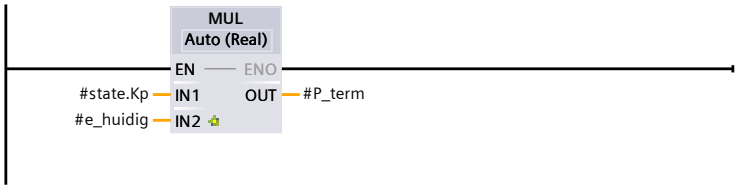
Network 5:



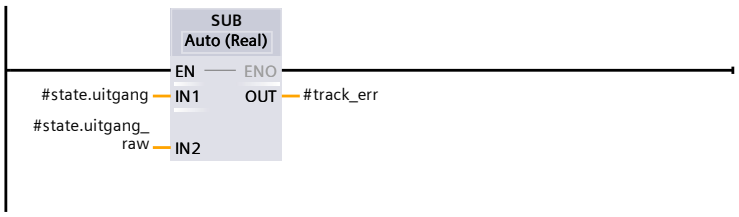
Network 6:



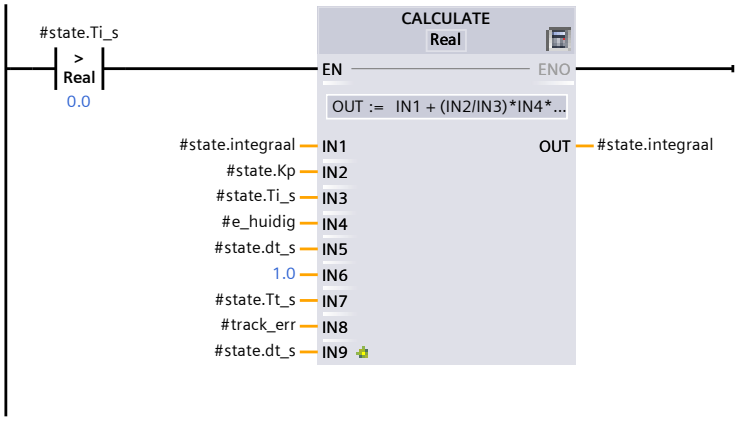
Network 7:



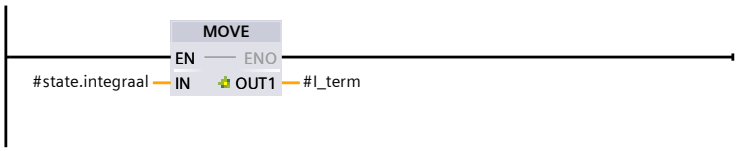
Network 8:



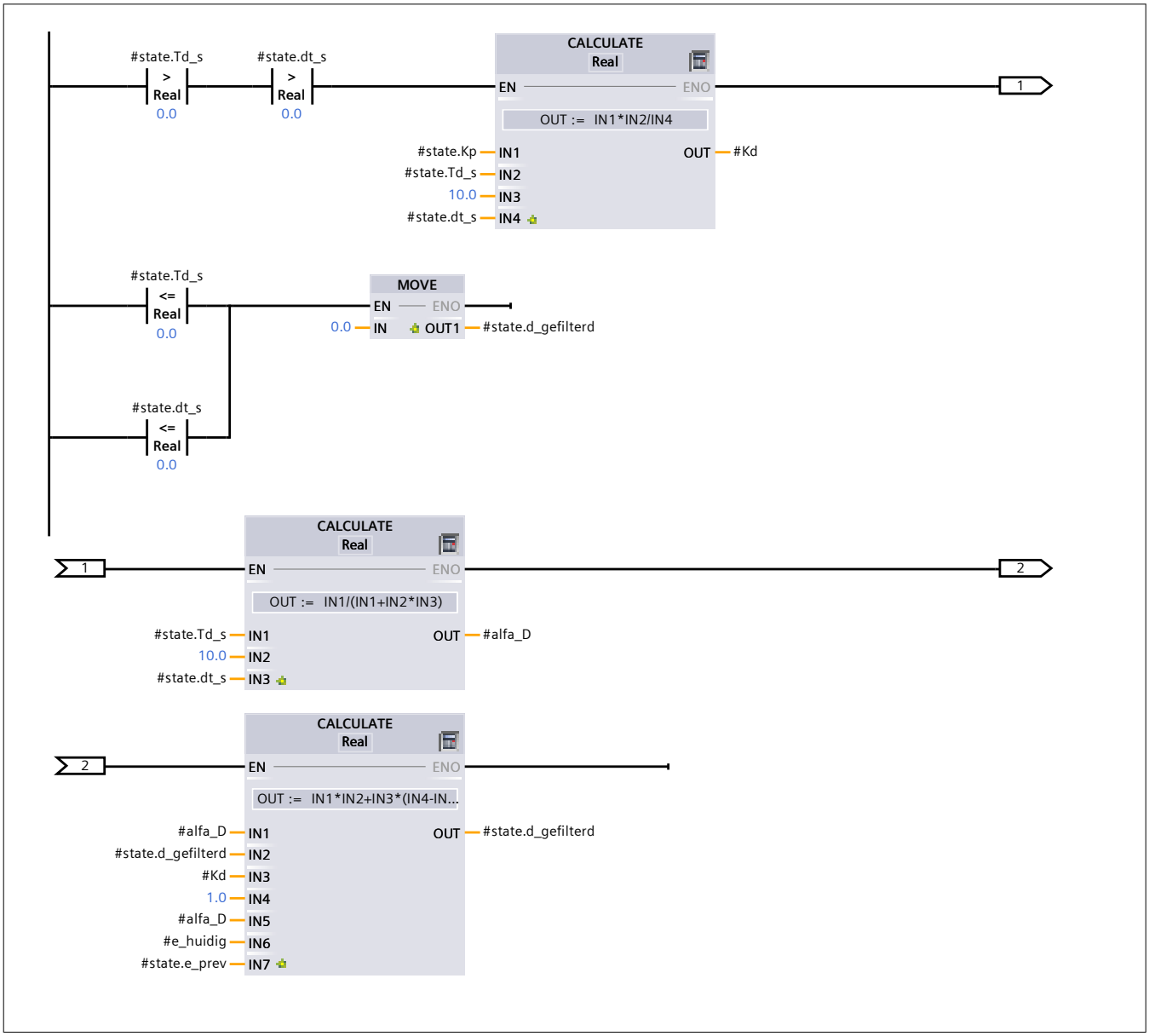
Network 9:



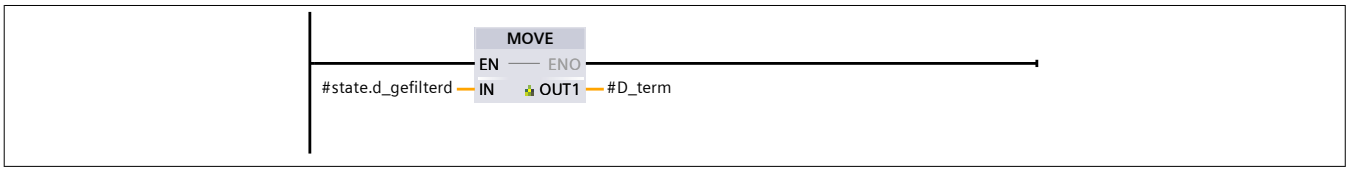
Network 10:



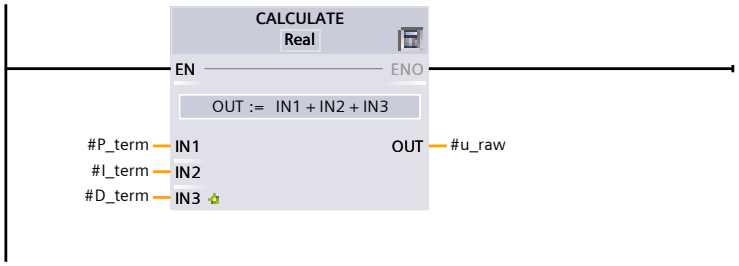
Network 11:



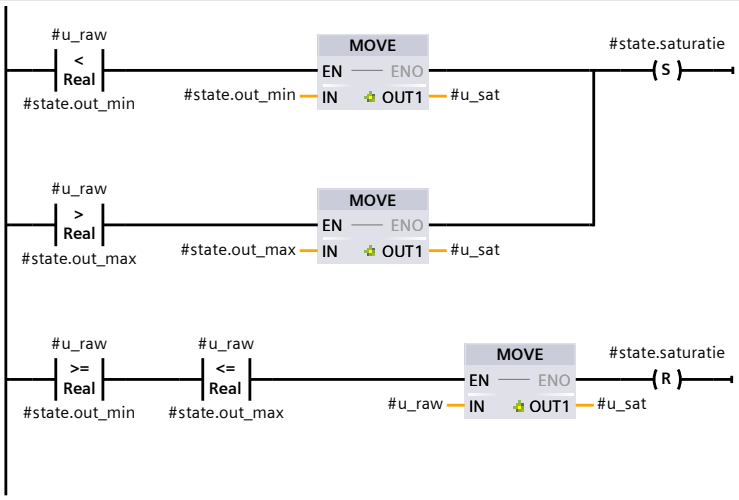
Network 12:



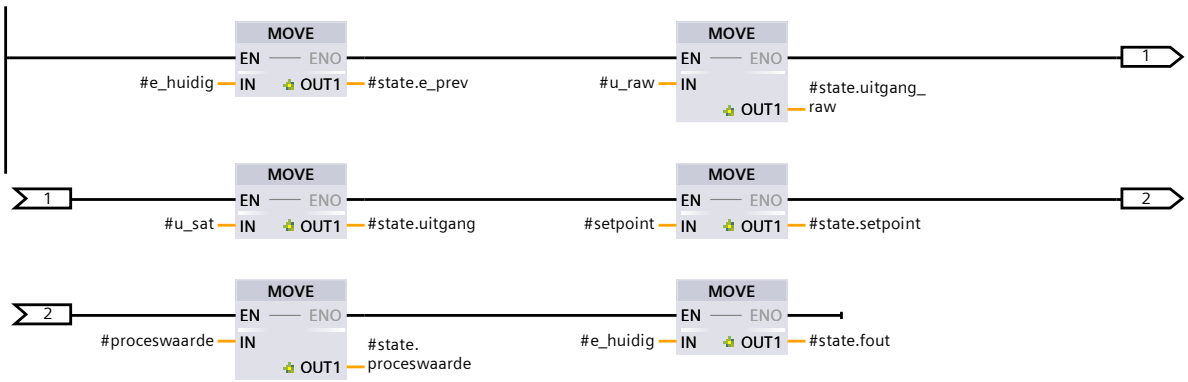
Network 13:



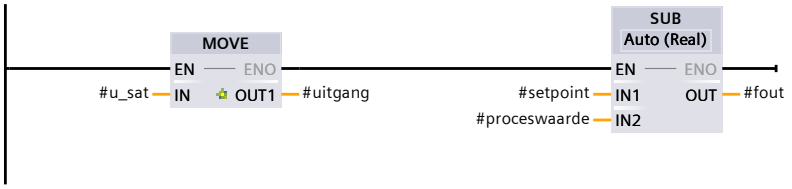
Network 14:



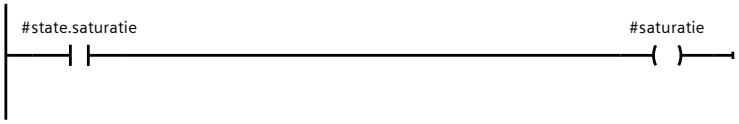
Network 15:



Network 16:



Network 17:



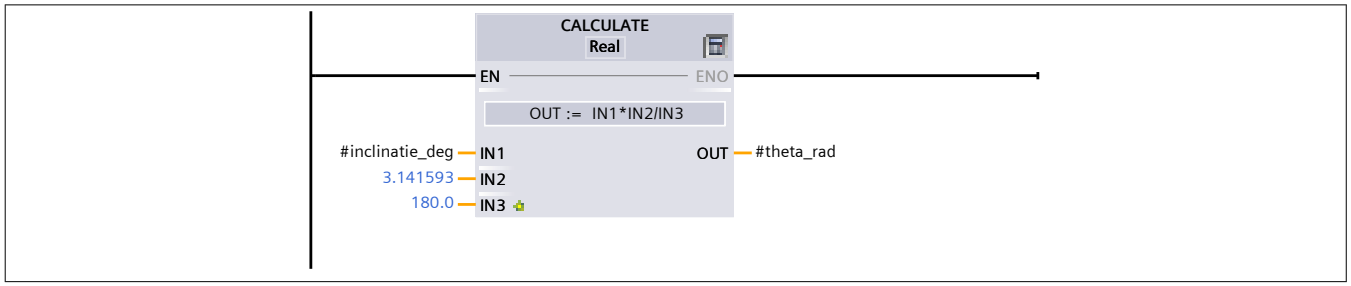
PLC_1 [CPU 1215C AC/DC/Rly] / Program blocks

FB_M1_HeightComp [FB4]

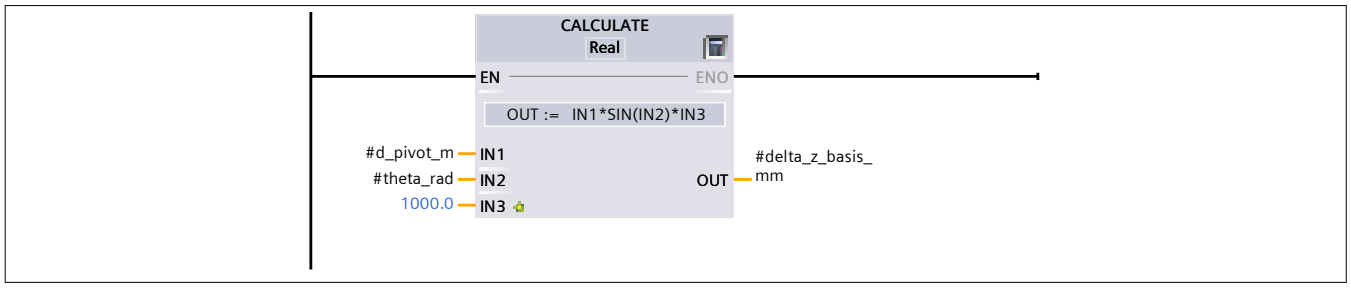
FB_M1_HeightComp Properties					
General					
Name	FB_M1_HeightComp	**Number**	4	**Type**	FB
Language	LAD	**Numbering**	Automatic		
Information					
Title		**Author**		**Comment**	
Family		**Version**	0.1	**User-defined ID**	

Name	Data type	Default value	Retain
Input			
inclinatie_deg	Real	0.0	Non-retain
pos_M1_mm	Real	0.0	Non-retain
enable	Bool	false	Non-retain
reset	Bool	false	Non-retain
doel_hoogte_mm	Real	0.0	Non-retain
d_pivot_m	Real	0.0	Non-retain
h_M1_nominaal_mm	Real	0.0	Non-retain
out_min_mm	Real	0.0	Non-retain
out_max_mm	Real	0.0	Non-retain
Output			
M1_setpoint_mm	Real	0.0	Non-retain
FF_bijdrage_mm	Real	0.0	Non-retain
FB_bijdrage_mm	Real	0.0	Non-retain
fout_mm	Real	0.0	Non-retain
op_positie	Bool	false	Non-retain
InOut			
pid_state	"UDT_PIDState"		
Static			
theta_rad	Real	0.0	Non-retain
delta_z_basis_mm	Real	0.0	Non-retain
delta_z_M1_mm	Real	0.0	Non-retain
FF_mm	Real	0.0	Non-retain
SP_voor_PID	Real	0.0	Non-retain
PID_uitgang	Real	0.0	Non-retain
SP_totaal	Real	0.0	Non-retain
pid_inst	"FB_PID_M0"		
pos_M1_mm-SP_totaal	Real	0.0	Non-retain
pid_inst_M1	"FB_PID_M1"		
Temp			
Constant			

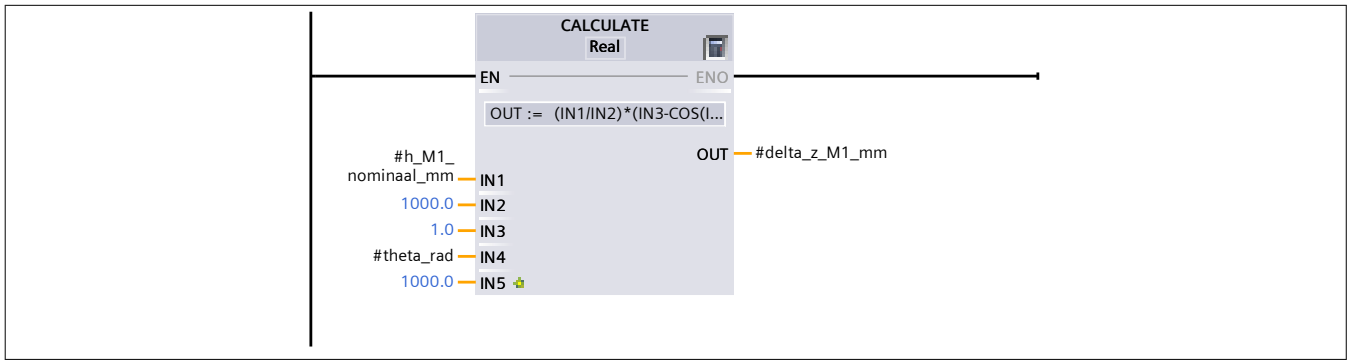
Network 1:



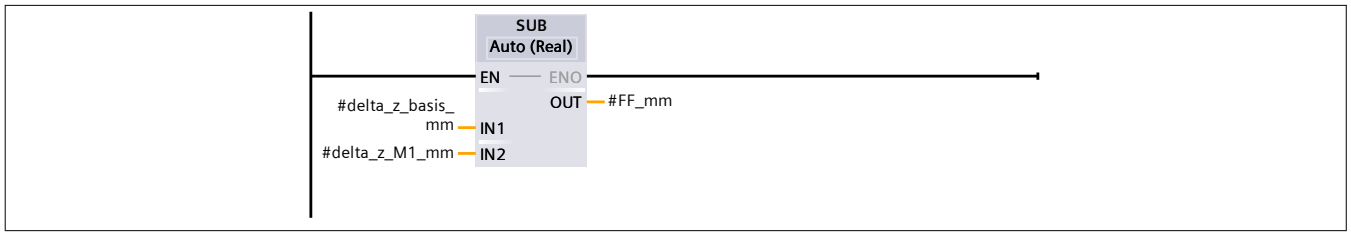
Network 2:



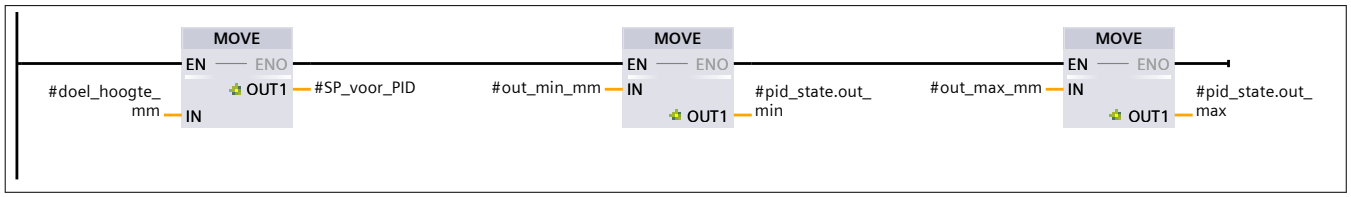
Network 3:



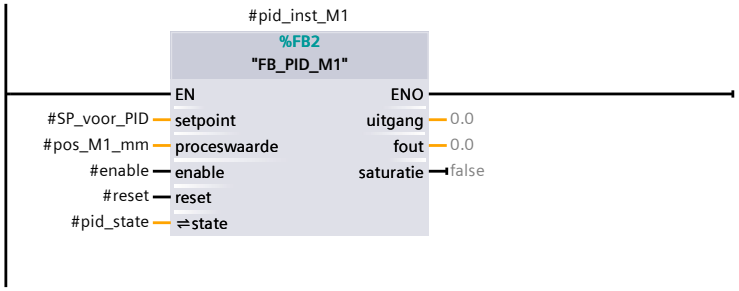
Network 4:



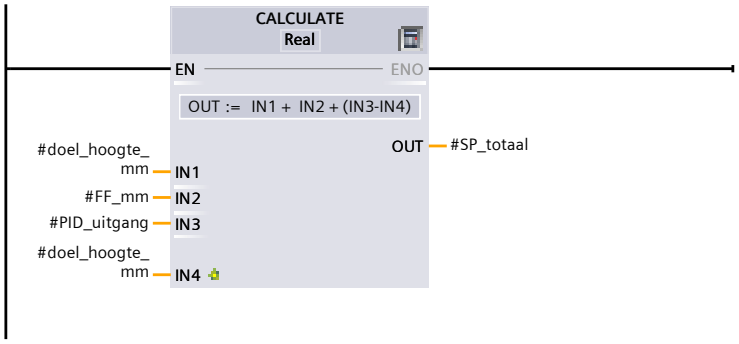
Network 5:



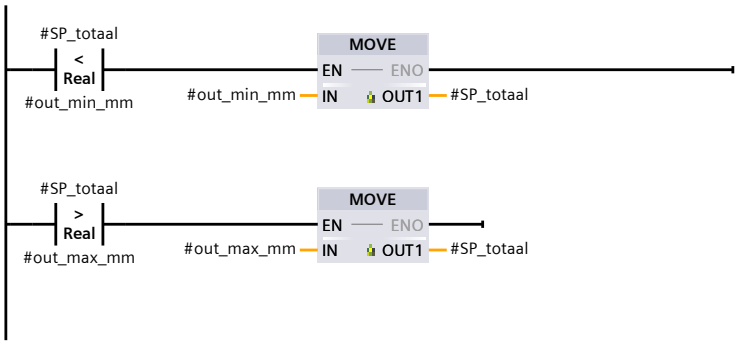
Network 6:



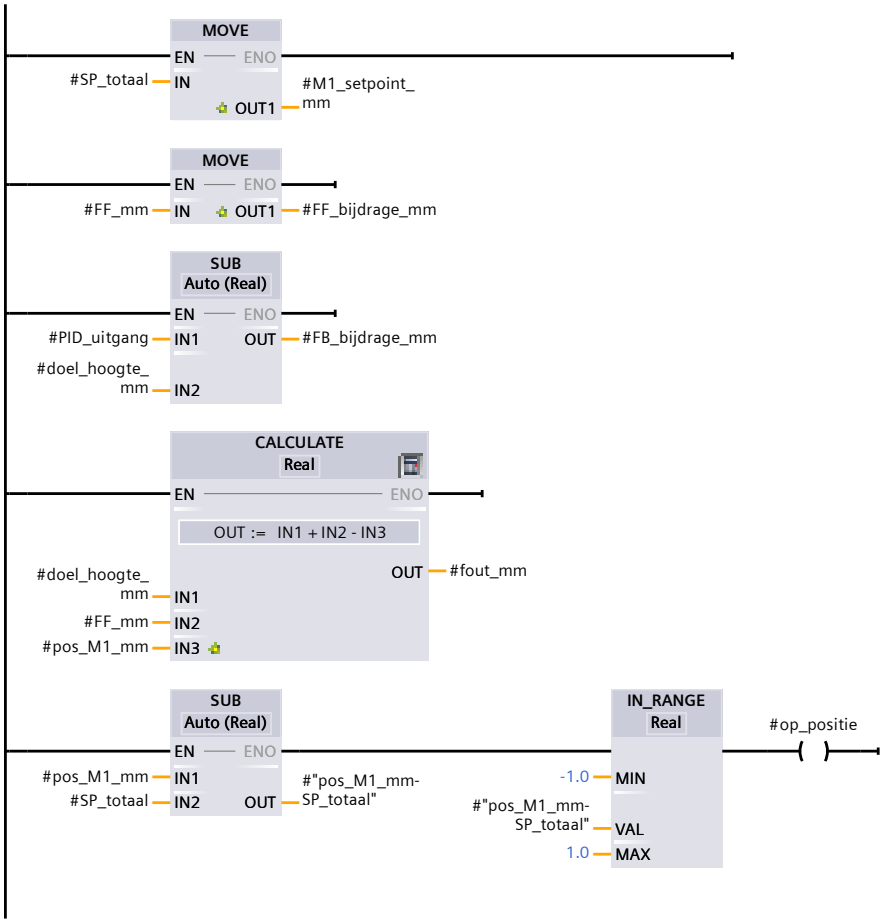
Network 7:



Network 8:



Network 9:


MOVE
EN
ENO
#SP_totaal
IN
OUT1
#M1_setpoint_mm
MOVE
EN
ENO
#FF_mm
IN
OUT1
#FF_bijdrage_mm
SUB
Auto (Real)
EN
ENO
#PID_uitgang
IN1
OUT
#FB_bijdrage_mm
#doel_hoogte_mm
IN2
CALCULATE
Real
EN
ENO
OUT := IN1 + IN2 - IN3
OUT
#fout_mm
#doel_hoogte_mm
IN1
#FF_mm
IN2
#pos_M1_mm
IN3
SUB
Auto (Real)
EN
ENO
#pos_M1_mm
IN1
#SP_totaal
IN2
OUT
#"pos_M1_mm-SP_totaal"
IN_RANGE
Real
-1.0
MIN
#"pos_M1_mm-SP_totaal"
VAL
1.0
MAX
#op_positie

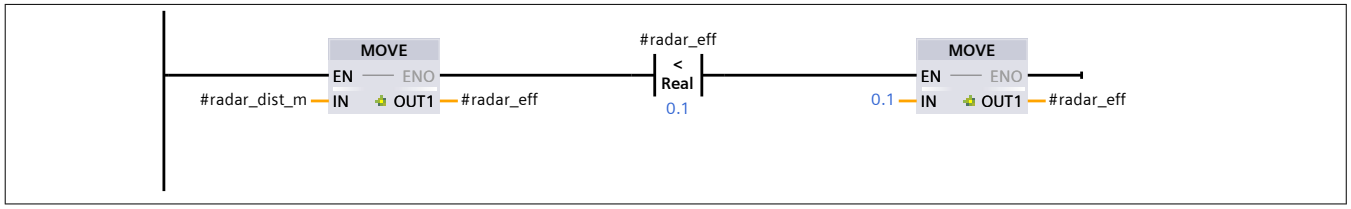
PLC_1 [CPU 1215C AC/DC/Rly] / Program blocks

FB_M2_AngleComp [FB5]

FB_M2_AngleComp Properties					
General					
Name	FB_M2_AngleComp	**Number**	5	**Type**	FB
Language	LAD	**Numbering**	Automatic		
Information					
Title		**Author**		**Comment**	
Family		**Version**	0.1	**User-defined ID**	

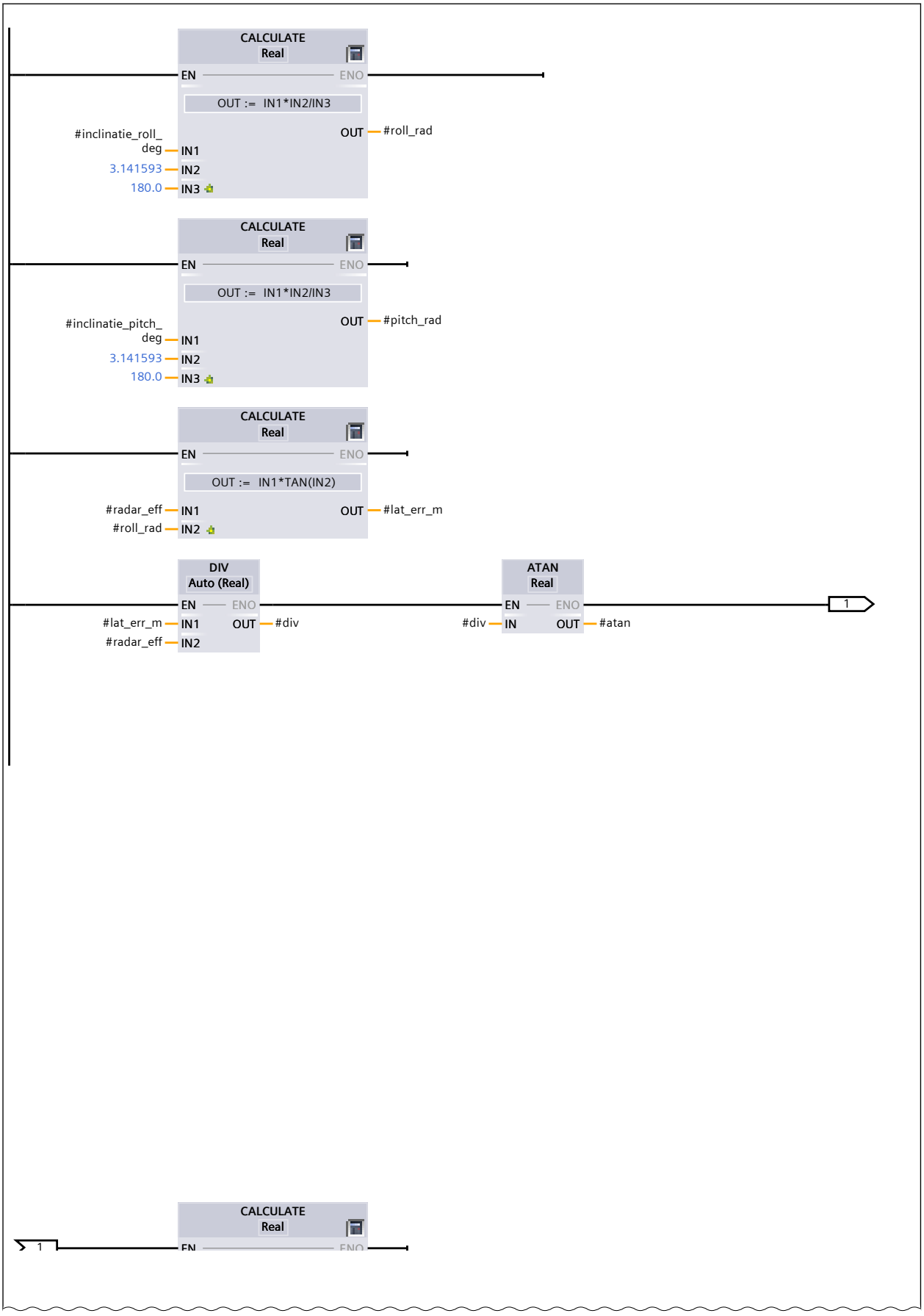
Name	Data type	Default value	Retain
Input			
inclinatie_pitch_deg	Real	0.0	Non-retain
inclinatie_roll_deg	Real	0.0	Non-retain
hoek_M2_deg	Real	0.0	Non-retain
radar_dist_m	Real	0.0	Non-retain
laser_dist_m	Real	0.0	Non-retain
enable	Bool	false	Non-retain
reset	Bool	false	Non-retain
doel_hoek_deg	Real	0.0	Non-retain
L_arm_m	Real	0.0	Non-retain
out_min_deg	Real	0.0	Non-retain
out_max_deg	Real	0.0	Non-retain
Output			
M2_setpoint_deg	Real	0.0	Non-retain
FF_bijdrage_deg	Real	0.0	Non-retain
FB_bijdrage_deg	Real	0.0	Non-retain
fout_deg	Real	0.0	Non-retain
lat_err_mm	Real	0.0	Non-retain
op_positie	Bool	false	Non-retain
InOut			
pid_state	"UDT_PIDState"		
Static			
roll_rad	Real	0.0	Non-retain
pitch_rad	Real	0.0	Non-retain
lat_err_m	Real	0.0	Non-retain
FF_lat_deg	Real	0.0	Non-retain
FF_pitch_deg	Real	0.0	Non-retain
FF_totaal_deg	Real	0.0	Non-retain
SP_totaal	Real	0.0	Non-retain
radar_eff	Real	0.0	Non-retain
pid_inst	"FB_PID_M0"		
div	Real	0.0	Non-retain
atan	Real	0.0	Non-retain
sub	Real	0.0	Non-retain
pid_inst_M2	"FB_PID_M0"		
Temp			
Constant			

Network 1:



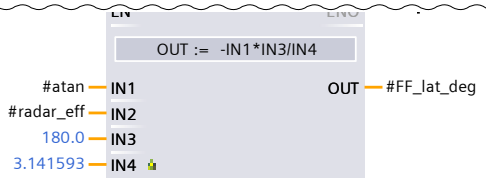
Network 2:

Network 2: (1.1 / 2.1)

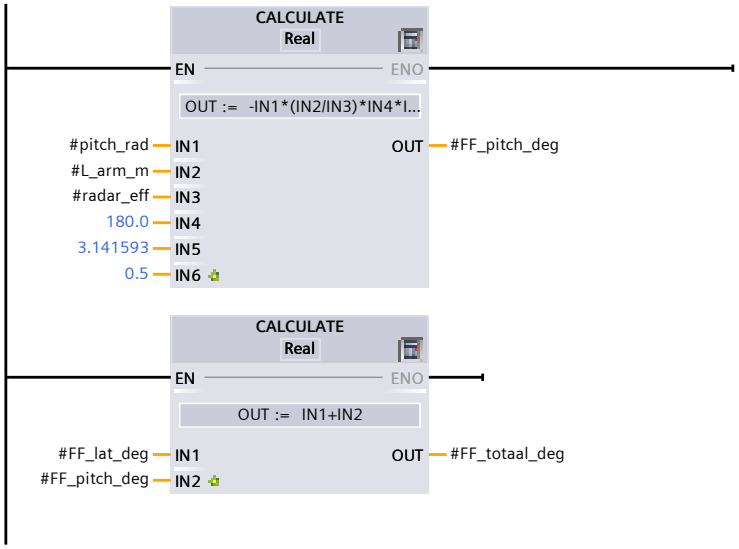


Network 2: (2.1 / 2.1)

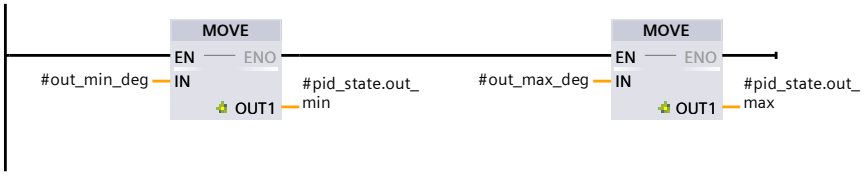
1.1 (Page8 - 3)



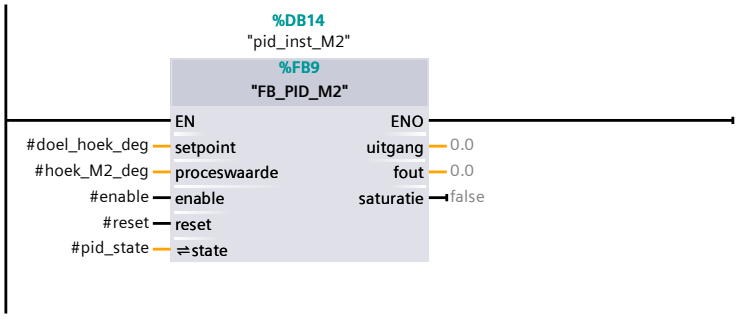
Network 3:



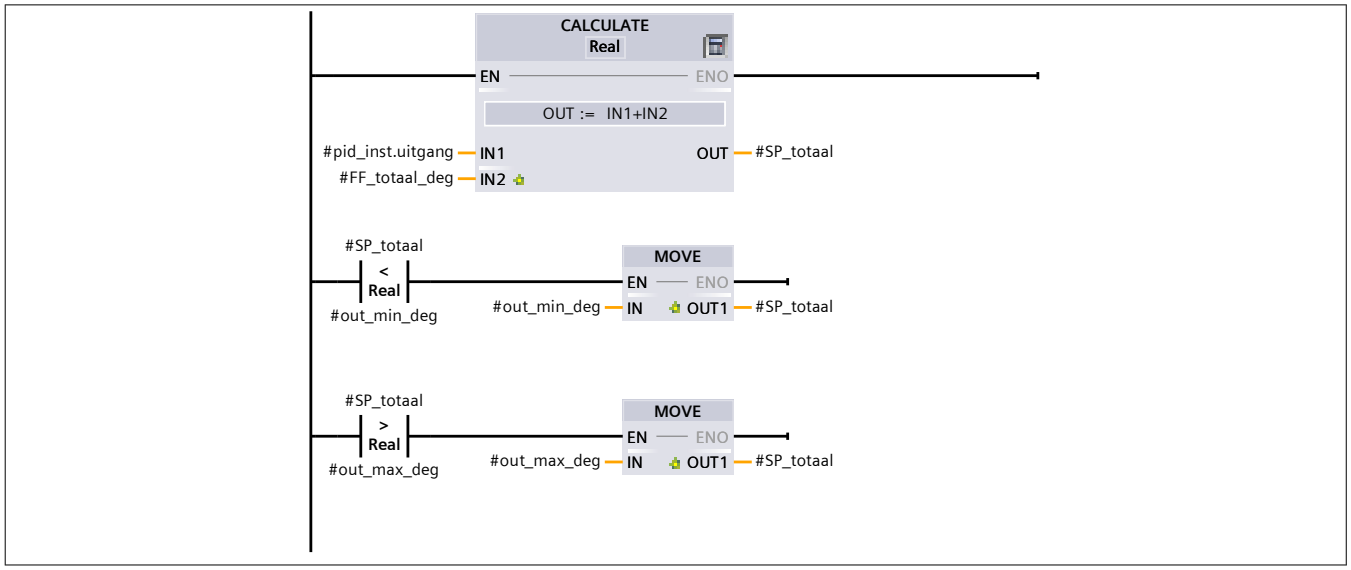
Network 4:



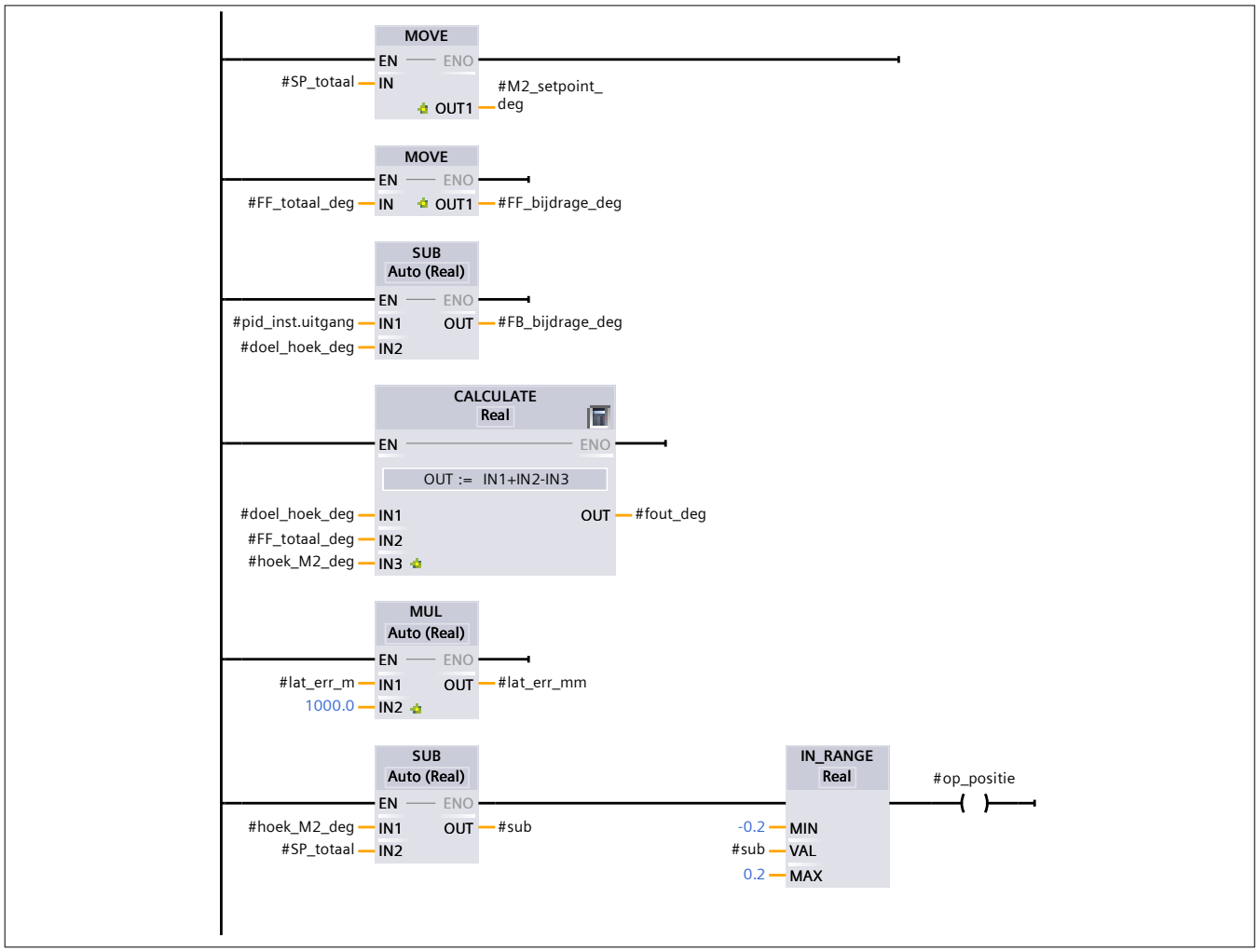
Network 5:



Network 6:


CALCULATE
Real
EN
ENO
OUT := IN1+IN2
#pid_inst.uitgang
IN1
OUT
#SP_totaal
#FF_totaal_deg
IN2
#SP_totaal
<
Real
#out_min_deg
MOVE
EN
ENO
#out_min_deg
IN
OUT1
#SP_totaal
#SP_totaal
>
Real
#out_max_deg
MOVE
EN
ENO
#out_max_deg
IN
OUT1
#SP_totaal

Network 7:


MOVE
EN
ENO
#SP_totaal
IN
OUT1
#M2_setpoint_
deg
MOVE
EN
ENO
#FF_totaal_deg
IN
OUT1
#FF_bijdrage_deg
SUB
Auto (Real)
EN
ENO
#pid_inst.uitgang
IN1
OUT
#FB_bijdrage_deg
#doel_hoek_deg
IN2
CALCULATE
Real
EN
ENO
OUT := IN1+IN2-IN3
#doel_hoek_deg
IN1
OUT
#fout_deg
#FF_totaal_deg
IN2
#hoek_M2_deg
IN3
MUL
Auto (Real)
EN
ENO
#lat_err_m
IN1
OUT
#lat_err_mm
1000.0
IN2
SUB
Auto (Real)
EN
ENO
#hoek_M2_deg
IN1
OUT
#sub
#SP_totaal
IN2
IN_RANGE
Real
-0.2
MIN
#sub
VAL
0.2
MAX
#op_positie

PLC_1 [CPU 1215C AC/DC/Rly] / Program blocks

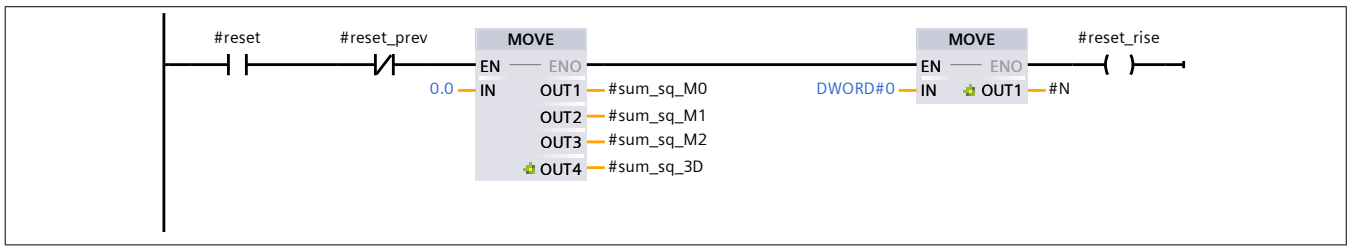
FB_MSE_Calculator [FB6]

FB_MSE_Calculator Properties					
General					
Name	FB_MSE_Calculator	**Number**	6	**Type**	FB
Language	LAD	**Numbering**	Automatic		
Information					
Title		**Author**		**Comment**	
Family		**Version**	0.1	**User-defined ID**	

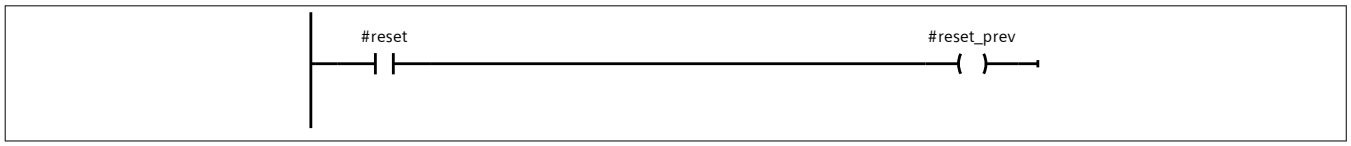
Name	Data type	Default value	Retain
Input			
err_M0_m	Real	0.0	Non-retain
err_M1_mm	Real	0.0	Non-retain
err_M2_deg	Real	0.0	Non-retain
ea_x_err_mm	Real	0.0	Non-retain
ea_y_err_mm	Real	0.0	Non-retain
ea_z_err_mm	Real	0.0	Non-retain
enable	Bool	false	Non-retain
reset	Bool	false	Non-retain
bereik_M0_m	Real	0.0	Non-retain
bereik_M1_mm	Real	0.0	Non-retain
bereik_M2_deg	Real	0.0	Non-retain
Output			
mse_M0	Real	0.0	Non-retain
mse_M1	Real	0.0	Non-retain
mse_M2	Real	0.0	Non-retain
mse_3D	Real	0.0	Non-retain
rmse_M0	Real	0.0	Non-retain
rmse_M1	Real	0.0	Non-retain
rmse_M2	Real	0.0	Non-retain
rmse_3D	Real	0.0	Non-retain
nrmse_M0	Real	0.0	Non-retain
nrmse_M1	Real	0.0	Non-retain
nrmse_M2	Real	0.0	Non-retain
dominante_motor	Int	0	Non-retain
n_samples	DWord	16#0	Non-retain
InOut			
Static			
sum_sq_M0	Real	0.0	Non-retain
sum_sq_M1	Real	0.0	Non-retain
sum_sq_M2	Real	0.0	Non-retain
sum_sq_3D	Real	0.0	Non-retain
N	DWord	16#0	Non-retain
err_3D_sq	Real	0.0	Non-retain
reset_prev	Bool	false	Non-retain
max_nrmse	Real	0.0	Non-retain
ber_M0	Real	0.0	Non-retain
ber_M1	Real	0.0	Non-retain
ber_M2	Real	0.0	Non-retain

Name	Data type	Default value	Retain
reset_rise	Bool	false	Non-retain
has_samples	Bool	false	Non-retain
Temp			
Constant			

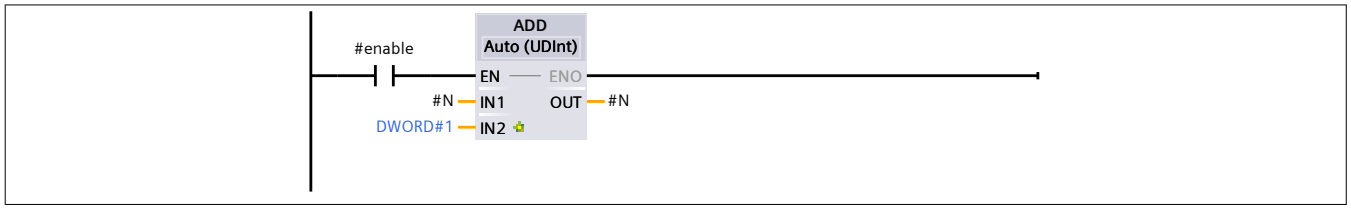
Network 1:



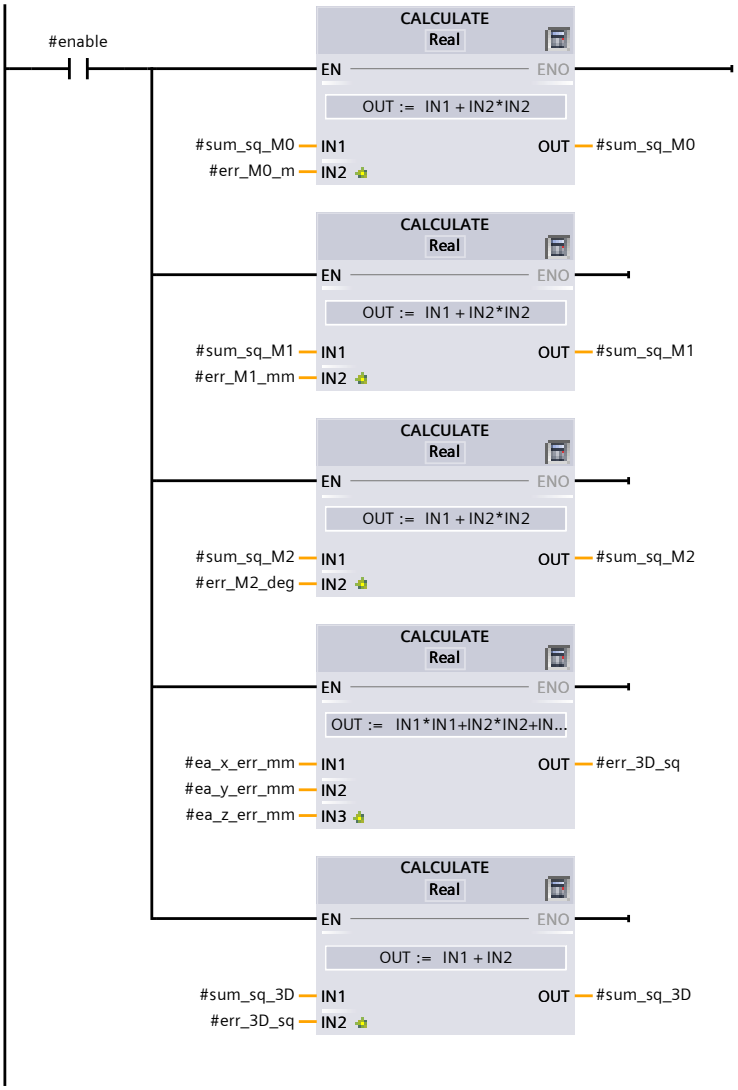
Network 2:



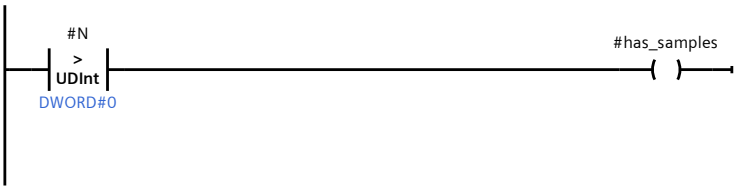
Network 3:



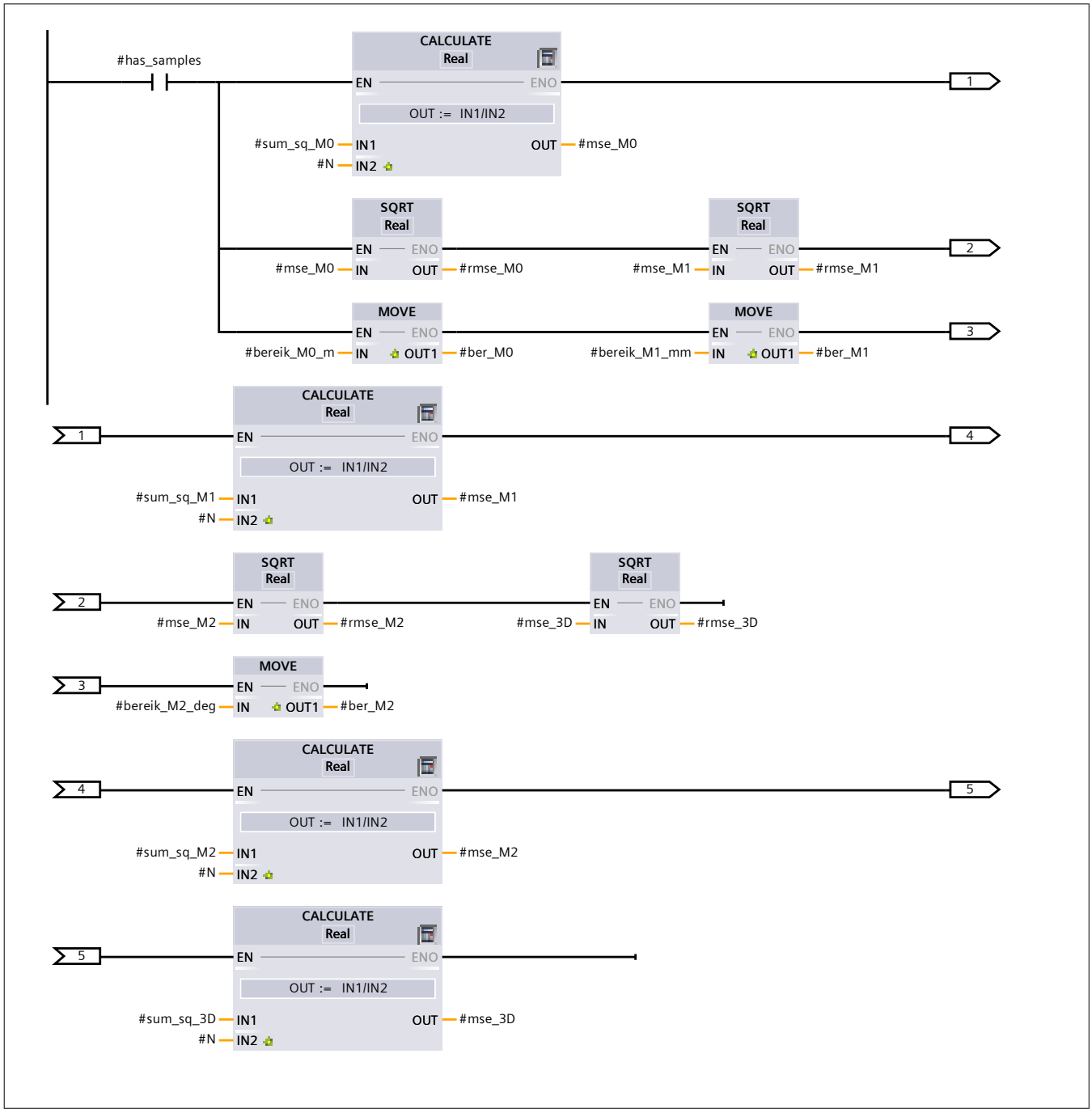
Network 4:



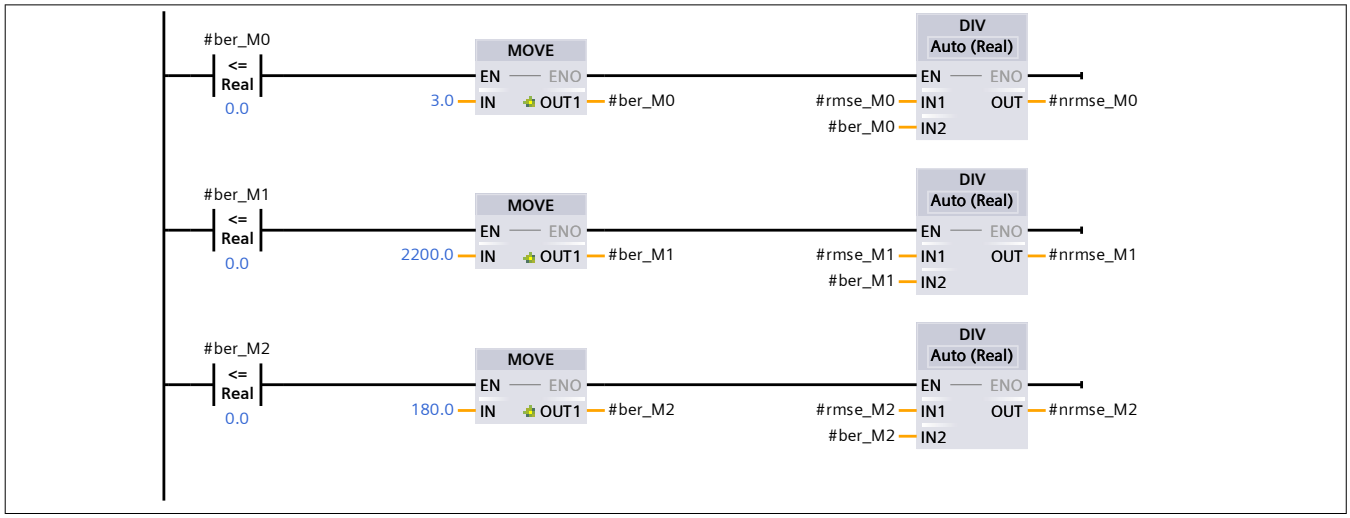
Network 5:



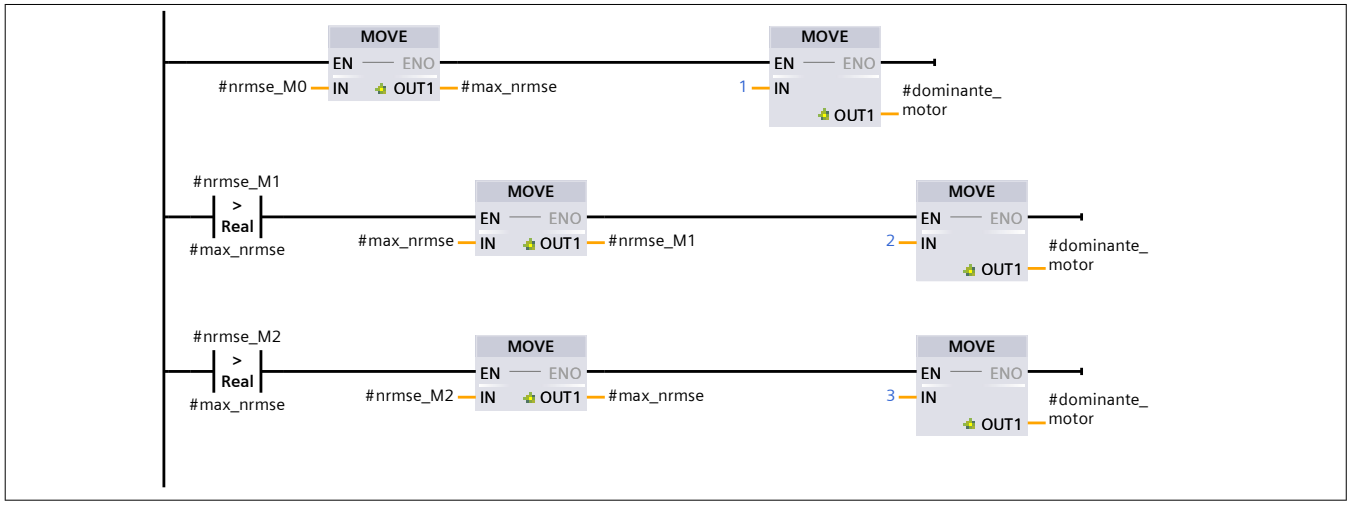
Network 6:


#has_samples
CALCULATE
Real
EN
ENO
OUT := IN1/IN2
#sum_sq_M0
IN1
#N
IN2
OUT
#mse_M0
1
SQRT
Real
#mse_M0
IN
#rmse_M0
#mse_M1
#rmse_M1
2
MOVE
#bereik_M0_m
OUT1
#ber_M0
#bereik_M1_mm
#ber_M1
3
#sum_sq_M1
#mse_M1
4
#mse_M2
#rmse_M2
#mse_3D
#rmse_3D
#bereik_M2_deg
#ber_M2
#sum_sq_M2
#mse_M2
5
#sum_sq_3D
#mse_3D

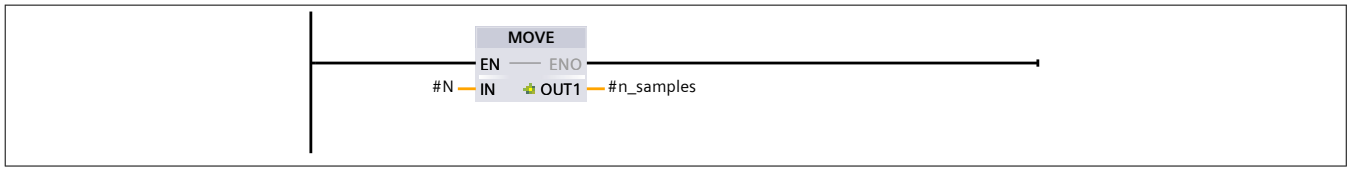
Network 7:



Network 8:



Network 9:



PLC_1 [CPU 1215C AC/DC/Rly] / Program blocks

FB_PathFollower [FB7]

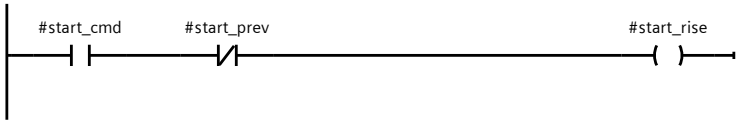
FB_PathFollower Properties					
General					
Name	FB_PathFollower	**Number**	7	**Type**	FB
Language	LAD	**Numbering**	Automatic		
Information					
Title		**Author**		**Comment**	
Family		**Version**	0.1	**User-defined ID**	

Name	Data type	Default value	Retain
Input			
dist_laser_m	Real	0.0	Non-retain
pos_M1_mm	Real	0.0	Non-retain
hoek_M2_deg	Real	0.0	Non-retain
inclinatie_deg	Real	0.0	Non-retain
start_cmd	Bool	false	Non-retain
e_stop	Bool	false	Non-retain
reset_cmd	Bool	false	Non-retain
doel_afstand_m	Real	0.0	Non-retain
doel_hoogte_mm	Real	0.0	Non-retain
doel_hoek_deg	Real	0.0	Non-retain
start_hoogte_mm	Real	0.0	Non-retain
start_hoek_deg	Real	0.0	Non-retain
M1_op_positie	Bool	false	Non-retain
M2_op_positie	Bool	false	Non-retain
M0_max_snelheid	Real	0.0	Non-retain
M0_rem_afstand	Real	0.0	Non-retain
M0_pos_tolerantie	Real	0.0	Non-retain
systeem_tijd_ms	DWord	16#0	Non-retain
Output			
M0_enable	Bool	false	Non-retain
M0_snelheid_pct	Real	0.0	Non-retain
M1_enable	Bool	false	Non-retain
M1_target_mm	Real	0.0	Non-retain
M2_enable	Bool	false	Non-retain
M2_target_deg	Real	0.0	Non-retain
mse_enable	Bool	false	Non-retain
mse_reset	Bool	false	Non-retain
toestand	Int	0	Non-retain
toestand_naam	String[30]	''	Non-retain
fout_code	Int	0	Non-retain
doel_bereikt	Bool	false	Non-retain
InOut			
Static			
sm_toestand	Int	0	Non-retain
sm_toestand_prev	Int	-1	Non-retain
sm_timer_ms	DWord	16#0	Non-retain
sm_entry	Bool	false	Non-retain
start_prev	Bool	false	Non-retain

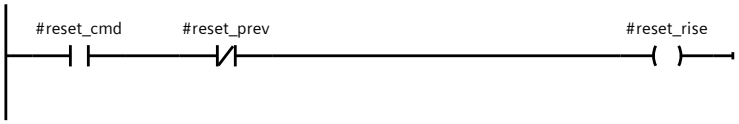
Name	Data type	Default value	Retain
reset_prev	Bool	false	Non-retain
mse_reset_intern	Bool	false	Non-retain
M0_snelheid_berekend	Real	0.0	Non-retain
rem_factor	Real	0.0	Non-retain
home_timeout_ms	DWord	DWORD#5000	Non-retain
M1_huidig_target	Real	0.0	Non-retain
M2_huidig_target	Real	0.0	Non-retain
start_rise	Bool	false	Non-retain
reset_rise	Bool	false	Non-retain
sequence_0	Bool	false	Non-retain
sequence_1	Bool	false	Non-retain
sequence_2	Bool	false	Non-retain
sequence_3	Bool	false	Non-retain
sequence_4	Bool	false	Non-retain
sequence_5	Bool	false	Non-retain
sequence_6	Bool	false	Non-retain
sequence_7	Bool	false	Non-retain
sequence_8	Bool	false	Non-retain
sequence_9	Bool	false	Non-retain
tijd_delta_ms	DWord	16#0	Non-retain
home_timeout	Bool	false	Non-retain
dist_in_remgebied	Bool	false	Non-retain
rem_te_klein_01	Bool	false	Non-retain
doel_nabij	Bool	false	Non-retain
cond_dist_gt0	Bool	false	Non-retain
cond_rem_gt0	Bool	false	Non-retain
rem_te_klein_005	Bool	false	Non-retain
rem_te_groot_10	Bool	false	Non-retain
final_timeout	Bool	false	Non-retain
fout_aanwezig	Bool	false	Non-retain
Temp			
Constant			

Network 1:

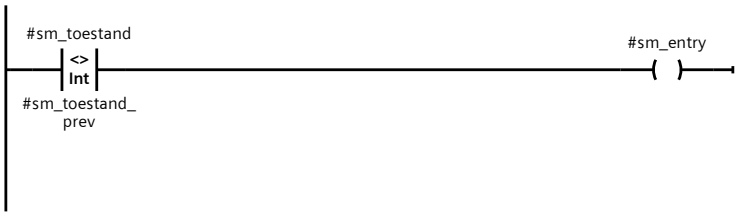
Network 2:



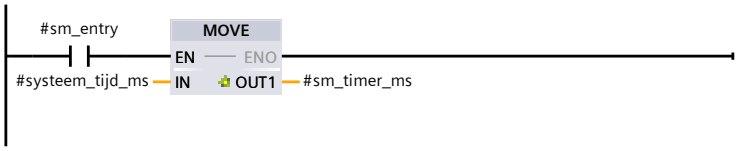
Network 3:



Network 4:

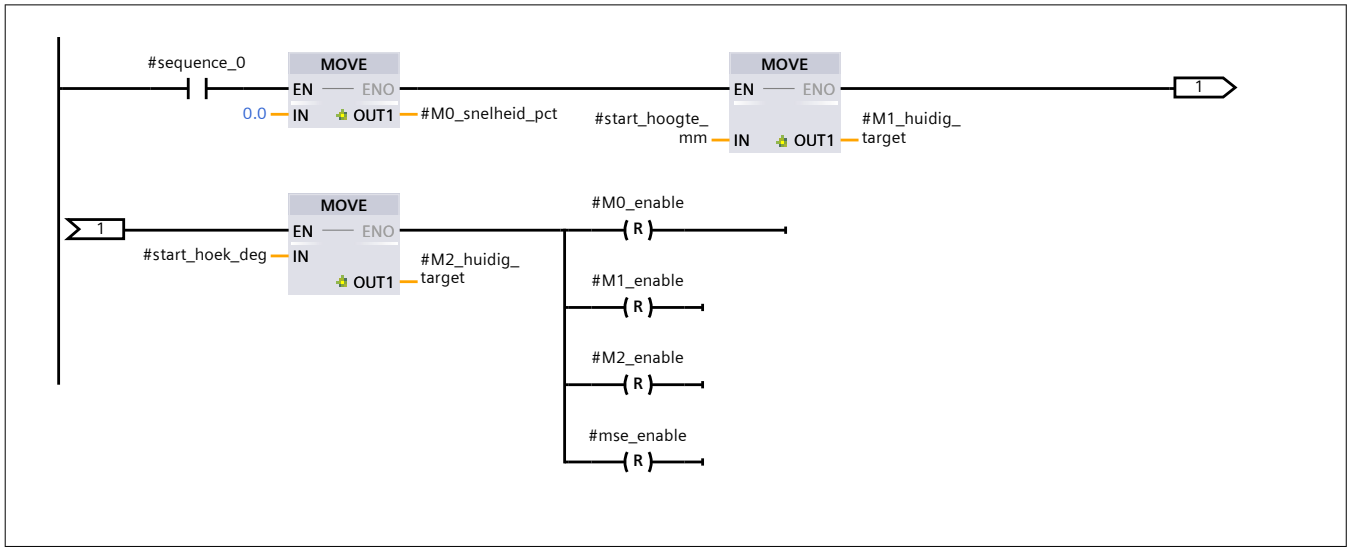


Network 5:

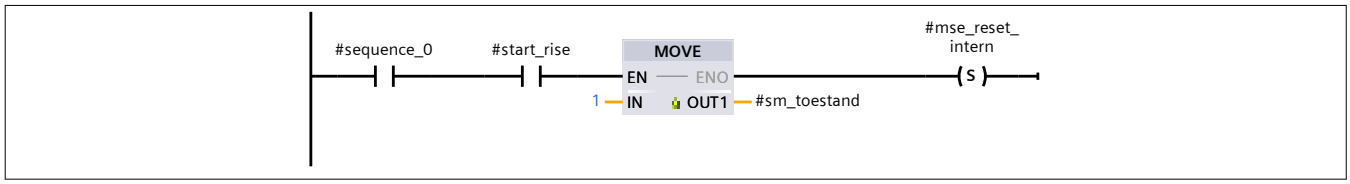


Network 6:

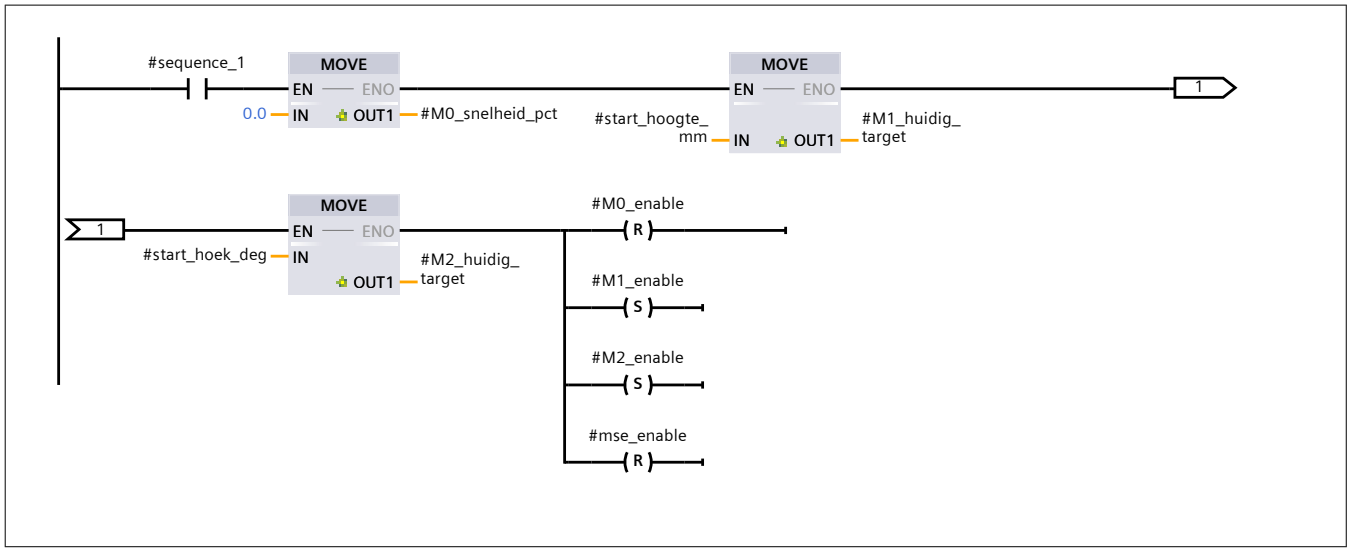
Network 7:



Network 8:



Network 9:



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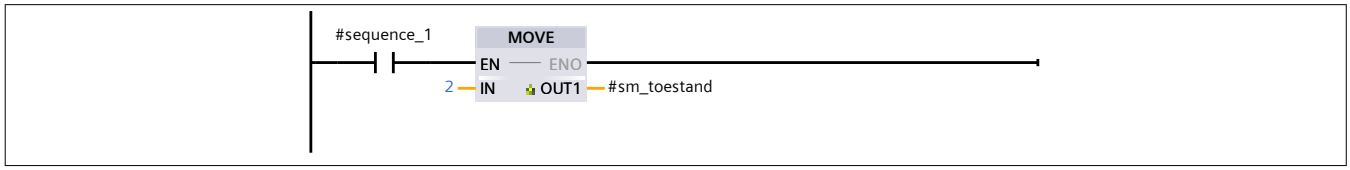
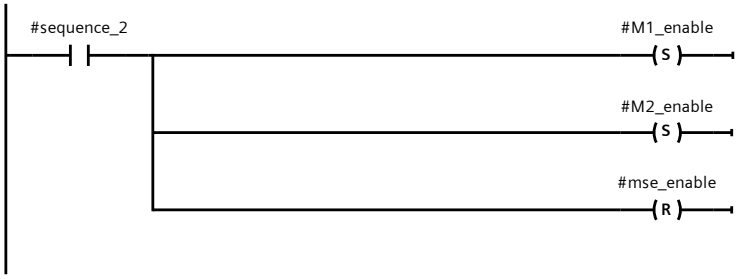


Network 11:



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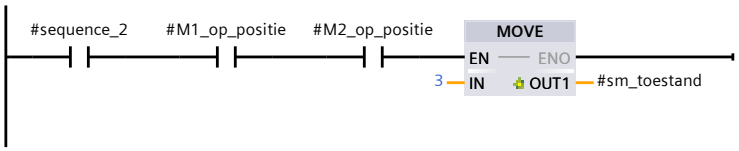

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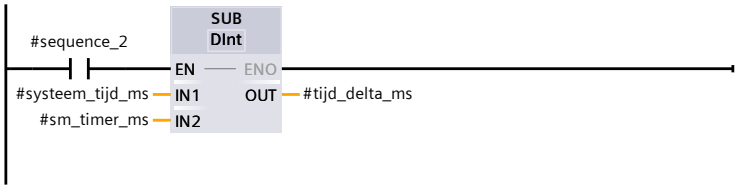
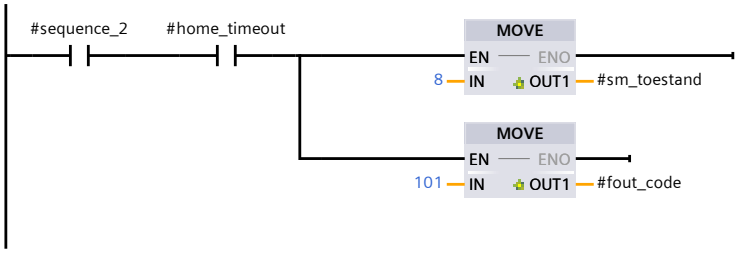


Network 14:

Network 15:



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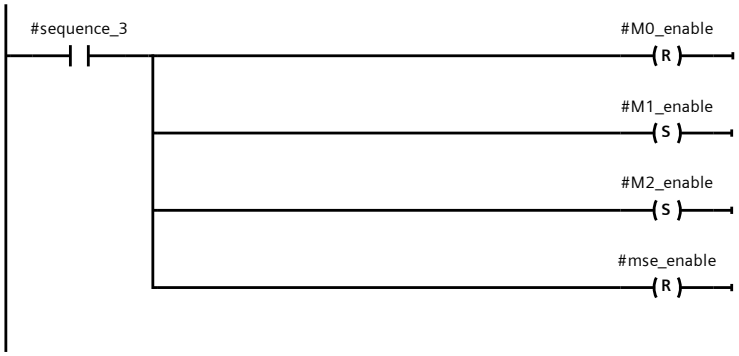
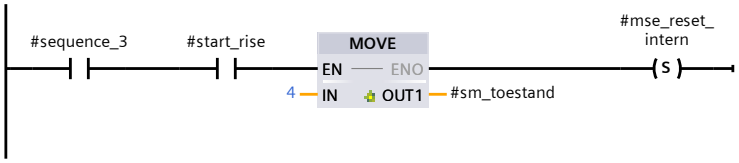


Network 17:



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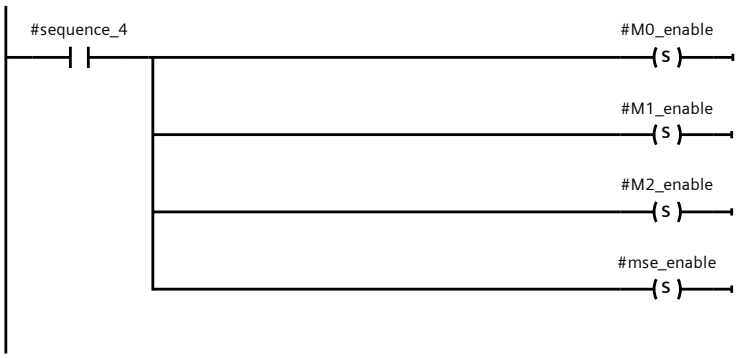

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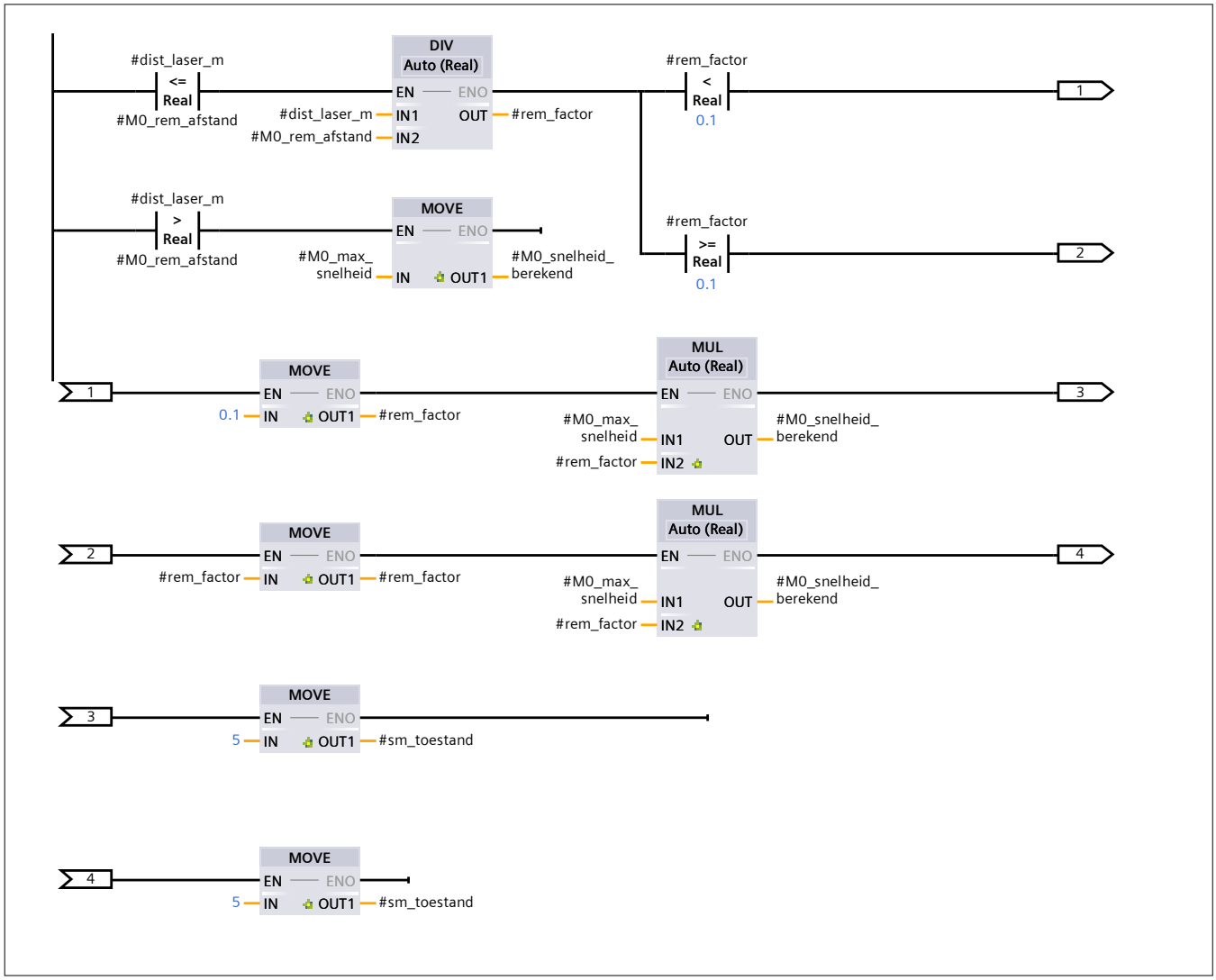

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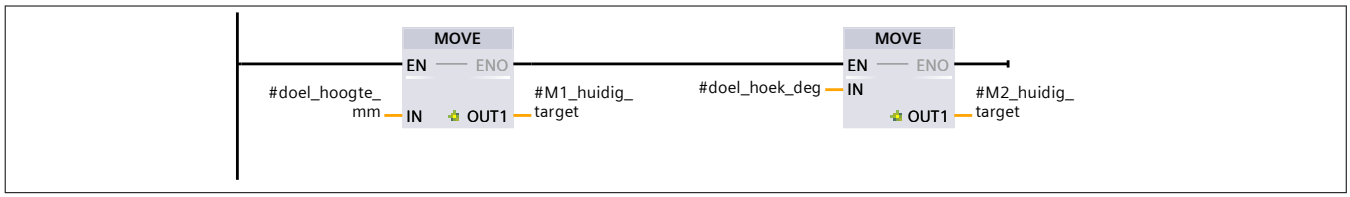

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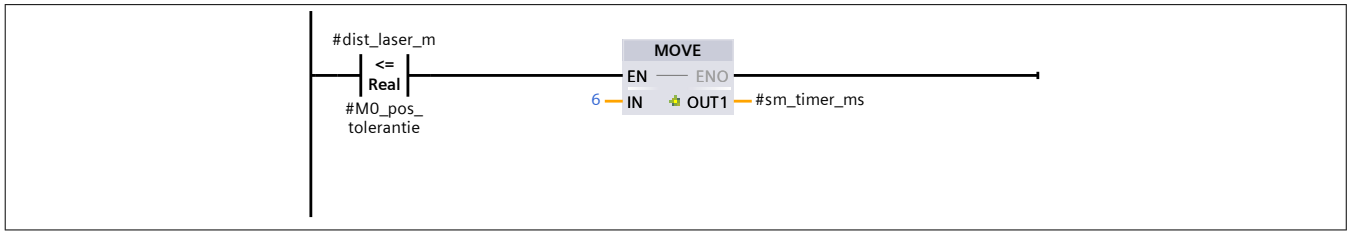

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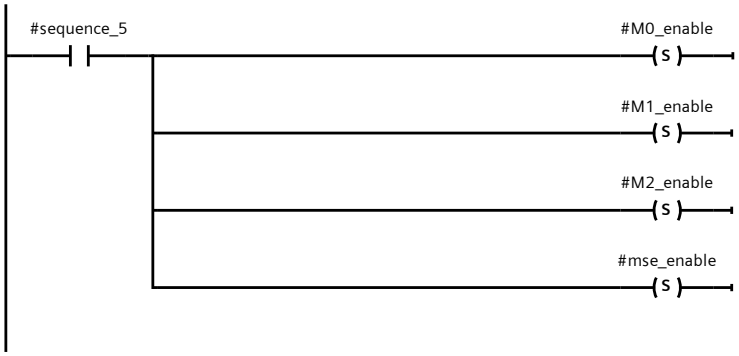
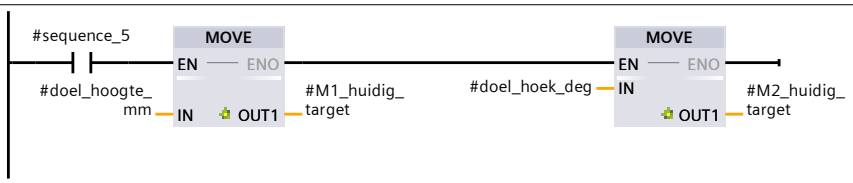
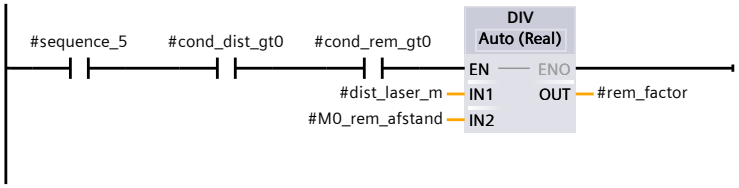


Network 23:



Network 24:

Network 25:



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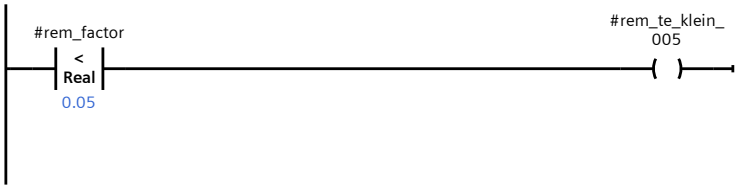
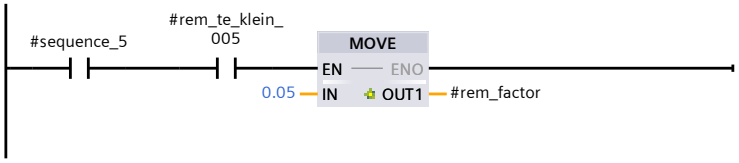
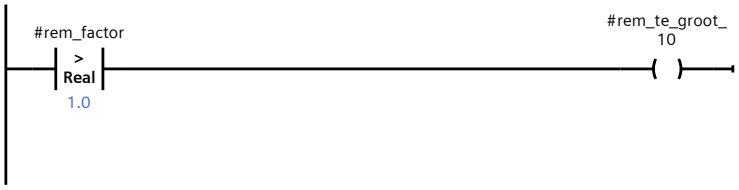


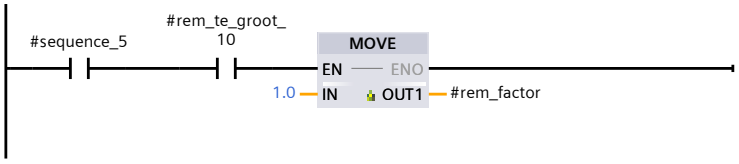
Network 27:



Network 28:



Network 29:



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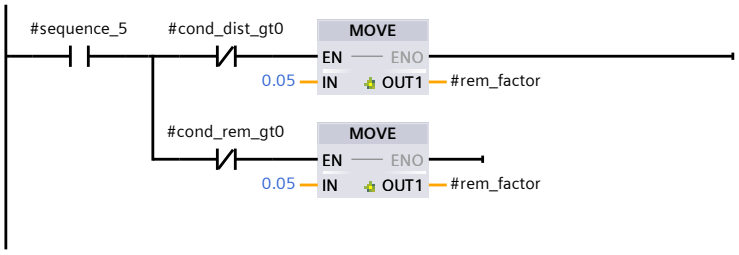

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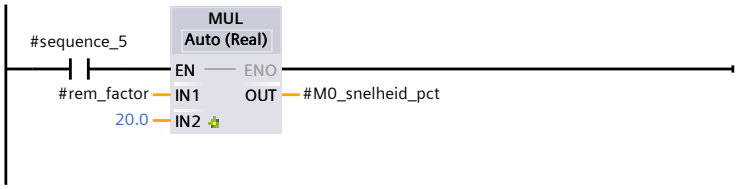

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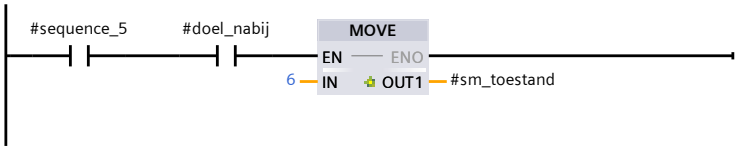

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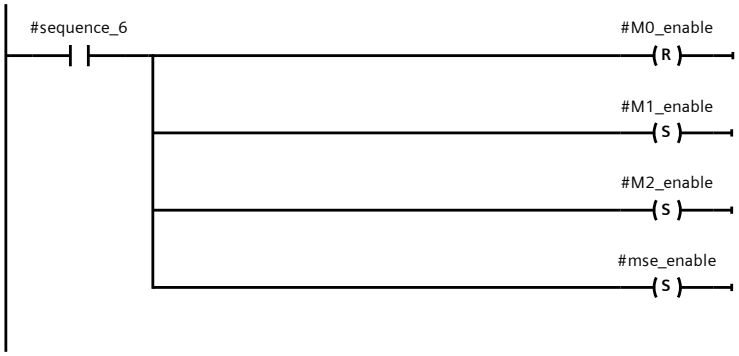

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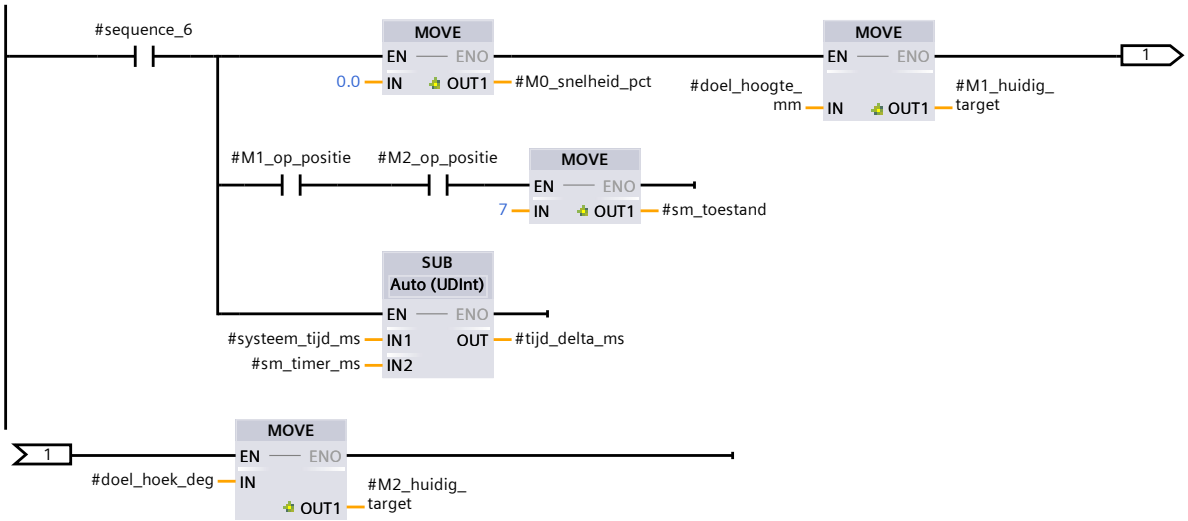

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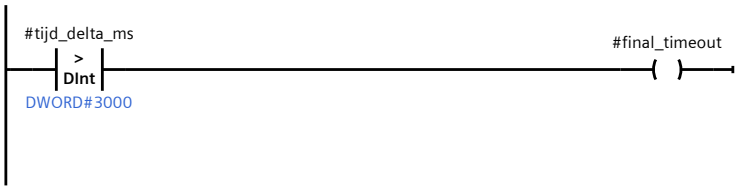

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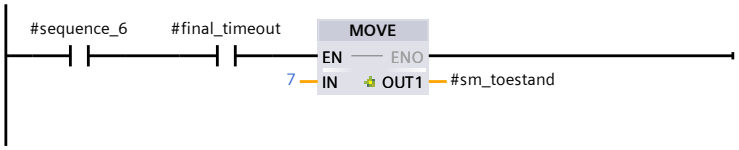

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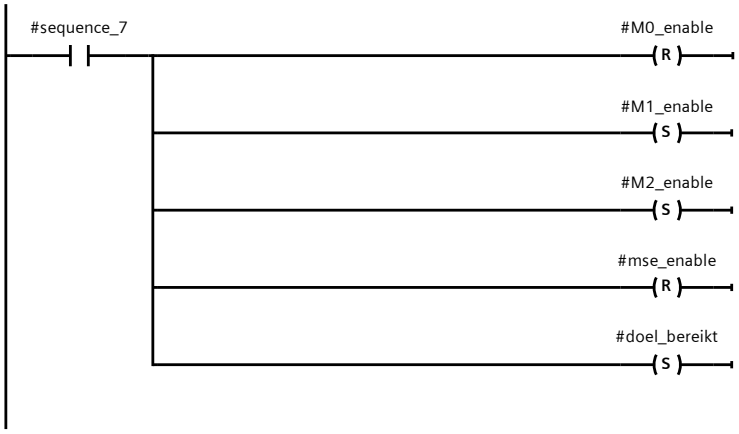

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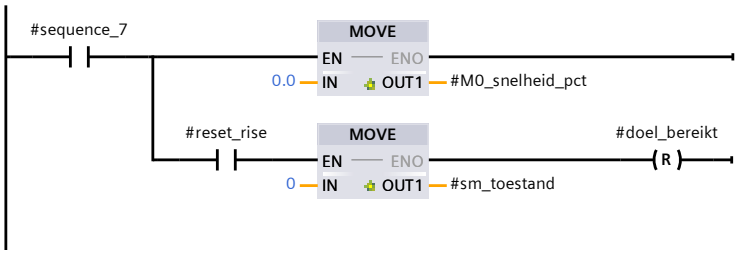

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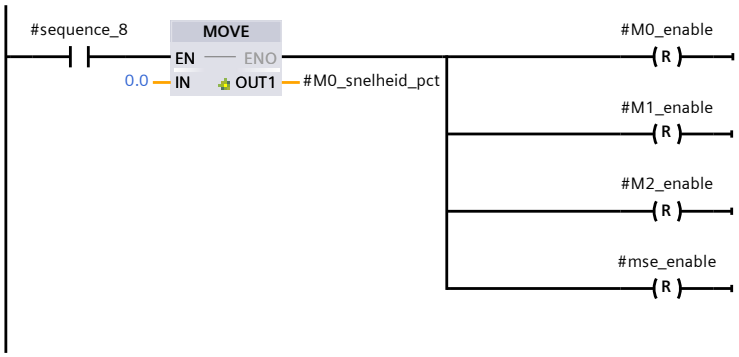

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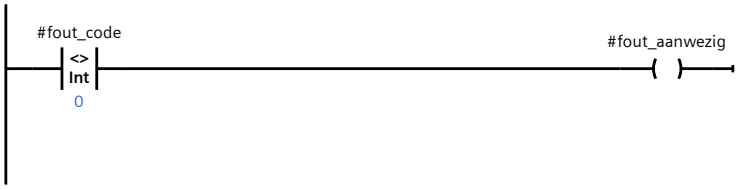

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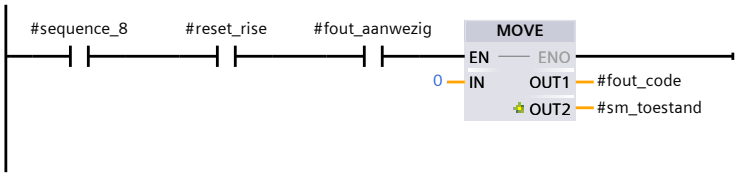
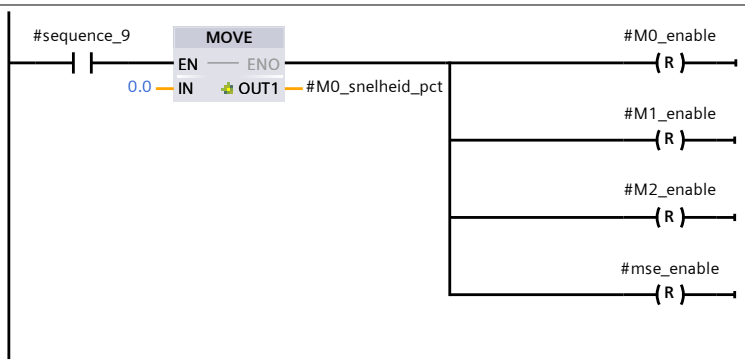


Network 42:



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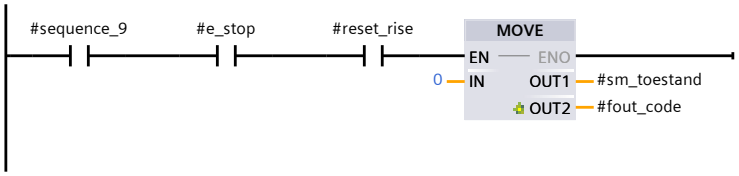
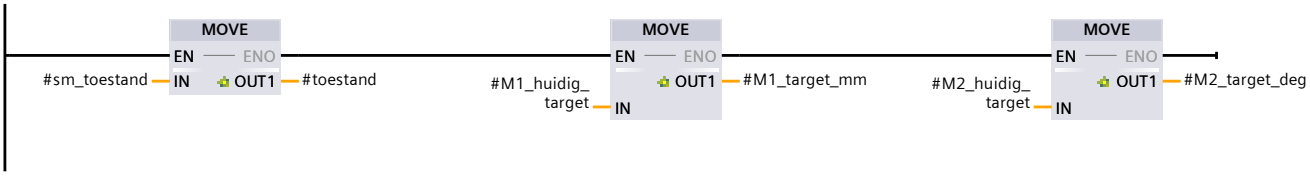
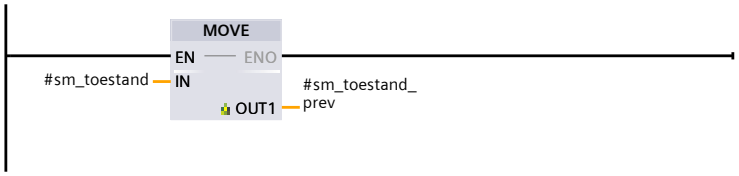


Network 44:

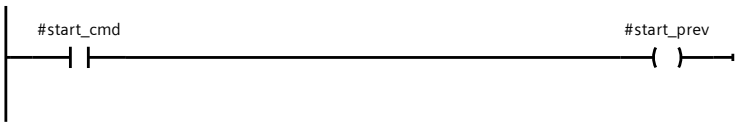


Network 45:

Network 46:



Network 47:



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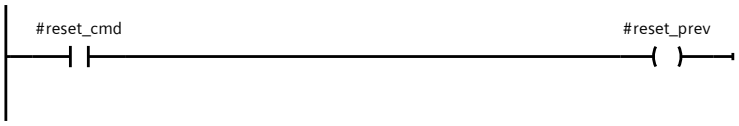


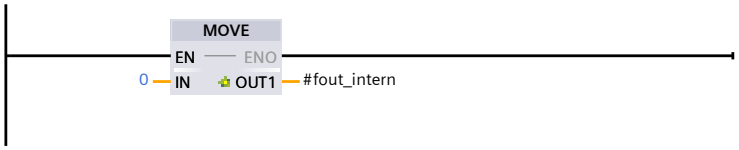
PLC_1 [CPU 1215C AC/DC/Rly] / Program blocks

FB_Safety [FB8]

FB_Safety Properties					
General					
Name	FB_Safety	**Number**	8	**Type**	FB
Language	LAD	**Numbering**	Automatic		
Information					
Title		**Author**		**Comment**	
Family		**Version**	0.1	**User-defined ID**	

Name	Data type	Default value	Retain
Input			
e_stop_hw	Bool	false	Non-retain
eindsch_M1_boven	Bool	false	Non-retain
eindsch_M1_onder	Bool	false	Non-retain
eindsch_M2_plus	Bool	false	Non-retain
eindsch_M2_min	Bool	false	Non-retain
pos_M1_mm	Real	0.0	Non-retain
hoek_M2_deg	Real	0.0	Non-retain
sensor_watchdog	DWord	16#0	Non-retain
sensor_watchdog_ref	DWord	16#0	Non-retain
systeem_tijd_ms	DWord	16#0	Non-retain
Output			
veilig	Bool	false	Non-retain
e_stop_actief	Bool	false	Non-retain
fout_M1_boven	Bool	false	Non-retain
fout_M1_onder	Bool	false	Non-retain
fout_M2_plus	Bool	false	Non-retain
fout_M2_min	Bool	false	Non-retain
fout_communicatie	Bool	false	Non-retain
fout_code	Int	0	Non-retain
InOut			
Static			
M1_sw_boven_mm	Real	2090.0	Non-retain
M1_sw_onder_mm	Real	110.0	Non-retain
M2_sw_plus_deg	Real	85.5	Non-retain
M2_sw_min_deg	Real	-85.5	Non-retain
wd_prev	DWord	DWord#0	Non-retain
wd_timeout_ms	DWord	DWord#150	Non-retain
wd_timer_ms	DWord	16#0	Non-retain
wd_foutteller	Int	0	Non-retain
fout_intern	Int	0	Non-retain
subst	DWord	16#0	Non-retain
Temp			
Constant			

Network 1:



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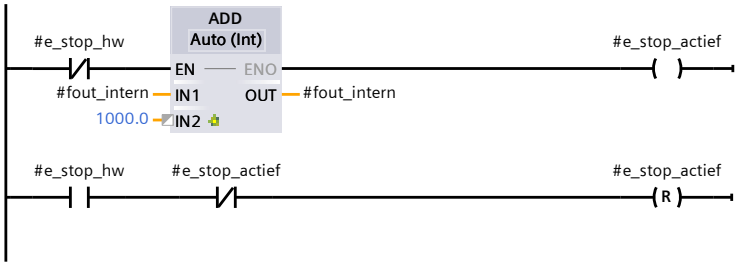
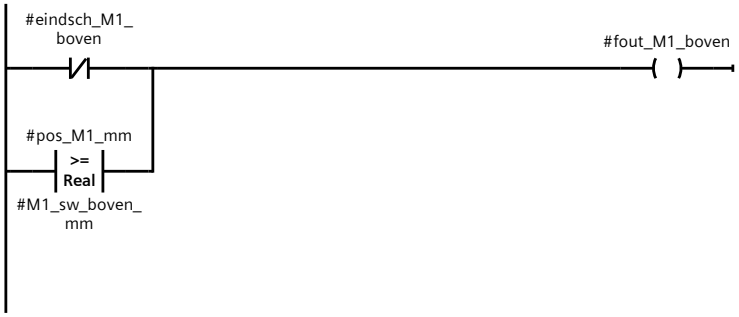
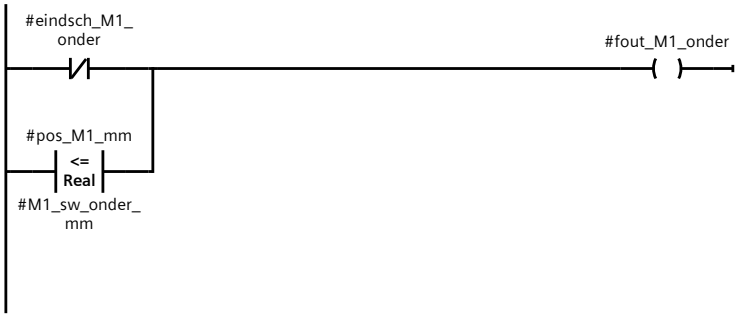


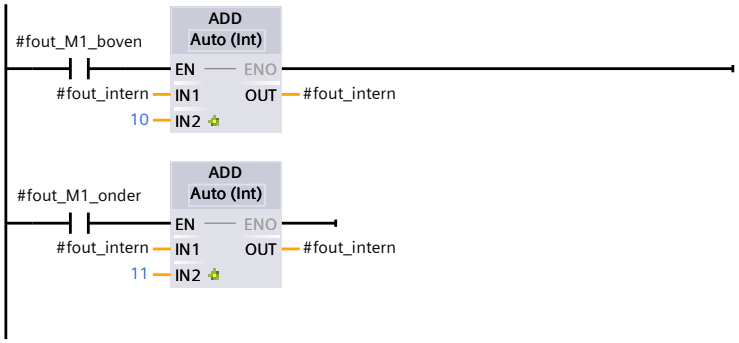
Network 3:



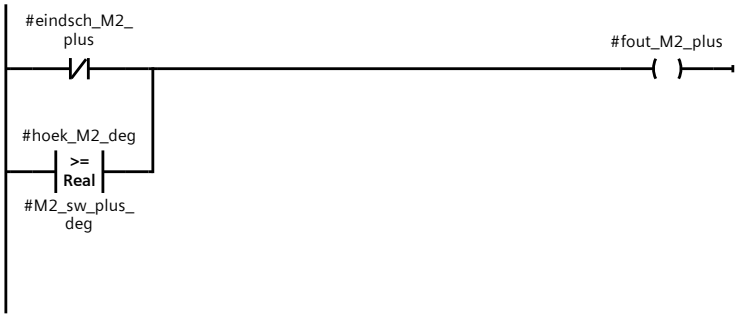
Network 4:



Network 5:

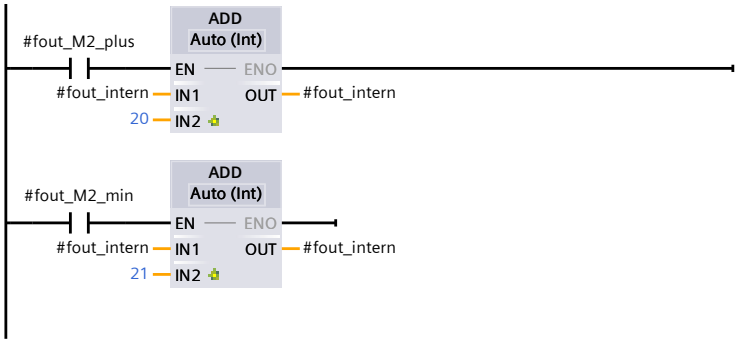


Network 6:

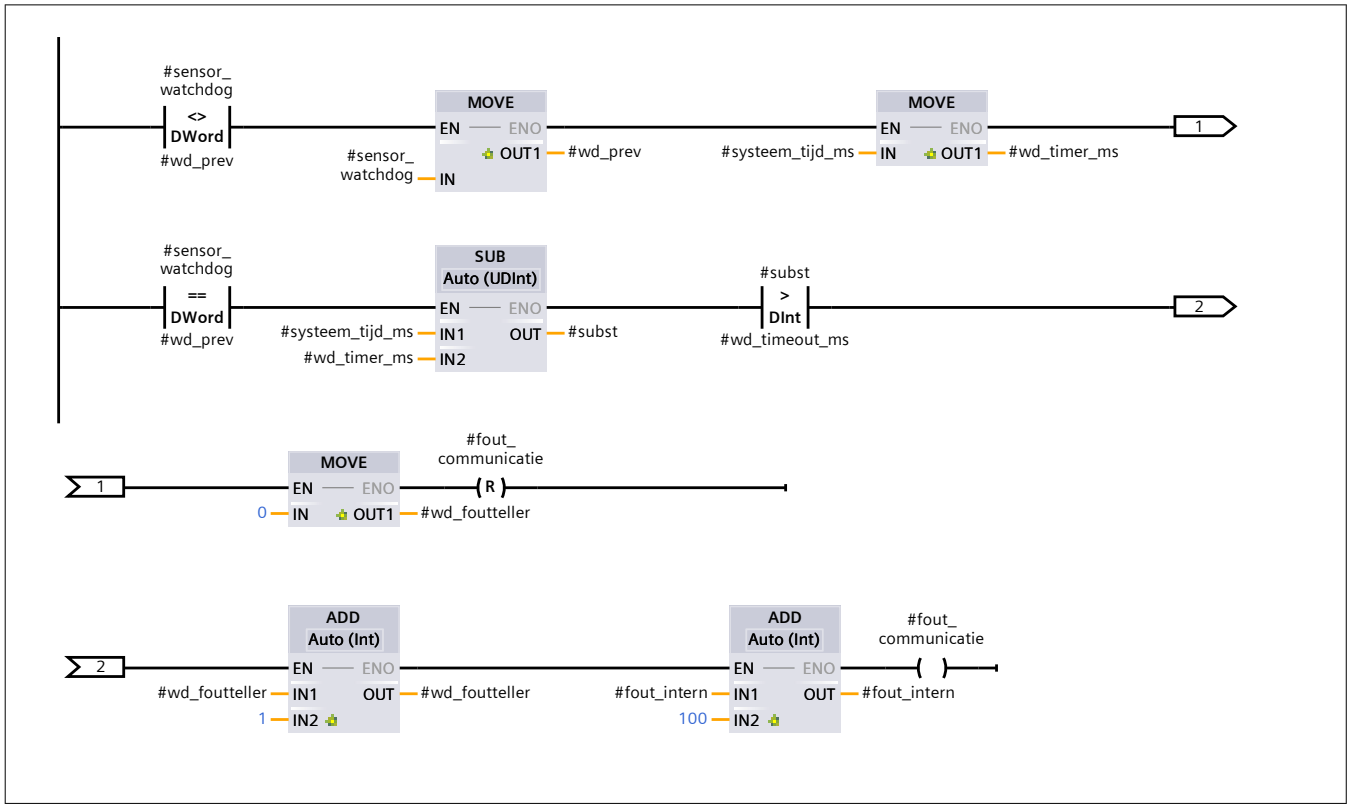


Network 7:

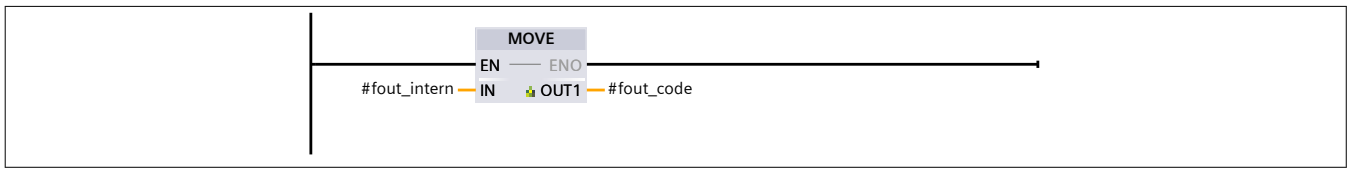
Network 8:



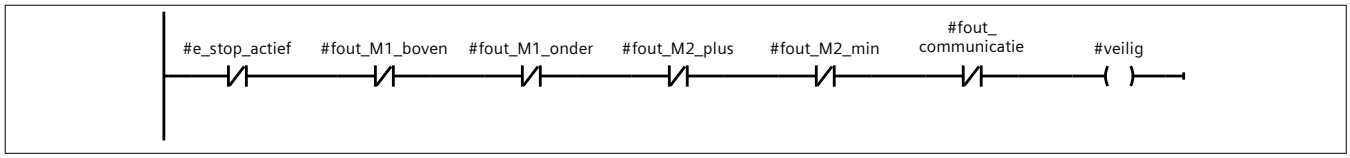
Network 9:



Network 10:



Network 11:



PLC_1 [CPU 1215C AC/DC/Rly] / Program blocks

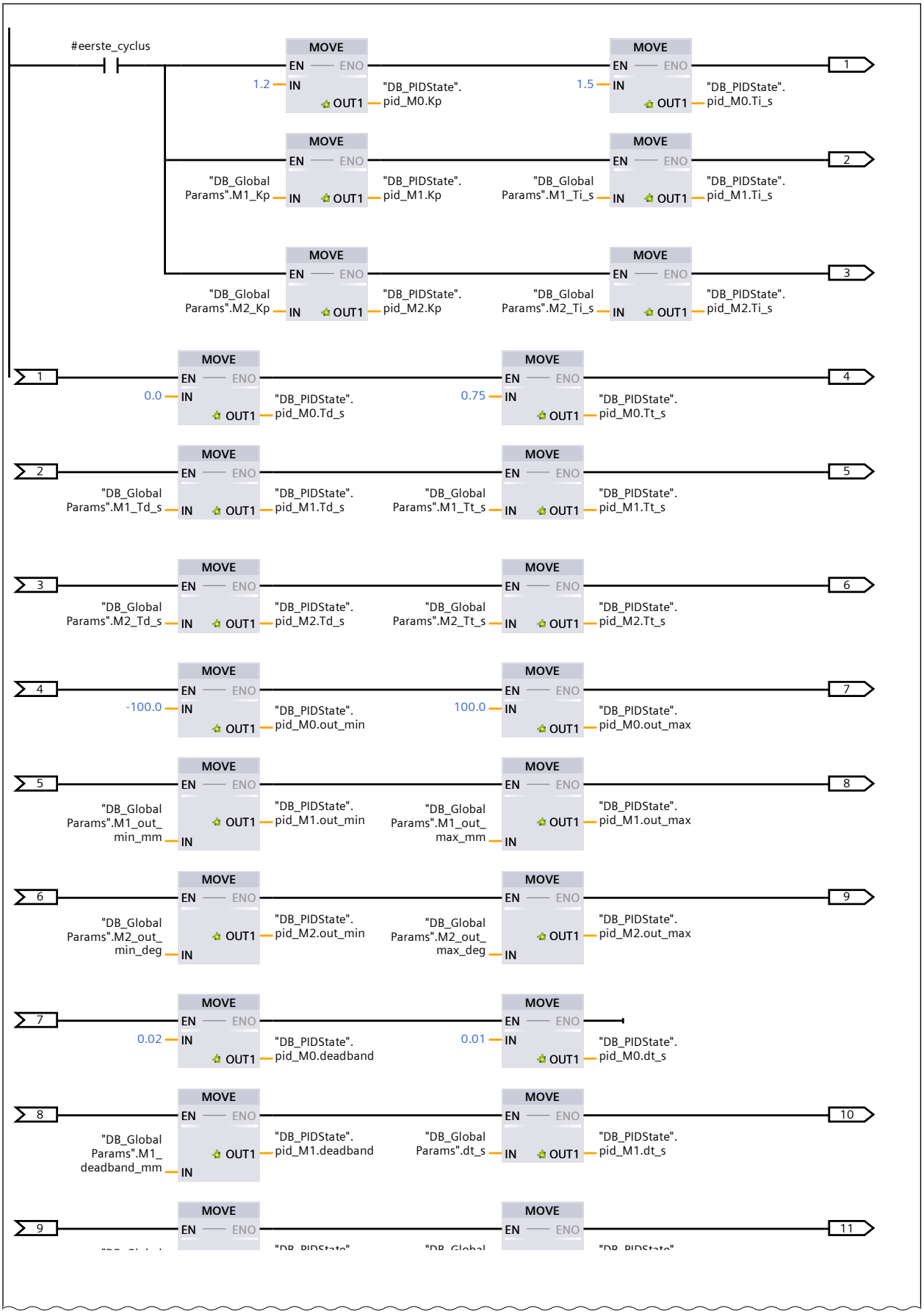
10ms cyclische interrupt — hoofdcontrolecyclus [OB35]

10ms cyclische interrupt — hoofdcontrolecyclus Properties					
General					
Name	10ms cyclische interrupt — hoofdcontrolecyclus	**Number**	35	**Type**	OB
Language	LAD	**Numbering**	Manual		
Information					
Title		**Author**		**Comment**	
Family		**Version**	0.1	**User-defined ID**	

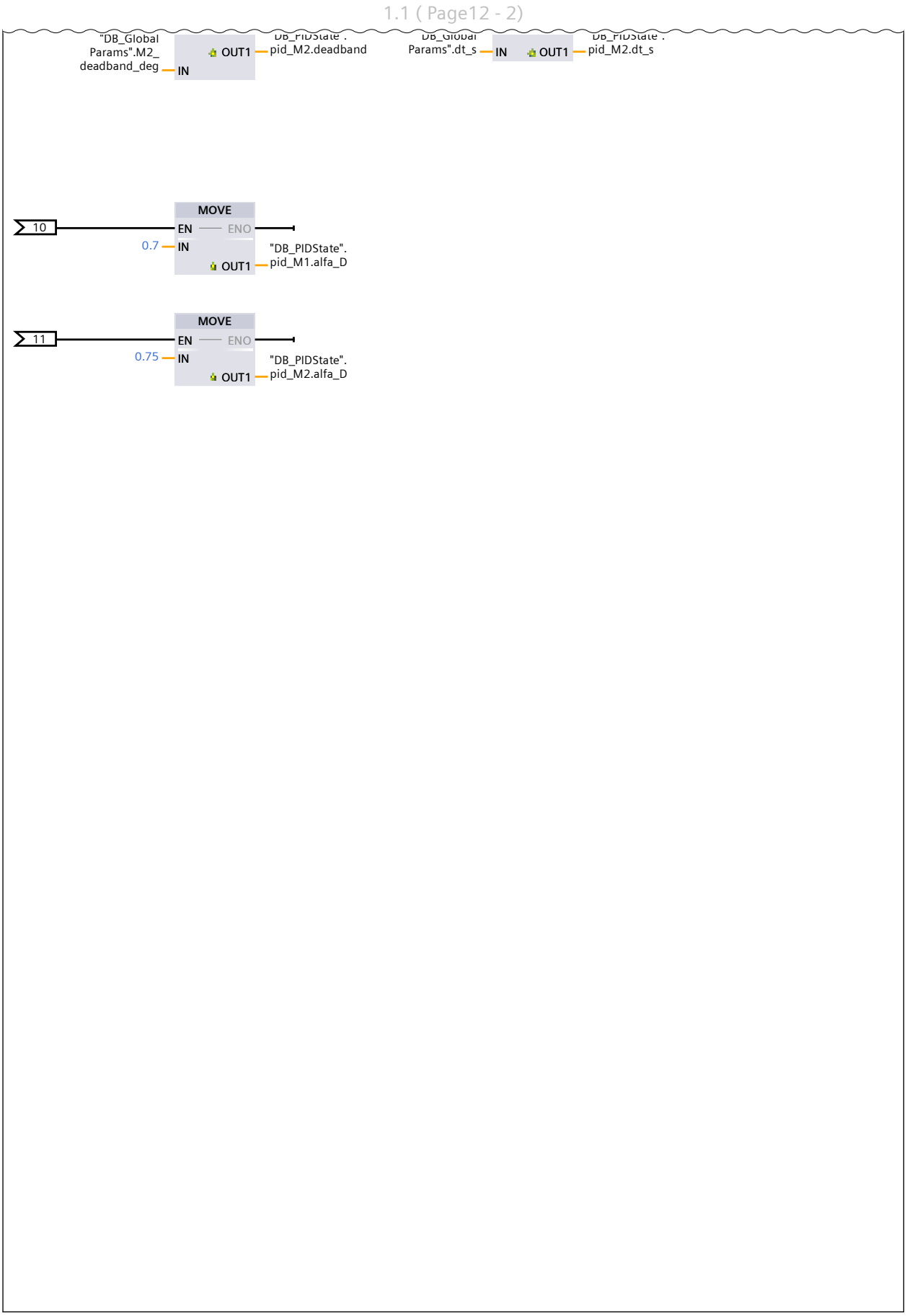
Name	Data type	Default value
Input		
Initial_Call	Bool	
Event_Count	Int	
Temp		
M1_sp_comp_mm	Real	
M2_sp_comp_deg	Real	
ea_x_doel_mm	Real	
ea_y_doel_mm	Real	
ea_z_doel_mm	Real	
ea_err_x	Real	
ea_err_y	Real	
ea_err_z	Real	
incl_gefilterd_prev	Real	
log_interval_teller	Int	
snap_idx	Int	
norm_tijd_s	Real	
M0_pid_uitgang	Real	
M0_fout	Real	
log_teller	Int	
tijd_start_ms	DWord	
eerste_cyclus	Bool	
pidstate	"UDT_PIDState"	
abs(pid)	Real	
abs(pathFollower)	Real	
Constant		

Network 1:

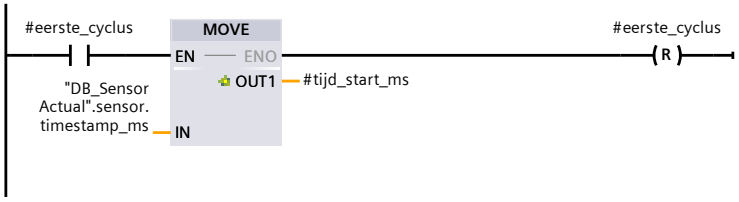
Network 1: (1.1 / 2.1)



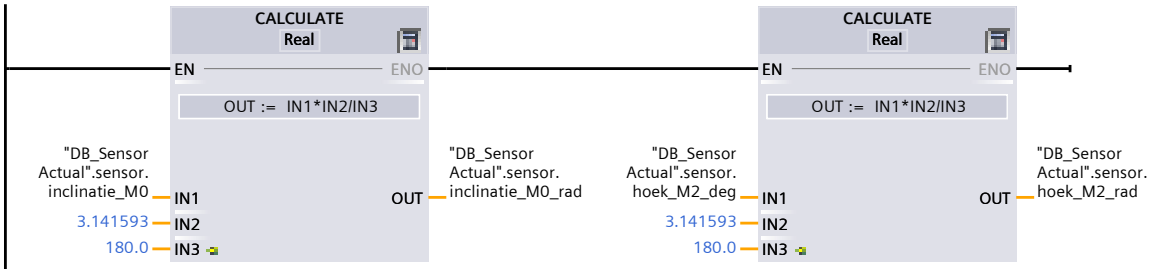
Network 1: (2.1 / 2.1)



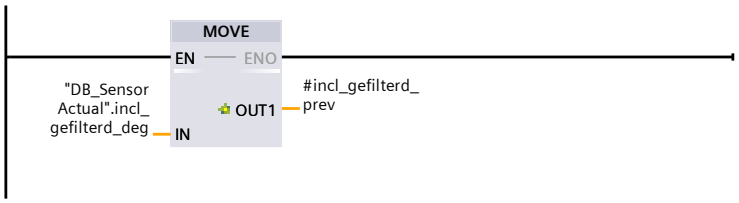
Network 2:



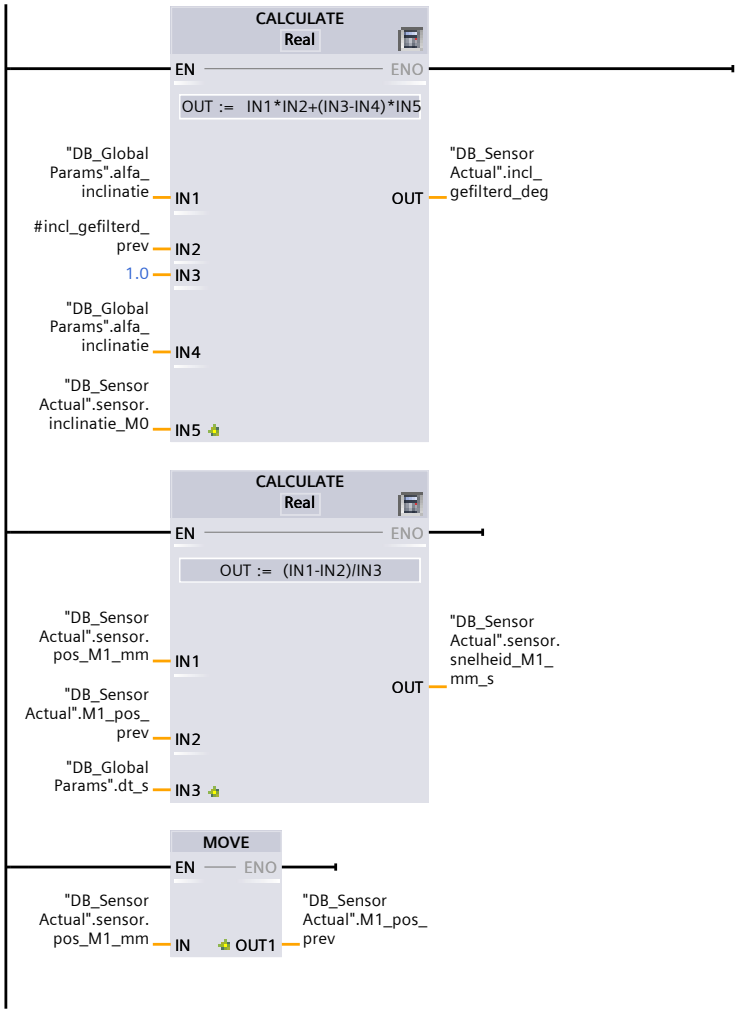
Network 3:



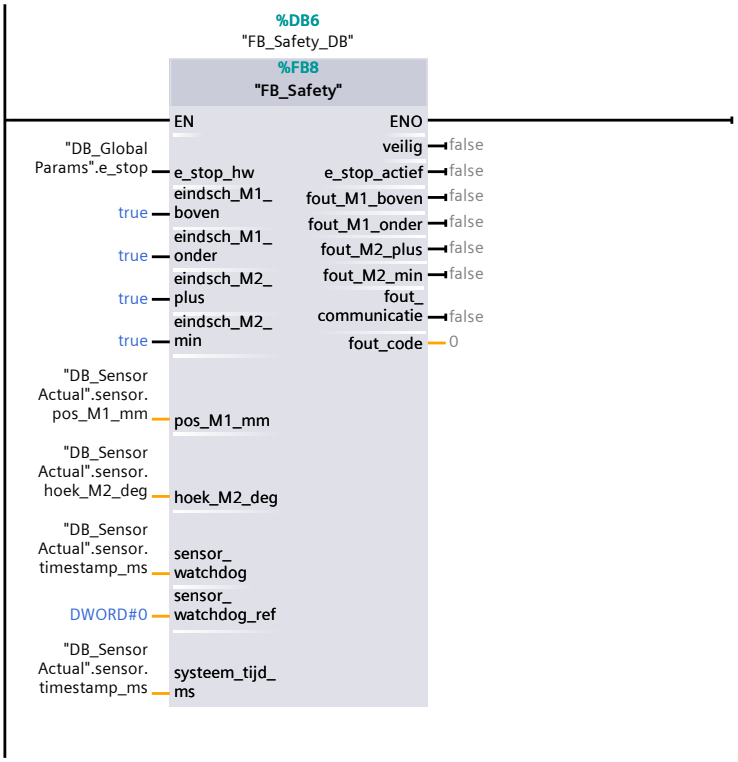
Network 4:



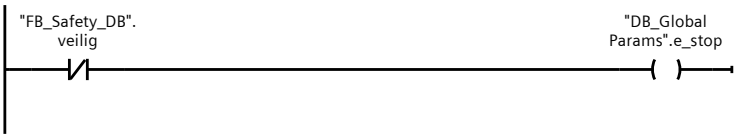
Network 5:


CALCULATE
Real
EN
ENO
OUT := IN1*IN2+(IN3-IN4)*IN5
"DB_Global Params".alfa_ inclinatie
IN1
OUT
"DB_Sensor Actual".incl_ gefilterd_deg
#incl_gefilterd_ prev
IN2
1.0
IN3
"DB_Global Params".alfa_ inclinatie
IN4
"DB_Sensor Actual".sensor. inclinatie_M0
IN5
CALCULATE
Real
EN
ENO
OUT := (IN1-IN2)/IN3
"DB_Sensor Actual".sensor. pos_M1_mm
IN1
OUT
"DB_Sensor Actual".sensor. snelheid_M1_ mm_s
"DB_Sensor Actual".M1_pos_ prev
IN2
"DB_Global Params".dt_s
IN3
MOVE
EN
ENO
"DB_Sensor Actual".sensor. pos_M1_mm
IN
OUT1
"DB_Sensor Actual".M1_pos_ prev

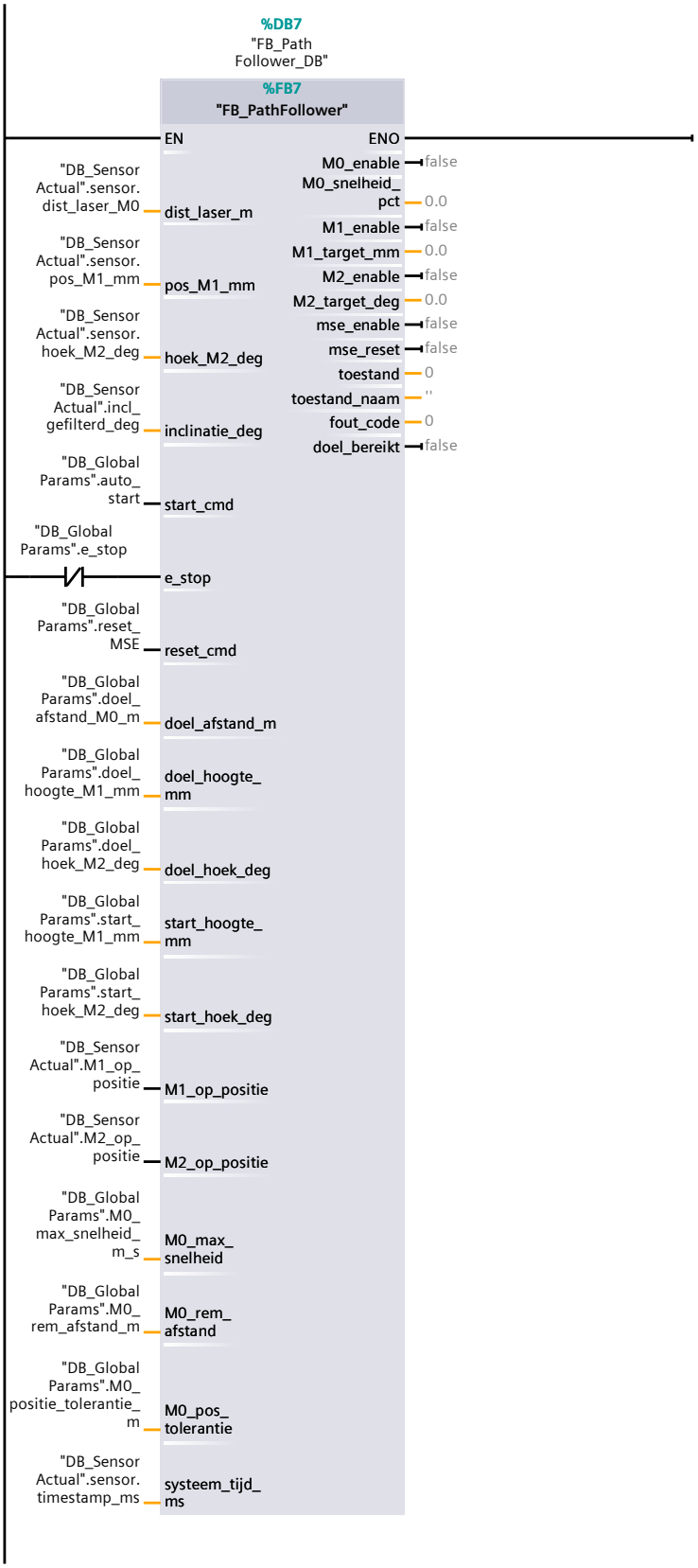
Network 6:



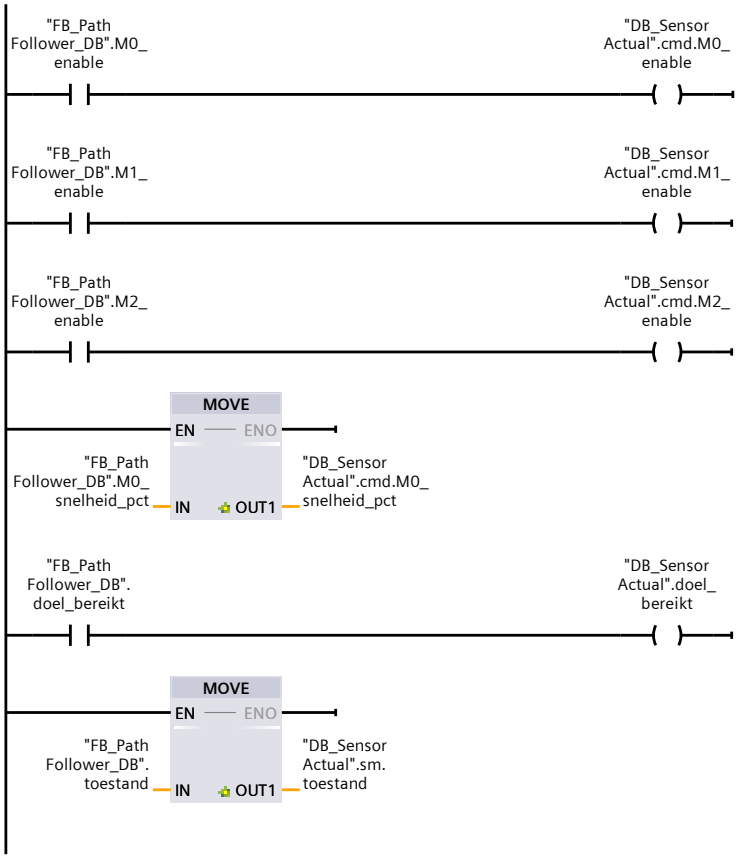
Network 7:



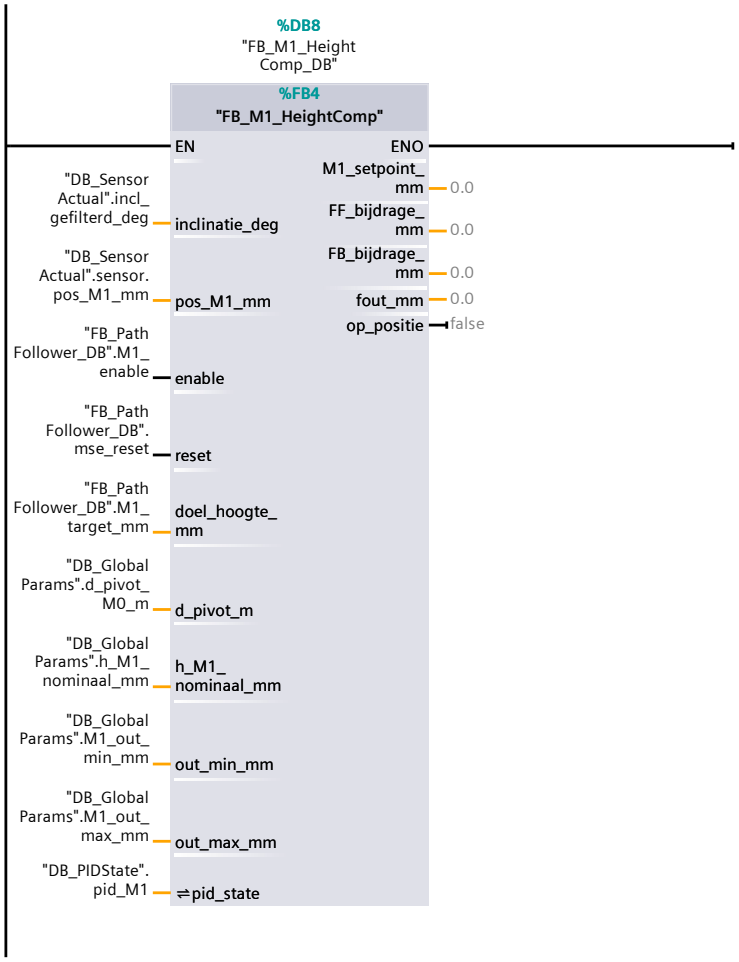
Network 8:


%DB7
"FB_Path
Follower_DB"
%FB7
"FB_PathFollower"
EN
ENO
"DB_Sensor Actual".sensor. dist_laser_M0
dist_laser_m
"DB_Sensor Actual".sensor. pos_M1_mm
pos_M1_mm
"DB_Sensor Actual".sensor. hoek_M2_deg
hoek_M2_deg
"DB_Sensor Actual".incl_ gefilterd_deg
inclinatie_deg
"DB_Global Params".auto_ start
start_cmd
"DB_Global Params".e_stop
e_stop
"DB_Global Params".reset_ MSE
reset_cmd
"DB_Global Params".doel_ afstand_M0_m
doel_afstand_m
"DB_Global Params".doel_ hoogte_M1_mm
doel_hoogte_ mm
"DB_Global Params".doel_ hoek_M2_deg
doel_hoek_deg
"DB_Global Params".start_ hoogte_M1_mm
start_hoogte_ mm
"DB_Global Params".start_ hoek_M2_deg
start_hoek_deg
"DB_Sensor Actual".M1_op_ positie
M1_op_positie
"DB_Sensor Actual".M2_op_ positie
M2_op_positie
"DB_Global Params".M0_ max_snelheid_ m_s
M0_max_ snelheid
"DB_Global Params".M0_ rem_afstand_m
M0_rem_ afstand
"DB_Global Params".M0_ positie_tolerantie_ m
M0_pos_ tolerantie
"DB_Sensor Actual".sensor. timestamp_ms
systeem_tijd_ ms
M0_enable false
M0_snelheid_ pct 0.0
M1_enable false
M1_target_mm 0.0
M2_enable false
M2_target_deg 0.0
mse_enable false
mse_reset false
toestand 0
toestand_naam ''
fout_code 0
doel_bereikt false

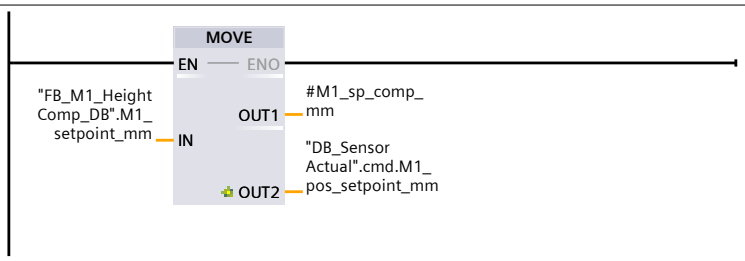
Network 9:


"FB_Path Follower_DB".M0_ enable
"DB_Sensor Actual".cmd.M0_ enable
"FB_Path Follower_DB".M1_ enable
"DB_Sensor Actual".cmd.M1_ enable
"FB_Path Follower_DB".M2_ enable
"DB_Sensor Actual".cmd.M2_ enable
MOVE
EN
ENO
"FB_Path Follower_DB".M0_ snelheid_pct
IN
OUT1
"DB_Sensor Actual".cmd.M0_ snelheid_pct
"FB_Path Follower_DB". doel_bereikt
"DB_Sensor Actual".doel_ bereikt
MOVE
EN
ENO
"FB_Path Follower_DB". toestand
IN
OUT1
"DB_Sensor Actual".sm. toestand

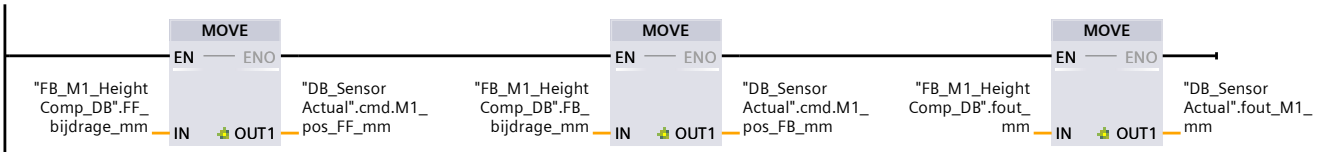
Network 10:



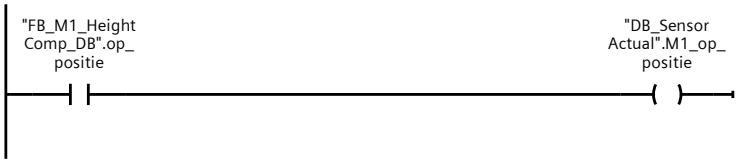
Network 11:



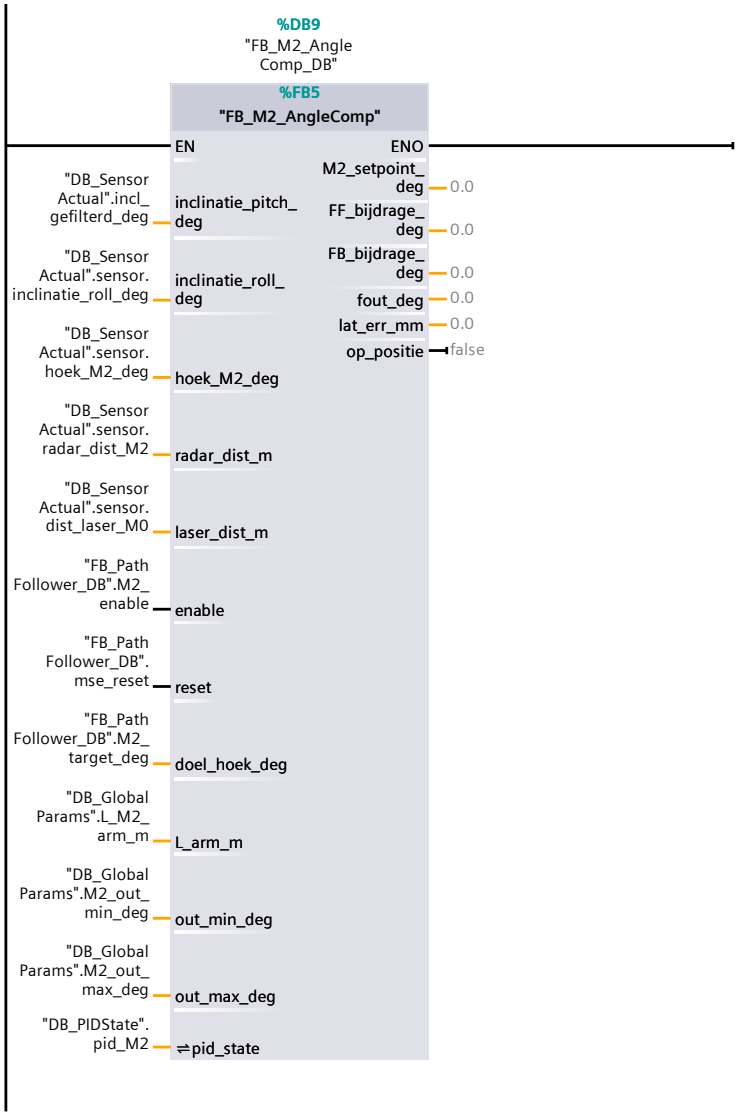
Network 12:



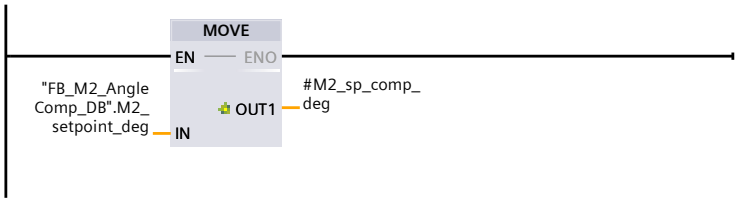
Network 13:



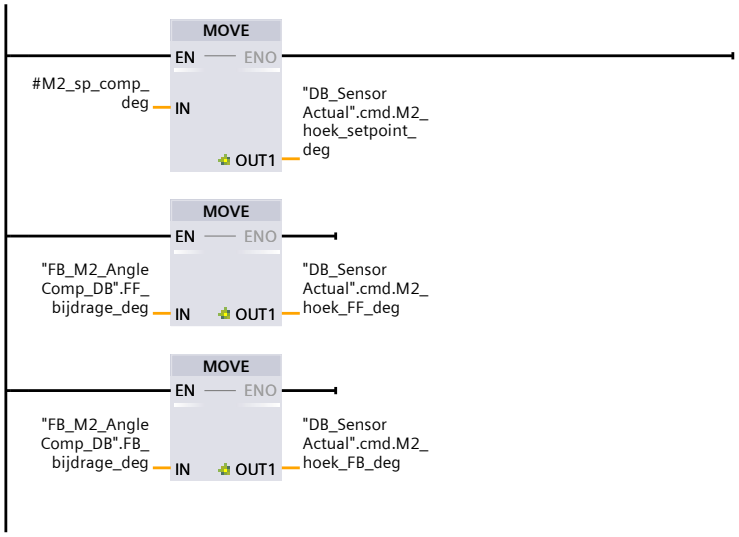
Network 14:



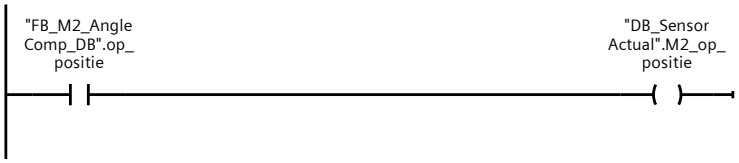
Network 15:



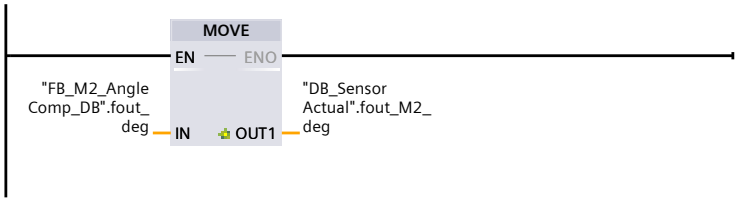
Network 16:



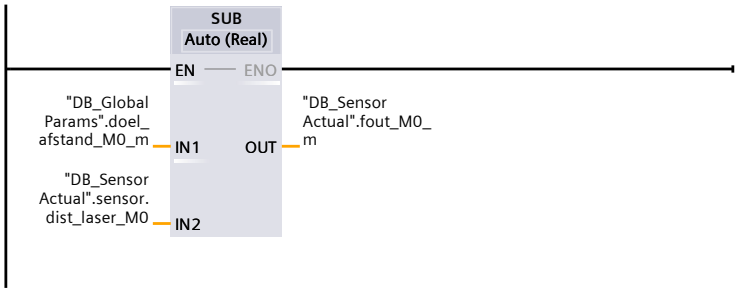
Network 17:



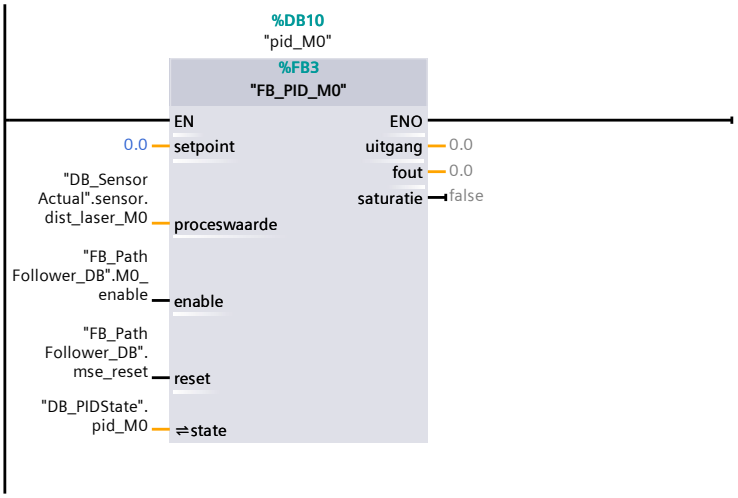
Network 18:



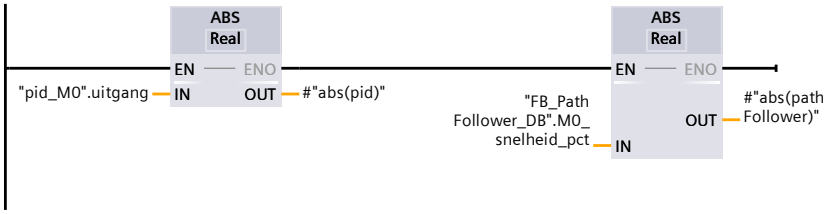
Network 19:



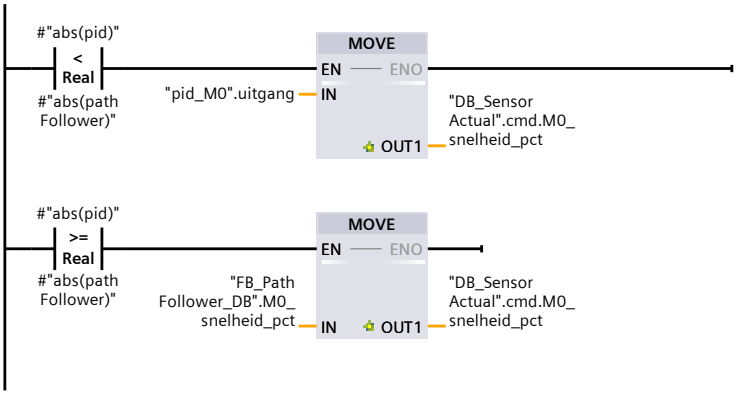
Network 20:



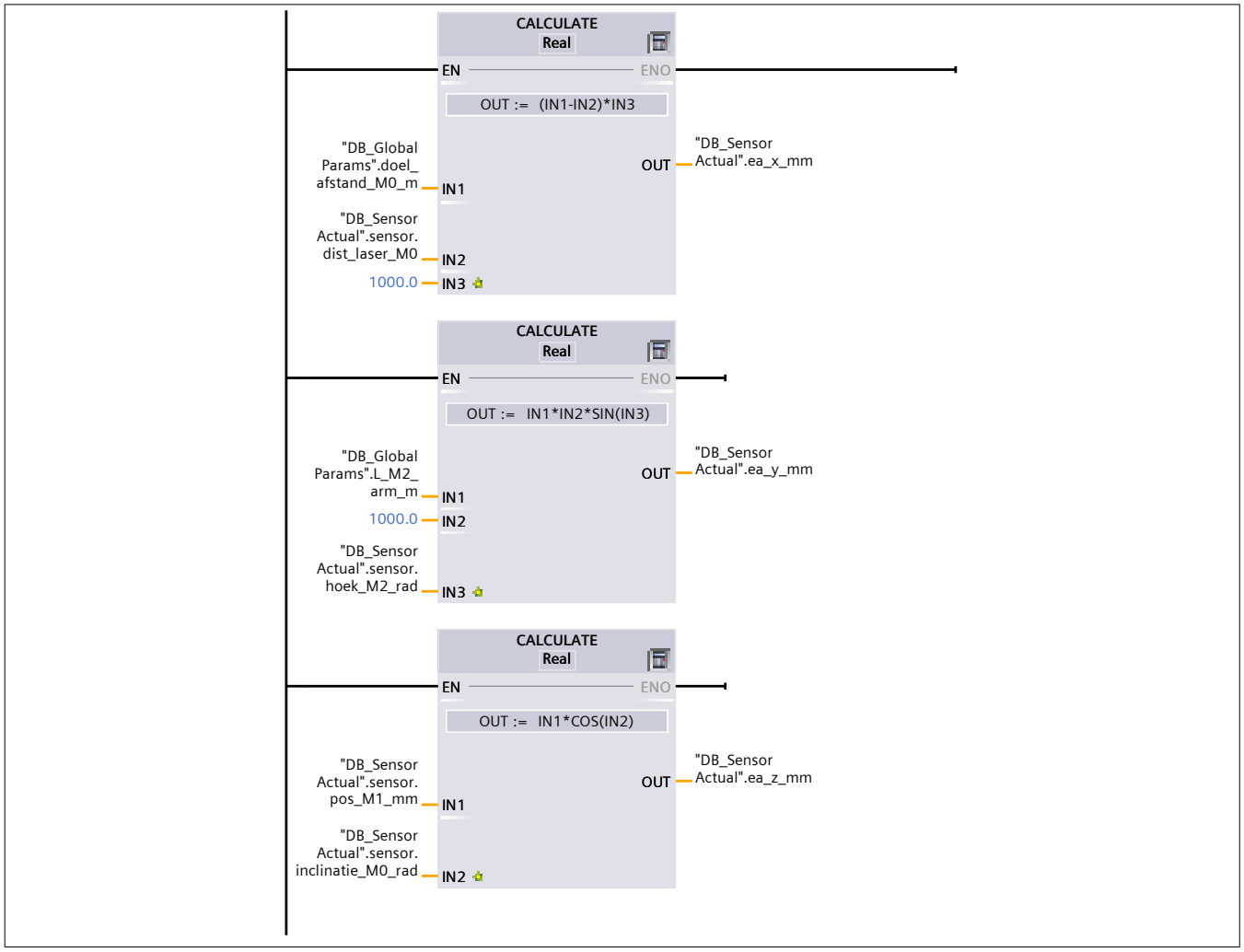
Network 21:



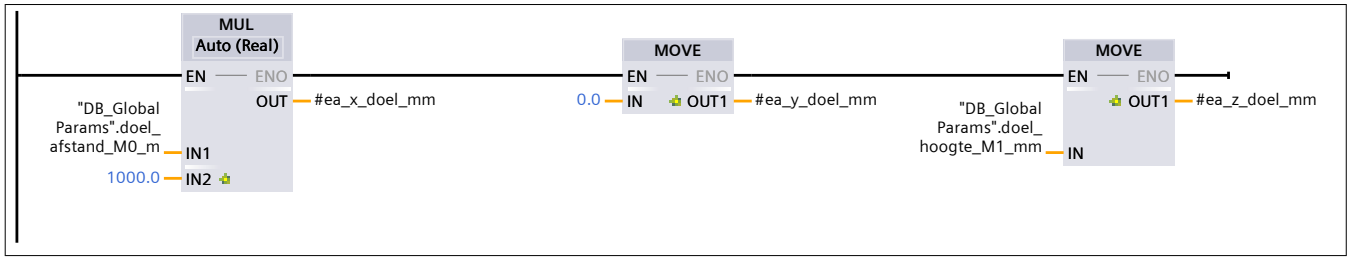
Network 22:



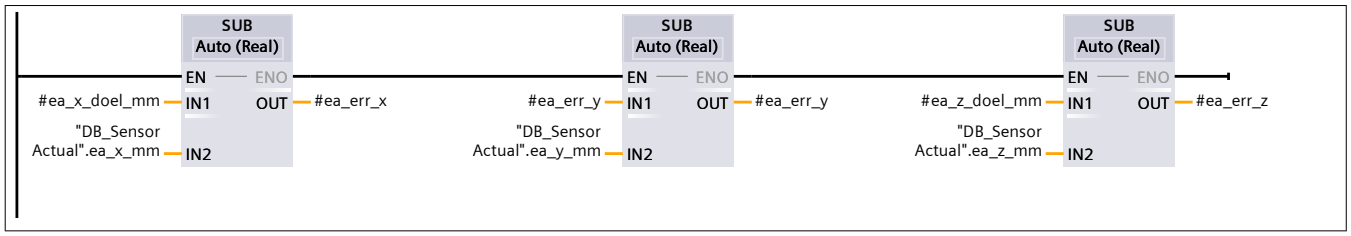
Network 23:



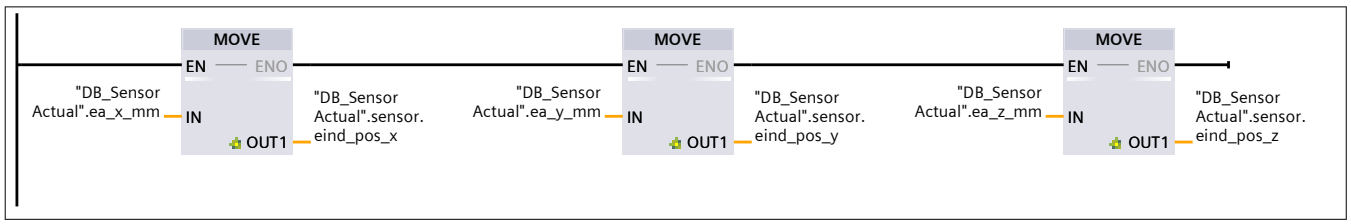
Network 24:



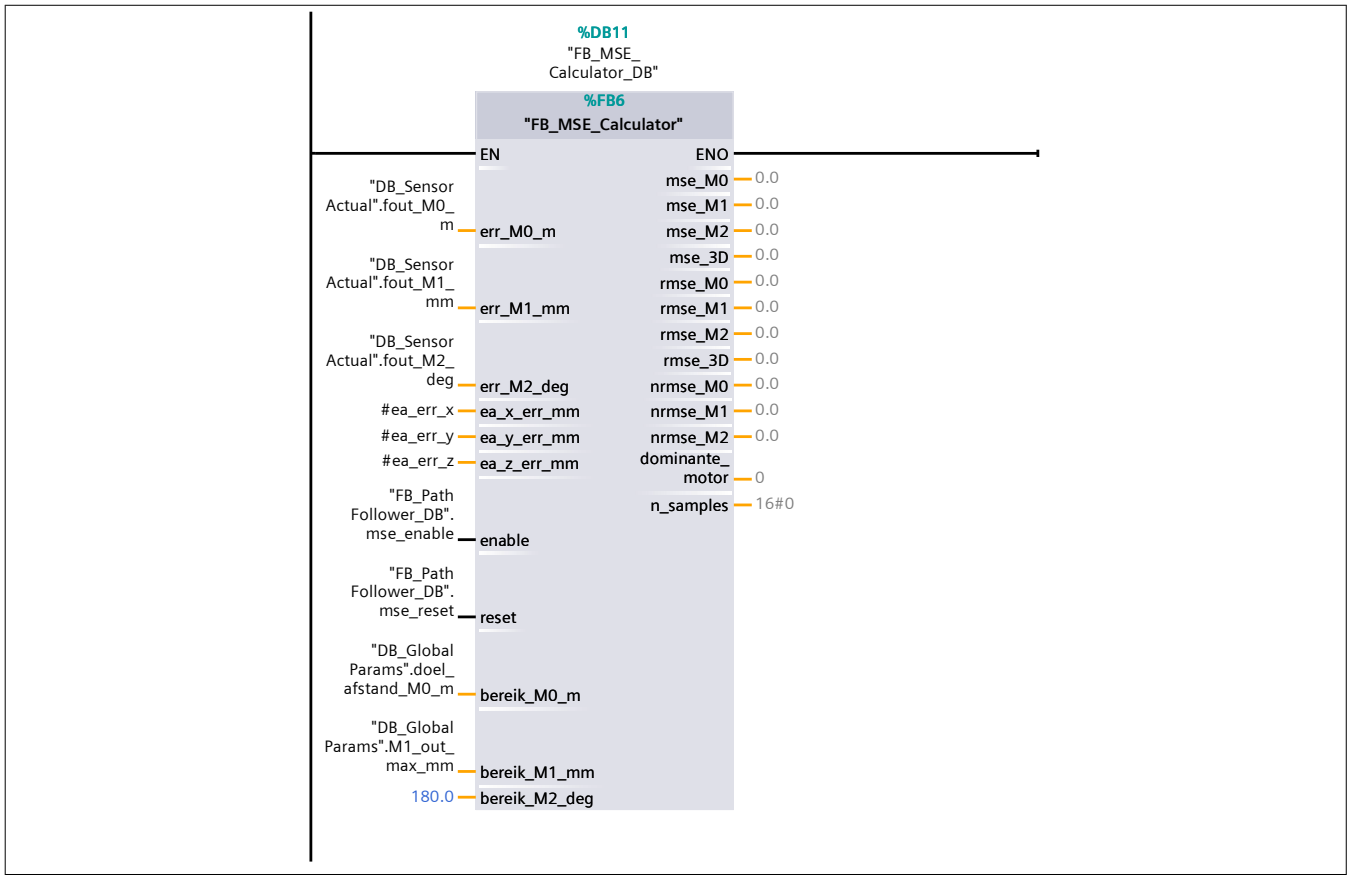
Network 25:



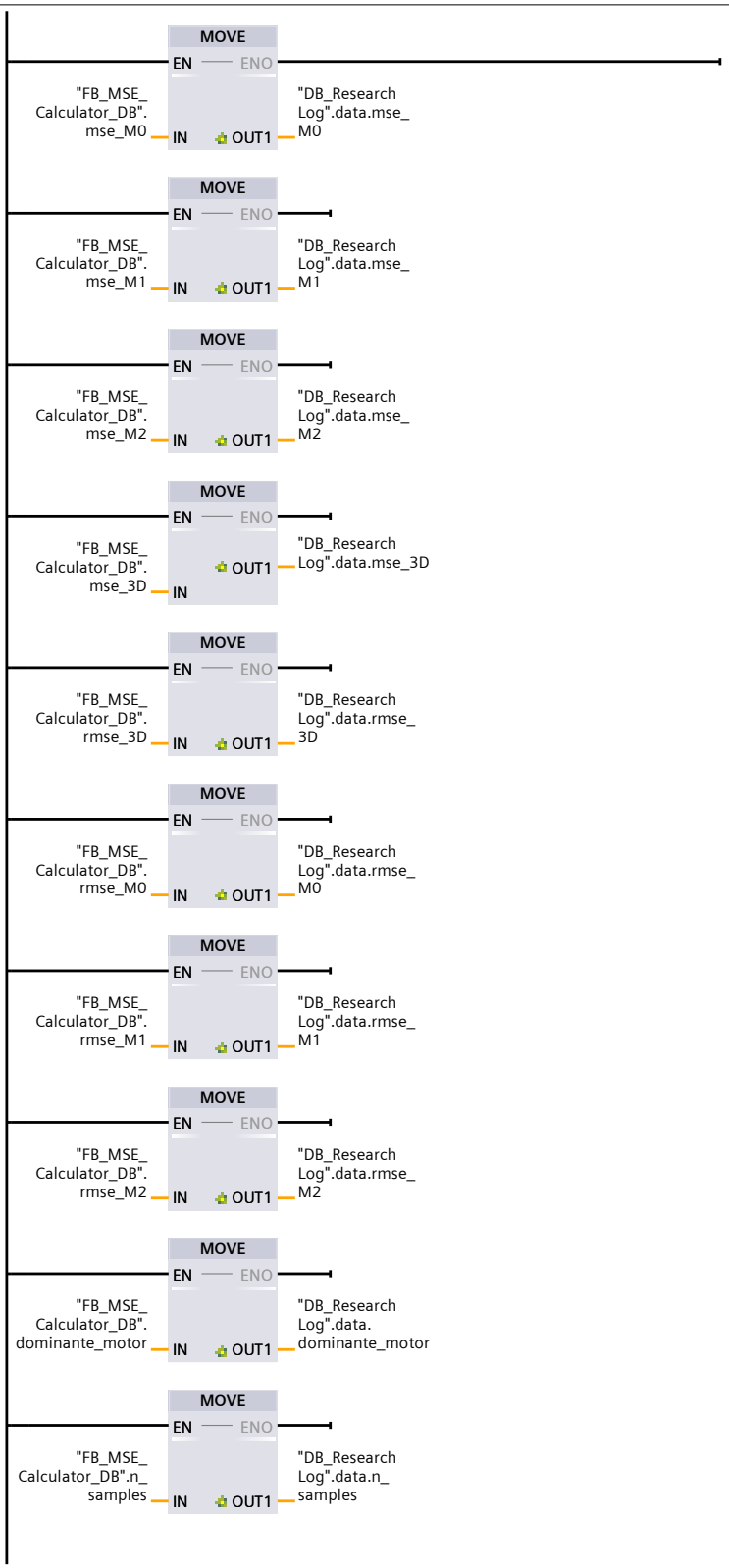
Network 26:



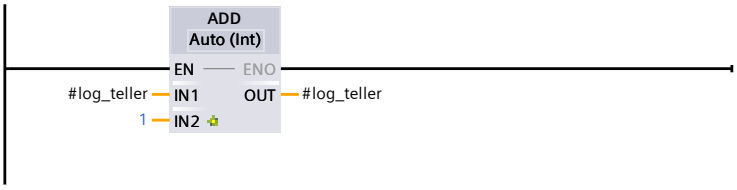
Network 27:



Network 28:

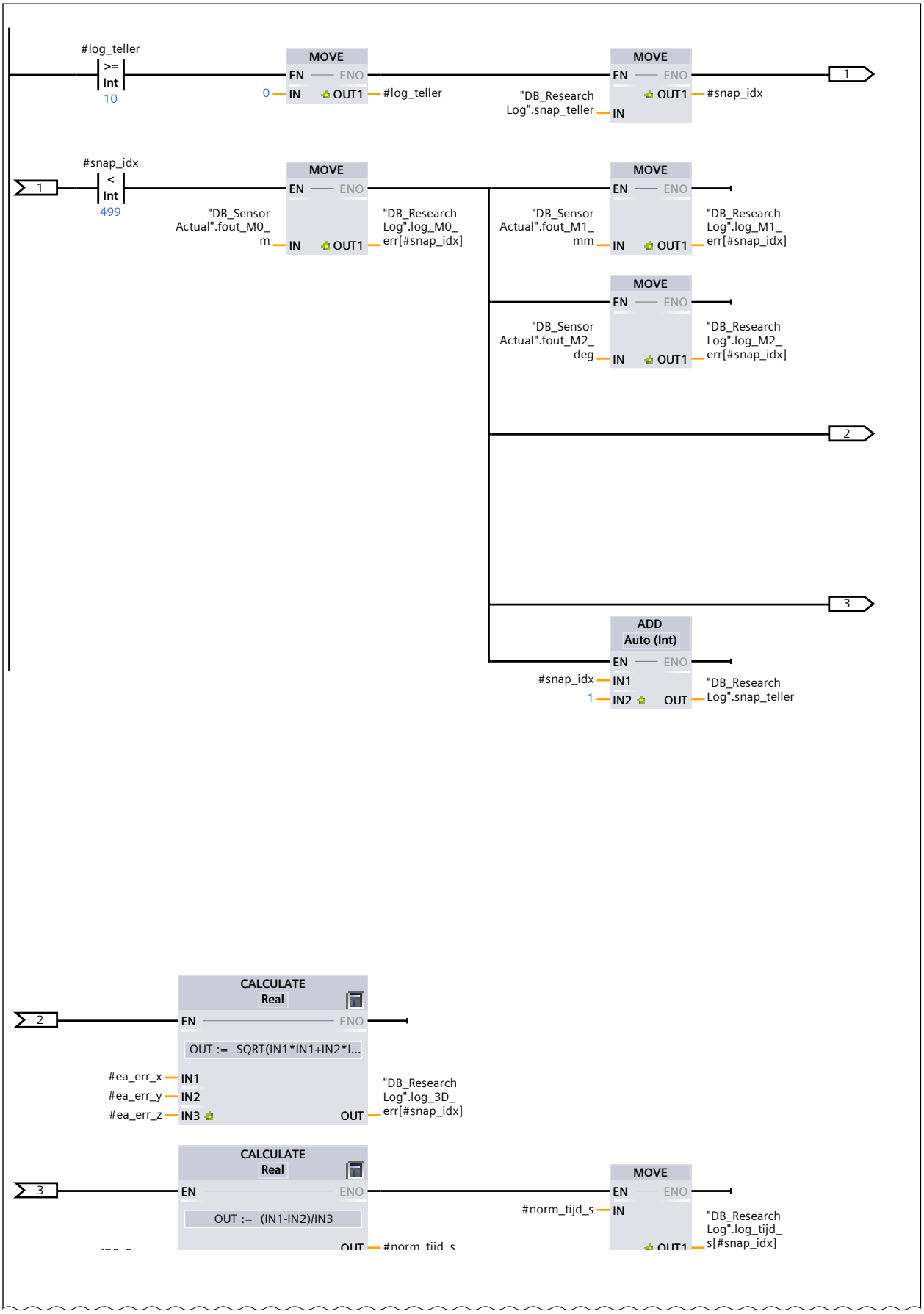

MOVE
EN
ENO
"FB_MSE_ Calculator_DB". mse_M0
IN
OUT1
"DB_Research Log".data.mse_ M0
MOVE
EN
ENO
"FB_MSE_ Calculator_DB". mse_M1
IN
OUT1
"DB_Research Log".data.mse_ M1
MOVE
EN
ENO
"FB_MSE_ Calculator_DB". mse_M2
IN
OUT1
"DB_Research Log".data.mse_ M2
MOVE
EN
ENO
"FB_MSE_ Calculator_DB". mse_3D
IN
OUT1
"DB_Research Log".data.mse_3D
MOVE
EN
ENO
"FB_MSE_ Calculator_DB". rmse_3D
IN
OUT1
"DB_Research Log".data.rmse_ 3D
MOVE
EN
ENO
"FB_MSE_ Calculator_DB". rmse_M0
IN
OUT1
"DB_Research Log".data.rmse_ M0
MOVE
EN
ENO
"FB_MSE_ Calculator_DB". rmse_M1
IN
OUT1
"DB_Research Log".data.rmse_ M1
MOVE
EN
ENO
"FB_MSE_ Calculator_DB". rmse_M2
IN
OUT1
"DB_Research Log".data.rmse_ M2
MOVE
EN
ENO
"FB_MSE_ Calculator_DB". dominante_motor
IN
OUT1
"DB_Research Log".data. dominante_motor
MOVE
EN
ENO
"FB_MSE_ Calculator_DB".n_ samples
IN
OUT1
"DB_Research Log".data.n_ samples

Network 29:



Network 30:

Network 30: (1.1 / 2.1)



Network 30: (2.1 / 2.1)

1.1 (Page12 - 17)

Network 31:

```
0001 IF "FB_PathFollower_DB".doel_bereikt AND NOT"DB_ResearchLog".resul-
     taat_klaar THEN
0002     "DB_ResearchLog".resultaat_klaar := TRUE;
0003     CASE "FB_MSE_Calculator_DB".dominante_motor
0004     OF
0005         1:
0006             "DB_ResearchLog".conclusie_tekst :=
0007             'M0 (AGV rijmotor) veroorzaakt de grootste fout. Aanbeveling:
     verbeter M0 positionering OF verlaag rijsnelheid.';
0008         2:
0009             "DB_ResearchLog".conclusie_tekst :=
0010             'M1 (vertikale as) veroorzaakt de grootste fout. Aanbeveling: ver-
     hoog M1 PID Kp, controleer FF-geometrie.';
0011         3:
0012             "DB_ResearchLog".conclusie_tekst :=
0013             'M2 (SCARA arm) veroorzaakt de grootste fout. Aanbeveling: ver-
     hoog M2 PID Kp, controleer laterale FF.';
0014         ELSE
0015             "DB_ResearchLog".conclusie_tekst :=
0016             'Onvoldoende data voor conclusie. Herhaal meting.';
0017     END_CASE;
0018 END_IF;
```

PLC_1 [CPU 1215C AC/DC/Rly] / Program blocks

FB_PID_DB [DB3]

FB_PID_DB Properties					
General					
Name	FB_PID_DB	**Number**	3	**Type**	DB
Language	DB	**Numbering**	Automatic		
Information					
Title		**Author**		**Comment**	
Family		**Version**	0.1	**User-defined ID**	

Name	Data type	Start value	Retain
Input			
setpoint	Real	0.0	False
proceswaarde	Real	0.0	False
enable	Bool	false	False
reset	Bool	false	False
Output			
uitgang	Real	0.0	False
fout	Real	0.0	False
saturatie	Bool	false	False
InOut			
state	"UDT_PIDState"		False
Static			
e_huidig	Real	0.0	False
P_term	Real	0.0	False
I_term	Real	0.0	False
D_term	Real	0.0	False
u_raw	Real	0.0	False
u_sat	Real	0.0	False
track_err	Real	0.0	False
alfa_D	Real	0.0	False
Kd	Real	0.0	False
in_deadband	Bool	false	False
reset_prev	Bool	false	False

PLC_1 [CPU 1215C AC/DC/Rly] / Program blocks

DB_PIDState [DB4]

DB_PIDState Properties					
General					
Name	DB_PIDState	**Number**	4	**Type**	DB
Language	DB	**Numbering**	Automatic		
Information					
Title		**Author**		**Comment**	
Family		**Version**	0.1	**User-defined ID**	

Name	Data type	Start value	Retain
Static			
pid_M0	"UDT_PIDState"		False
pid_M1	"UDT_PIDState"		False
pid_M2	"UDT_PIDState"		False

PLC_1 [CPU 1215C AC/DC/Rly] / Program blocks

DB_SensorActual [DB5]

DB_SensorActual Properties					
General					
Name	DB_SensorActual	**Number**	5	**Type**	DB
Language	DB	**Numbering**	Automatic		
Information					
Title		**Author**		**Comment**	
Family		**Version**	0.1	**User-defined ID**	

Name	Data type	Start value	Retain
Static			
sensor	"UDT_SensorData"		False
cmd	"UDT_MotorCmd"		False
sm	"UDT_StateMachine"		False
setpoint_M1_mm	Real	0.0	False
setpoint_M2_deg	Real	0.0	False
setpoint_M0_m	Real	0.0	False
fout_M0_m	Real	0.0	False
fout_M1_mm	Real	0.0	False
fout_M2_deg	Real	0.0	False
ea_x_mm	Real	0.0	False
ea_y_mm	Real	0.0	False
ea_z_mm	Real	0.0	False
ea_x_doel_mm	Real	0.0	False
ea_y_doel_mm	Real	0.0	False
ea_z_doel_mm	Real	0.0	False
incl_gefilterd_deg	Real	0.0	False
M1_snelheid_filt	Real	0.0	False
M1_pos_prev	Real	0.0	False
M1_op_positie	Bool	false	False
M2_op_positie	Bool	false	False
systeem_gereed	Bool	false	False
doel_bereikt	Bool	false	False

PLC_1 [CPU 1215C AC/DC/Rly] / Program blocks

FB_Safety_DB [DB6]

FB_Safety_DB Properties					
General					
Name	FB_Safety_DB	**Number**	6	**Type**	DB
Language	DB	**Numbering**	Automatic		
Information					
Title		**Author**		**Comment**	
Family		**Version**	0.1	**User-defined ID**	

Name	Data type	Start value	Retain
Input			
e_stop_hw	Bool	false	False
eindsch_M1_boven	Bool	false	False
eindsch_M1_onder	Bool	false	False
eindsch_M2_plus	Bool	false	False
eindsch_M2_min	Bool	false	False
pos_M1_mm	Real	0.0	False
hoek_M2_deg	Real	0.0	False
sensor_watchdog	DWord	16#0	False
sensor_watchdog_ref	DWord	16#0	False
systeem_tijd_ms	DWord	16#0	False
Output			
veilig	Bool	false	False
e_stop_actief	Bool	false	False
fout_M1_boven	Bool	false	False
fout_M1_onder	Bool	false	False
fout_M2_plus	Bool	false	False
fout_M2_min	Bool	false	False
fout_communicatie	Bool	false	False
fout_code	Int	0	False
InOut			
Static			
M1_sw_boven_mm	Real	2090.0	False
M1_sw_onder_mm	Real	110.0	False
M2_sw_plus_deg	Real	85.5	False
M2_sw_min_deg	Real	-85.5	False
wd_prev	DWord	DWord#0	False
wd_timeout_ms	DWord	DWord#150	False
wd_timer_ms	DWord	16#0	False
wd_foutteller	Int	0	False
fout_intern	Int	0	False
subst	DWord	16#0	False

PLC_1 [CPU 1215C AC/DC/Rly] / Program blocks

FB_PathFollower_DB [DB7]

FB_PathFollower_DB Properties					
General					
Name	FB_PathFollower_DB	**Number**	7	**Type**	DB
Language	DB	**Numbering**	Automatic		
Information					
Title		**Author**		**Comment**	
Family		**Version**	0.1	**User-defined ID**	

Name	Data type	Start value	Retain
Input			
dist_laser_m	Real	0.0	False
pos_M1_mm	Real	0.0	False
hoek_M2_deg	Real	0.0	False
inclinatie_deg	Real	0.0	False
start_cmd	Bool	false	False
e_stop	Bool	false	False
reset_cmd	Bool	false	False
doel_afstand_m	Real	0.0	False
doel_hoogte_mm	Real	0.0	False
doel_hoek_deg	Real	0.0	False
start_hoogte_mm	Real	0.0	False
start_hoek_deg	Real	0.0	False
M1_op_positie	Bool	false	False
M2_op_positie	Bool	false	False
M0_max_snelheid	Real	0.0	False
M0_rem_afstand	Real	0.0	False
M0_pos_tolerantie	Real	0.0	False
systeem_tijd_ms	DWord	16#0	False
Output			
M0_enable	Bool	false	False
M0_snelheid_pct	Real	0.0	False
M1_enable	Bool	false	False
M1_target_mm	Real	0.0	False
M2_enable	Bool	false	False
M2_target_deg	Real	0.0	False
mse_enable	Bool	false	False
mse_reset	Bool	false	False
toestand	Int	0	False
toestand_naam	String[30]	''	False
fout_code	Int	0	False
doel_bereikt	Bool	false	False
InOut			
Static			
sm_toestand	Int	0	False
sm_toestand_prev	Int	-1	False
sm_timer_ms	DWord	16#0	False
sm_entry	Bool	false	False
start_prev	Bool	false	False

Name	Data type	Start value	Retain
reset_prev	Bool	false	False
mse_reset_intern	Bool	false	False
M0_snelheid_berekend	Real	0.0	False
rem_factor	Real	0.0	False
home_timeout_ms	DWord	DWORD#5000	False
M1_huidig_target	Real	0.0	False
M2_huidig_target	Real	0.0	False
start_rise	Bool	false	False
reset_rise	Bool	false	False
sequence_0	Bool	false	False
sequence_1	Bool	false	False
sequence_2	Bool	false	False
sequence_3	Bool	false	False
sequence_4	Bool	false	False
sequence_5	Bool	false	False
sequence_6	Bool	false	False
sequence_7	Bool	false	False
sequence_8	Bool	false	False
sequence_9	Bool	false	False
tijd_delta_ms	DWord	16#0	False
home_timeout	Bool	false	False
dist_in_remgebied	Bool	false	False
rem_te_klein_01	Bool	false	False
doel_nabij	Bool	false	False
cond_dist_gt0	Bool	false	False
cond_rem_gt0	Bool	false	False
rem_te_klein_005	Bool	false	False
rem_te_groot_10	Bool	false	False
final_timeout	Bool	false	False
fout_aanwezig	Bool	false	False

PLC_1 [CPU 1215C AC/DC/Rly] / Program blocks

FB_M1_HeightComp_DB [DB8]

FB_M1_HeightComp_DB Properties					
General					
Name	FB_M1_HeightComp_DB	**Number**	8	**Type**	DB
Language	DB	**Numbering**	Automatic		
Information					
Title		**Author**		**Comment**	
Family		**Version**	0.1	**User-defined ID**	

Name	Data type	Start value	Retain
Input			
inclinatie_deg	Real	0.0	False
pos_M1_mm	Real	0.0	False
enable	Bool	false	False
reset	Bool	false	False
doel_hoogte_mm	Real	0.0	False
d_pivot_m	Real	0.0	False
h_M1_nominaal_mm	Real	0.0	False
out_min_mm	Real	0.0	False
out_max_mm	Real	0.0	False
Output			
M1_setpoint_mm	Real	0.0	False
FF_bijdrage_mm	Real	0.0	False
FB_bijdrage_mm	Real	0.0	False
fout_mm	Real	0.0	False
op_positie	Bool	false	False
InOut			
pid_state	"UDT_PIDState"		False
Static			
theta_rad	Real	0.0	False
delta_z_basis_mm	Real	0.0	False
delta_z_M1_mm	Real	0.0	False
FF_mm	Real	0.0	False
SP_voor_PID	Real	0.0	False
PID_uitgang	Real	0.0	False
SP_totaal	Real	0.0	False
pid_inst	"FB_PID_M0"		False
pos_M1_mm-SP_totaal	Real	0.0	False
pid_inst_M1	"FB_PID_M1"		False

PLC_1 [CPU 1215C AC/DC/Rly] / Program blocks

FB_M2_AngleComp_DB [DB9]

FB_M2_AngleComp_DB Properties					
General					
Name	FB_M2_AngleComp_DB	**Number**	9	**Type**	DB
Language	DB	**Numbering**	Automatic		
Information					
Title		**Author**		**Comment**	
Family		**Version**	0.1	**User-defined ID**	

Name	Data type	Start value	Retain
Input			
inclinatie_pitch_deg	Real	0.0	False
inclinatie_roll_deg	Real	0.0	False
hoek_M2_deg	Real	0.0	False
radar_dist_m	Real	0.0	False
laser_dist_m	Real	0.0	False
enable	Bool	false	False
reset	Bool	false	False
doel_hoek_deg	Real	0.0	False
L_arm_m	Real	0.0	False
out_min_deg	Real	0.0	False
out_max_deg	Real	0.0	False
Output			
M2_setpoint_deg	Real	0.0	False
FF_bijdrage_deg	Real	0.0	False
FB_bijdrage_deg	Real	0.0	False
fout_deg	Real	0.0	False
lat_err_mm	Real	0.0	False
op_positie	Bool	false	False
InOut			
pid_state	"UDT_PIDState"		False
Static			
roll_rad	Real	0.0	False
pitch_rad	Real	0.0	False
lat_err_m	Real	0.0	False
FF_lat_deg	Real	0.0	False
FF_pitch_deg	Real	0.0	False
FF_totaal_deg	Real	0.0	False
SP_totaal	Real	0.0	False
radar_eff	Real	0.0	False
pid_inst	"FB_PID_M0"		False
div	Real	0.0	False
atan	Real	0.0	False
sub	Real	0.0	False
pid_inst_M2	"FB_PID_M0"		False

PLC_1 [CPU 1215C AC/DC/Rly] / Program blocks

pid_M0 [DB10]

pid_M0 Properties					
General					
Name	pid_M0	**Number**	10	**Type**	DB
Language	DB	**Numbering**	Automatic		
Information					
Title		**Author**		**Comment**	
Family		**Version**	0.1	**User-defined ID**	

Name	Data type	Start value	Retain
Input			
setpoint	Real	0.0	False
proceswaarde	Real	0.0	False
enable	Bool	false	False
reset	Bool	false	False
Output			
uitgang	Real	0.0	False
fout	Real	0.0	False
saturatie	Bool	false	False
InOut			
state	"UDT_PIDState"		False
Static			
e_huidig	Real	0.0	False
P_term	Real	0.0	False
I_term	Real	0.0	False
D_term	Real	0.0	False
u_raw	Real	0.0	False
u_sat	Real	0.0	False
track_err	Real	0.0	False
alfa_D	Real	0.0	False
Kd	Real	0.0	False
in_deadband	Bool	false	False
reset_prev	Bool	false	False
abs(e_huidig)	Real	0.0	False
abs(state.deadb)	Real	0.0	False

PLC_1 [CPU 1215C AC/DC/Rly] / Program blocks

FB_MSE_Calculator_DB [DB11]

FB_MSE_Calculator_DB Properties					
General					
Name	FB_MSE_Calculator_DB	**Number**	11	**Type**	DB
Language	DB	**Numbering**	Automatic		
Information					
Title		**Author**		**Comment**	
Family		**Version**	0.1	**User-defined ID**	

Name	Data type	Start value	Retain
Input			
err_M0_m	Real	0.0	False
err_M1_mm	Real	0.0	False
err_M2_deg	Real	0.0	False
ea_x_err_mm	Real	0.0	False
ea_y_err_mm	Real	0.0	False
ea_z_err_mm	Real	0.0	False
enable	Bool	false	False
reset	Bool	false	False
bereik_M0_m	Real	0.0	False
bereik_M1_mm	Real	0.0	False
bereik_M2_deg	Real	0.0	False
Output			
mse_M0	Real	0.0	False
mse_M1	Real	0.0	False
mse_M2	Real	0.0	False
mse_3D	Real	0.0	False
rmse_M0	Real	0.0	False
rmse_M1	Real	0.0	False
rmse_M2	Real	0.0	False
rmse_3D	Real	0.0	False
nrmse_M0	Real	0.0	False
nrmse_M1	Real	0.0	False
nrmse_M2	Real	0.0	False
dominante_motor	Int	0	False
n_samples	DWord	16#0	False
InOut			
Static			
sum_sq_M0	Real	0.0	False
sum_sq_M1	Real	0.0	False
sum_sq_M2	Real	0.0	False
sum_sq_3D	Real	0.0	False
N	DWord	16#0	False
err_3D_sq	Real	0.0	False
reset_prev	Bool	false	False
max_nrmse	Real	0.0	False
ber_M0	Real	0.0	False
ber_M1	Real	0.0	False
ber_M2	Real	0.0	False

Name	Data type	Start value	Retain
reset_rise	Bool	false	False
has_samples	Bool	false	False

PLC_1 [CPU 1215C AC/DC/Rly] / Program blocks

DB_ResearchLog [DB12]

DB_ResearchLog Properties					
General					
Name	DB_ResearchLog	**Number**	12	**Type**	DB
Language	DB	**Numbering**	Automatic		
Information					
Title		**Author**		**Comment**	
Family		**Version**	0.1	**User-defined ID**	

Name	Data type	Start value	Retain
Static			
data	"UDT_ResearchData"		False
snap_teller	Int	0	False
log_M0_err	Array[0..499] of Real		False
log_M1_err	Array[0..499] of Real		False
log_M2_err	Array[0..499] of Real		False
log_3D_err	Array[0..499] of Real		False
log_tijd_s	Array[0..499] of Real		False
resultaat_klaar	Bool	false	False
conclusie_tekst	String[200]	''	False

PLC_1 [CPU 1215C AC/DC/Rly] / Program blocks

pid_inst [DB13]

pid_inst Properties					
General					
Name	pid_inst	**Number**	13	**Type**	DB
Language	DB	**Numbering**	Automatic		
Information					
Title		**Author**		**Comment**	
Family		**Version**	0.1	**User-defined ID**	

Name	Data type	Start value	Retain
Input			
setpoint	Real	0.0	False
proceswaarde	Real	0.0	False
enable	Bool	false	False
reset	Bool	false	False
Output			
uitgang	Real	0.0	False
fout	Real	0.0	False
saturatie	Bool	false	False
InOut			
state	"UDT_PIDState"		False
Static			
e_huidig	Real	0.0	False
P_term	Real	0.0	False
I_term	Real	0.0	False
D_term	Real	0.0	False
u_raw	Real	0.0	False
u_sat	Real	0.0	False
track_err	Real	0.0	False
alfa_D	Real	0.0	False
Kd	Real	0.0	False
in_deadband	Bool	false	False
reset_prev	Bool	false	False

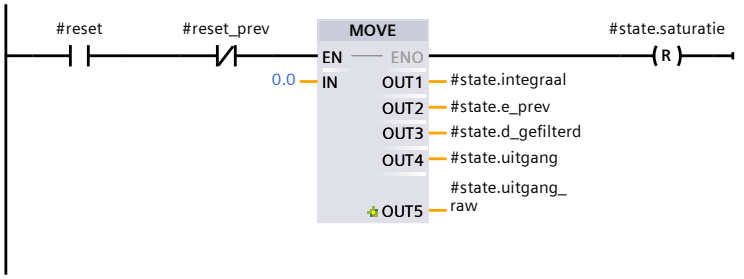
PLC_1 [CPU 1215C AC/DC/Rly] / Program blocks

FB_PID_M1 [FB2]

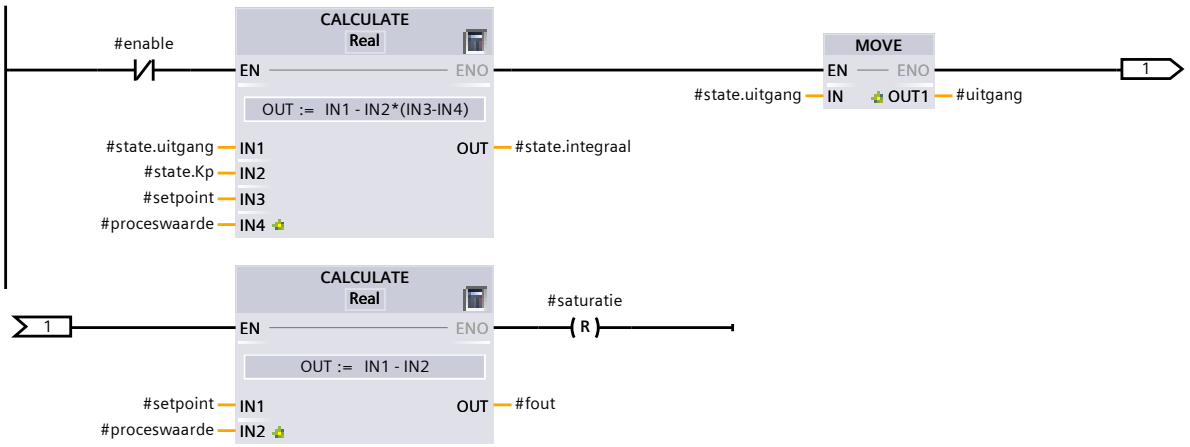
FB_PID_M1 Properties					
General					
Name	FB_PID_M1	**Number**	2	**Type**	FB
Language	LAD	**Numbering**	Automatic		
Information					
Title		**Author**		**Comment**	
Family		**Version**	0.1	**User-defined ID**	

Name	Data type	Default value	Retain
Input			
setpoint	Real	0.0	Non-retain
proceswaarde	Real	0.0	Non-retain
enable	Bool	false	Non-retain
reset	Bool	false	Non-retain
Output			
uitgang	Real	0.0	Non-retain
fout	Real	0.0	Non-retain
saturatie	Bool	false	Non-retain
InOut			
state	"UDT_PIDState"		
Static			
e_huidig	Real	0.0	Non-retain
P_term	Real	0.0	Non-retain
I_term	Real	0.0	Non-retain
D_term	Real	0.0	Non-retain
u_raw	Real	0.0	Non-retain
u_sat	Real	0.0	Non-retain
track_err	Real	0.0	Non-retain
alfa_D	Real	0.0	Non-retain
Kd	Real	0.0	Non-retain
in_deadband	Bool	false	Non-retain
reset_prev	Bool	false	Non-retain
abs(e_huidig)	Real	0.0	Non-retain
abs(state.deadb)	Real	0.0	Non-retain
Temp			
Constant			

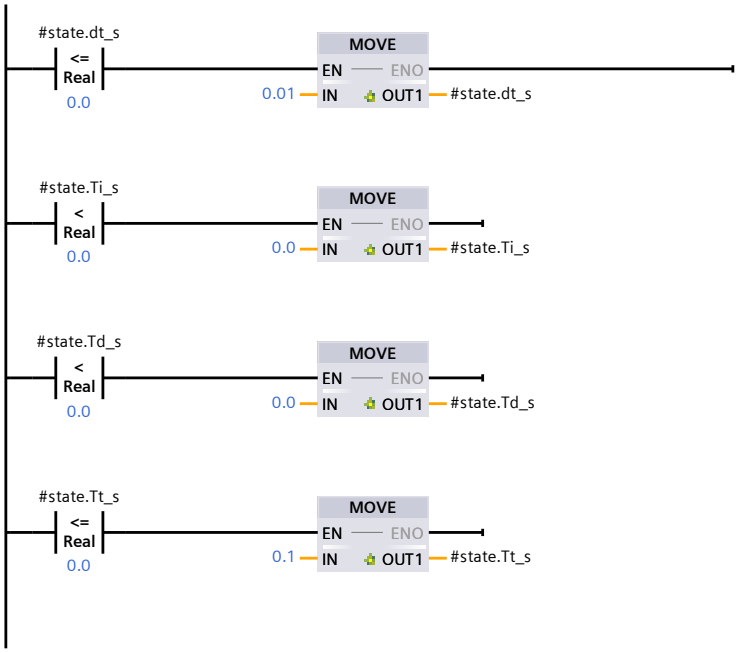
Network 1:



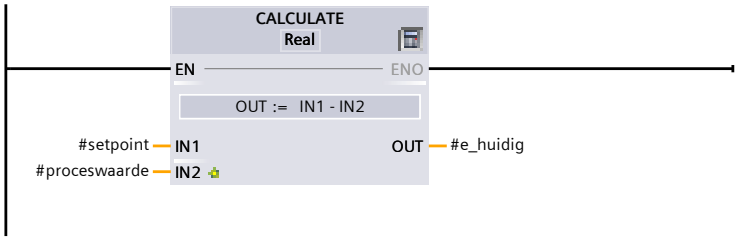
Network 2:



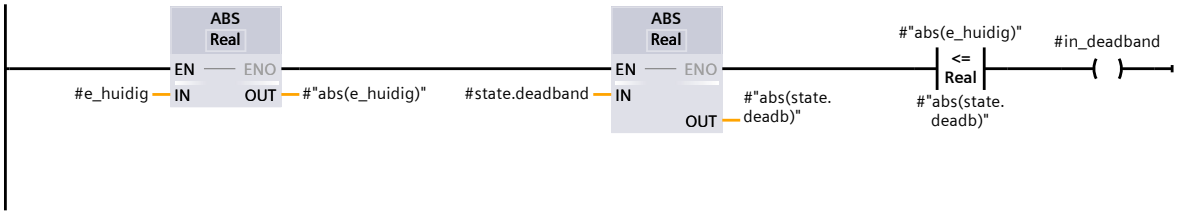
Network 3:



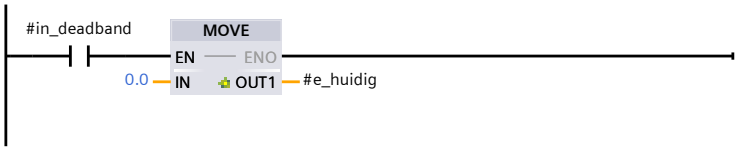
Network 4:



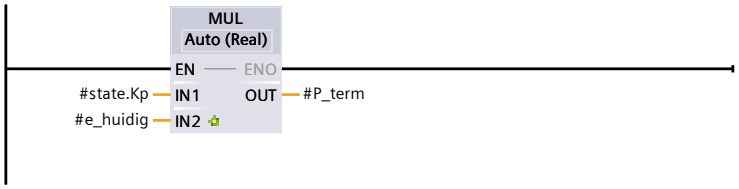
Network 5:



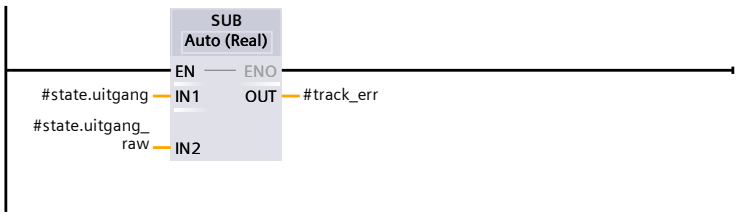
Network 6:



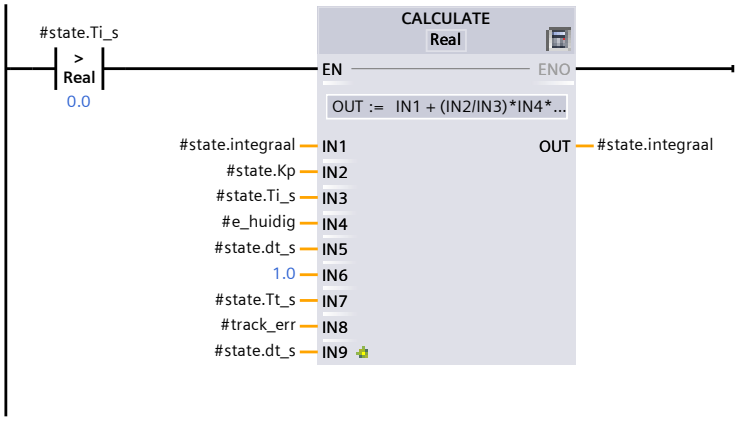
Network 7:



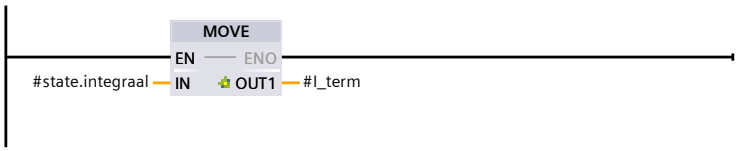
Network 8:



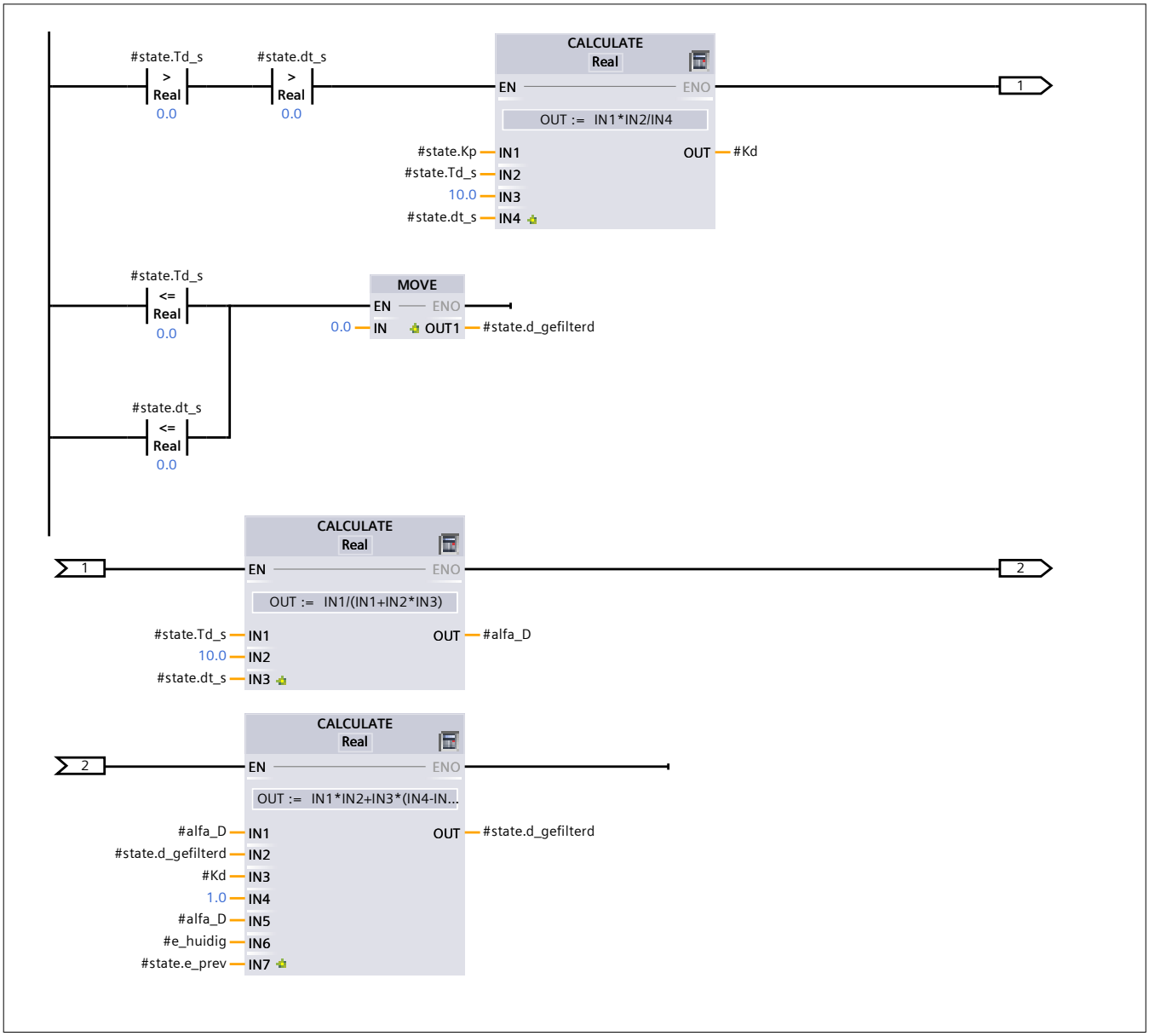
Network 9:



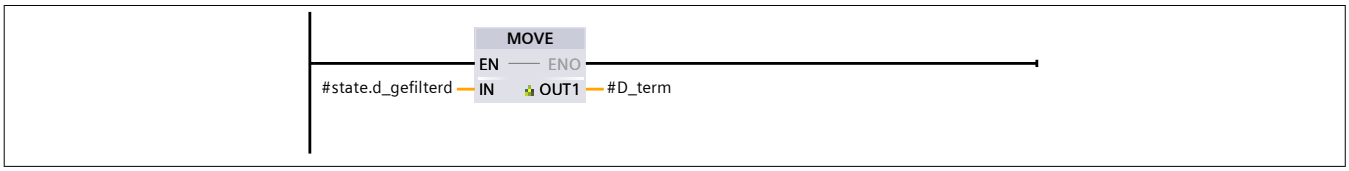
Network 10:



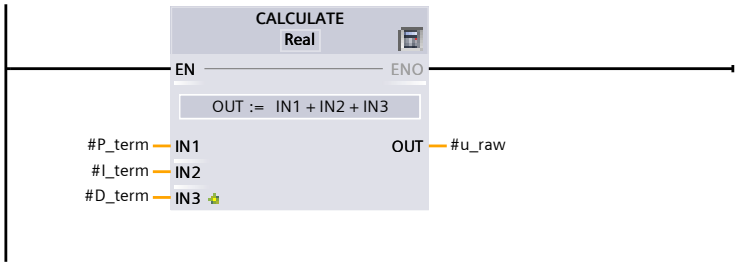
Network 11:



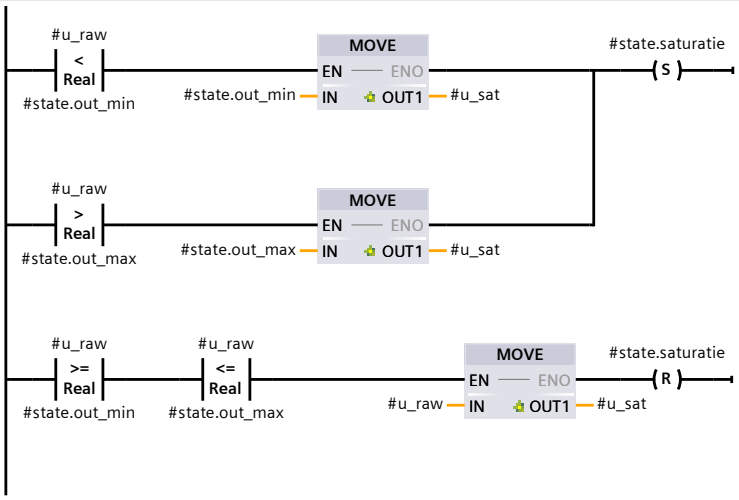
Network 12:



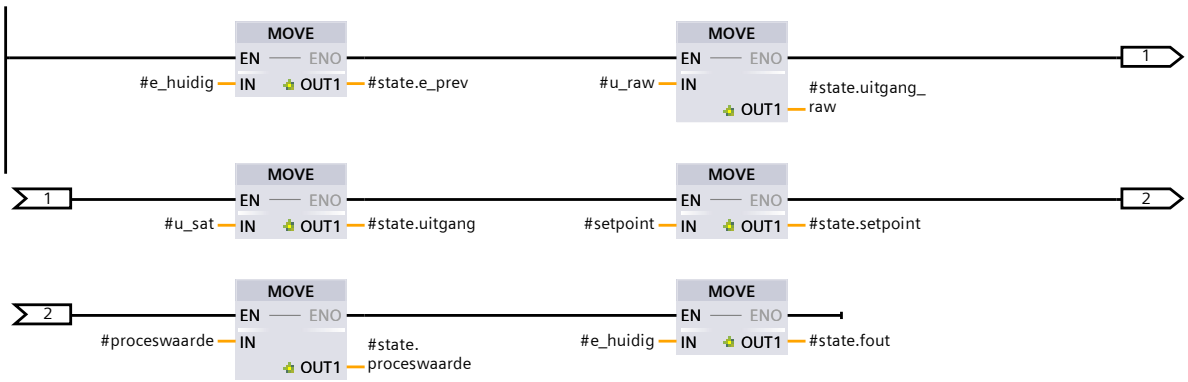
Network 13:



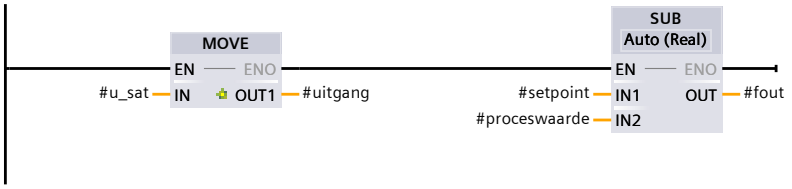
Network 14:



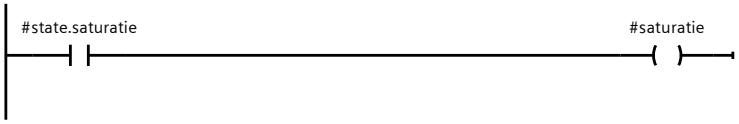
Network 15:



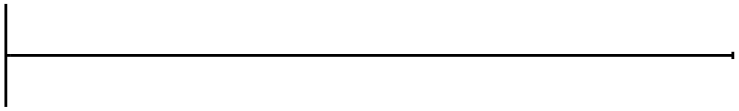
Network 16:



Network 17:



Network 18:



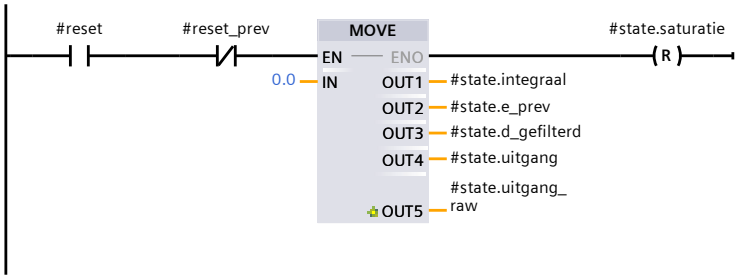
PLC_1 [CPU 1215C AC/DC/Rly] / Program blocks

FB_PID_M2 [FB9]

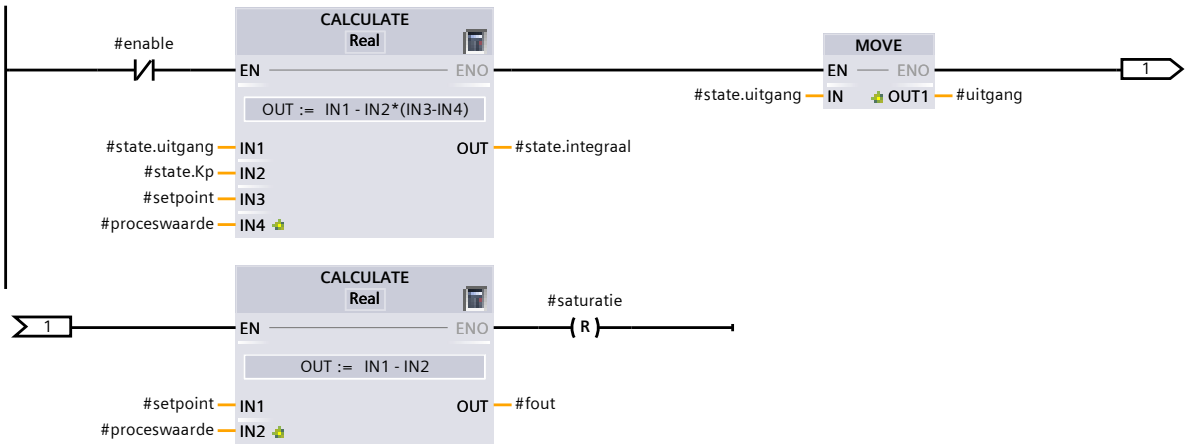
FB_PID_M2 Properties					
General					
Name	FB_PID_M2	**Number**	9	**Type**	FB
Language	LAD	**Numbering**	Automatic		
Information					
Title		**Author**		**Comment**	
Family		**Version**	0.1	**User-defined ID**	

Name	Data type	Default value	Retain
Input			
setpoint	Real	0.0	Non-retain
proceswaarde	Real	0.0	Non-retain
enable	Bool	false	Non-retain
reset	Bool	false	Non-retain
Output			
uitgang	Real	0.0	Non-retain
fout	Real	0.0	Non-retain
saturatie	Bool	false	Non-retain
InOut			
state	"UDT_PIDState"		
Static			
e_huidig	Real	0.0	Non-retain
P_term	Real	0.0	Non-retain
I_term	Real	0.0	Non-retain
D_term	Real	0.0	Non-retain
u_raw	Real	0.0	Non-retain
u_sat	Real	0.0	Non-retain
track_err	Real	0.0	Non-retain
alfa_D	Real	0.0	Non-retain
Kd	Real	0.0	Non-retain
in_deadband	Bool	false	Non-retain
reset_prev	Bool	false	Non-retain
abs(e_huidig)	Real	0.0	Non-retain
abs(state.deadb)	Real	0.0	Non-retain
Temp			
Constant			

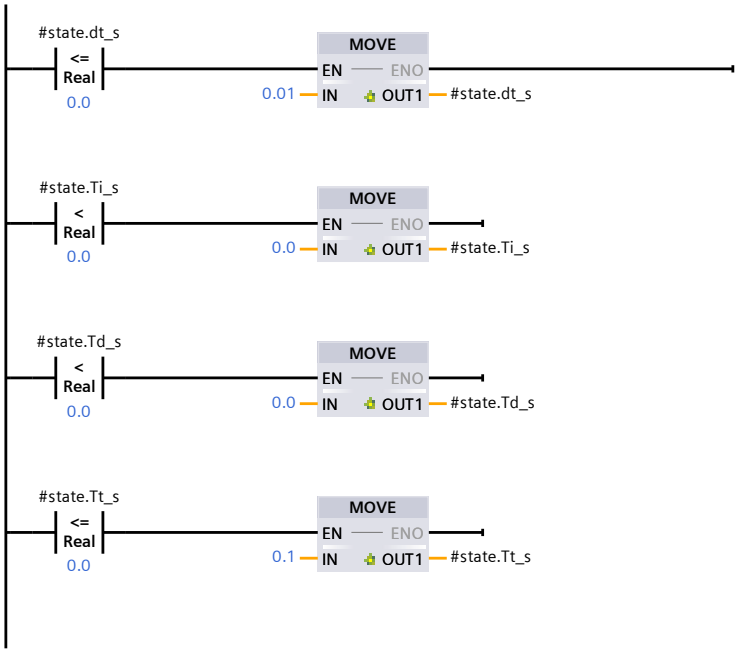
Network 1:



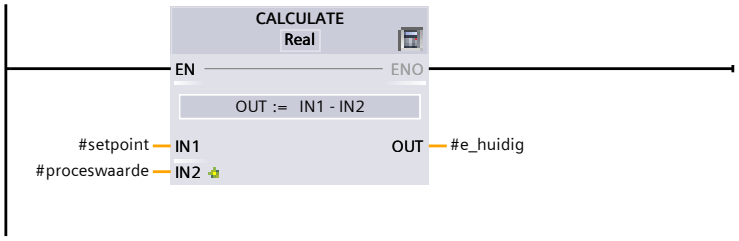
Network 2:



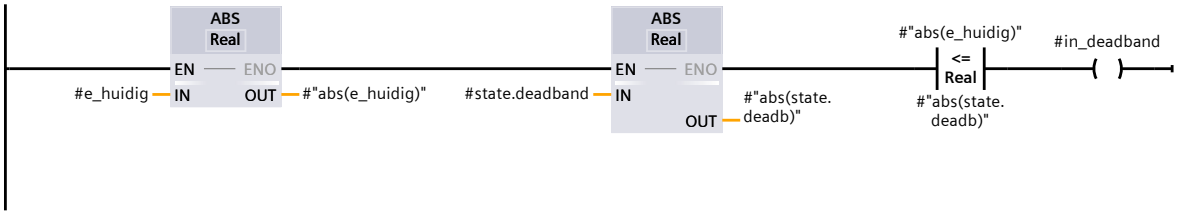
Network 3:



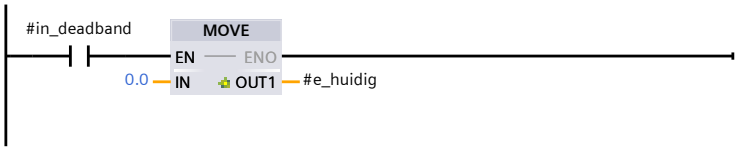
Network 4:



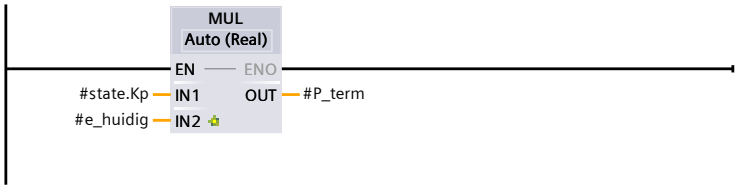
Network 5:



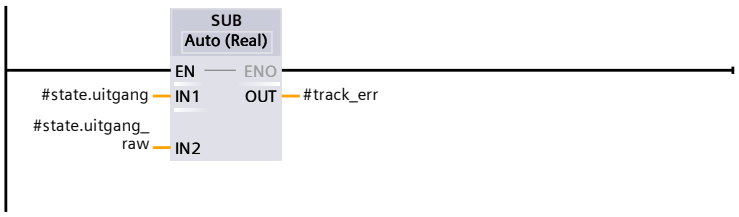
Network 6:



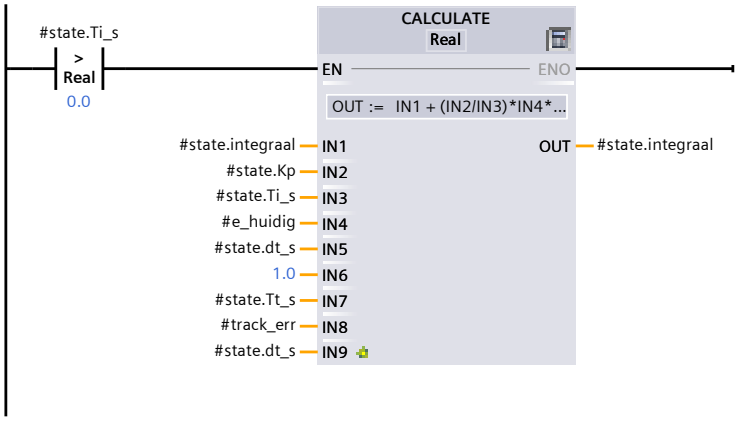
Network 7:



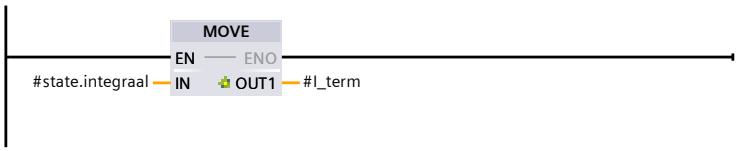
Network 8:



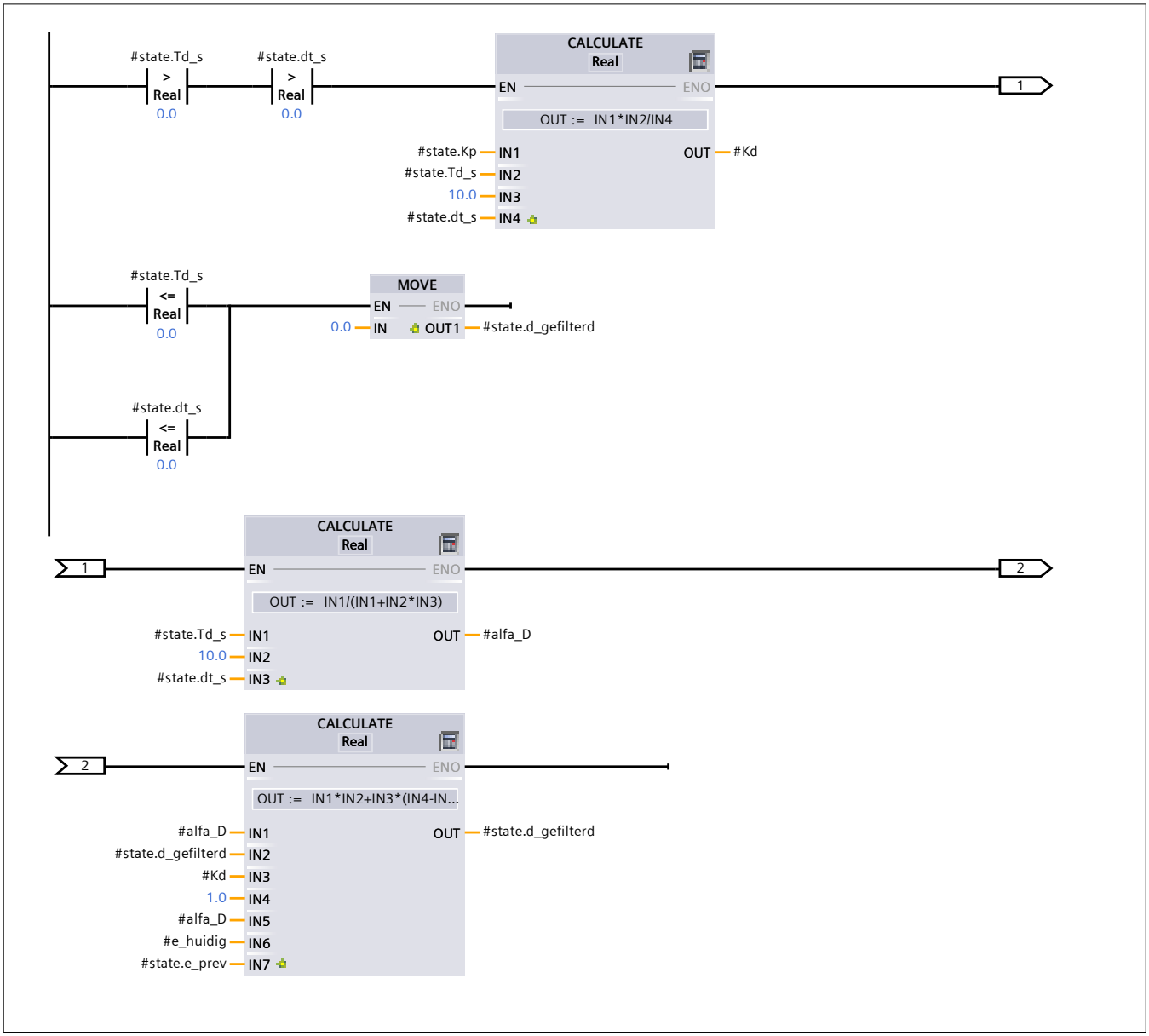
Network 9:



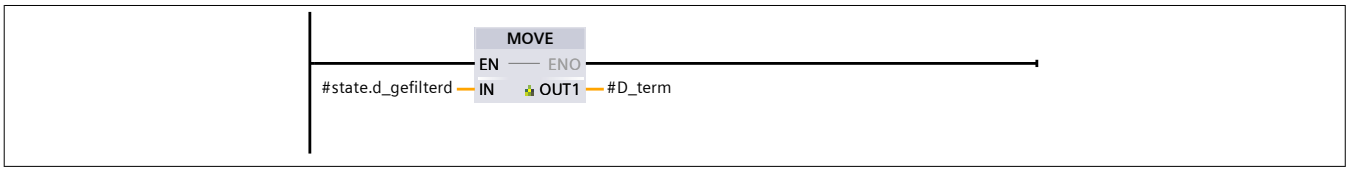
Network 10:



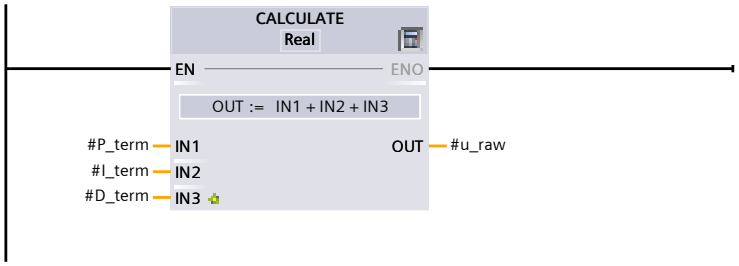
Network 11:



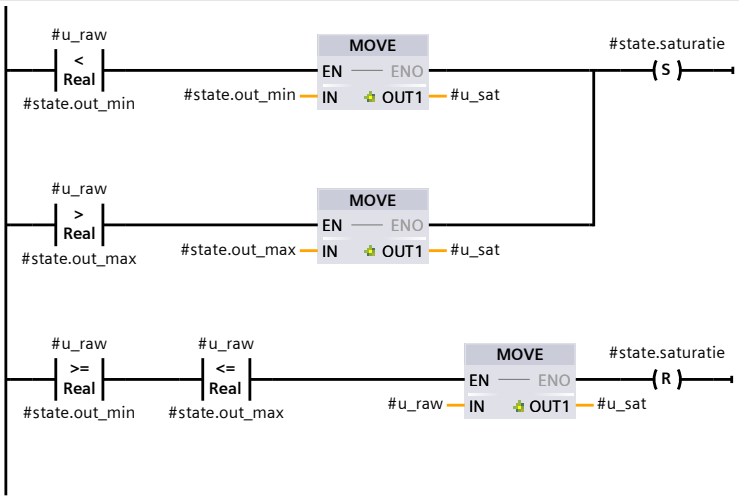
Network 12:



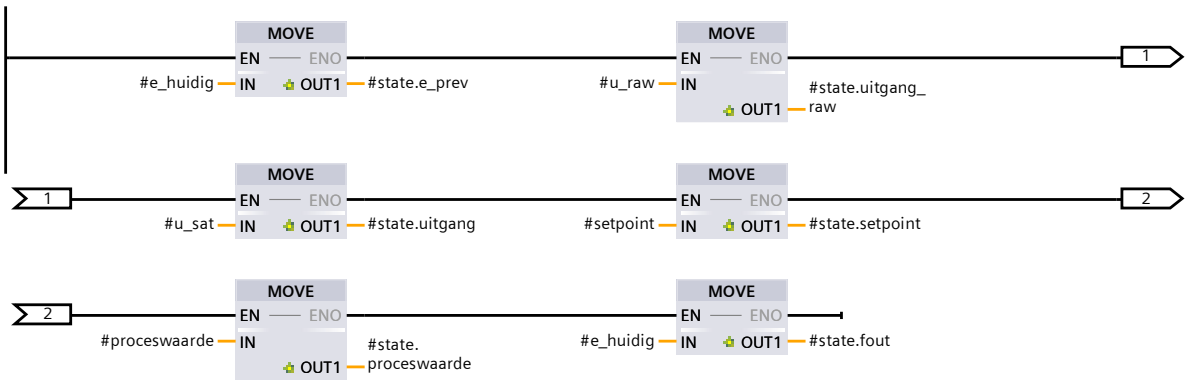
Network 13:



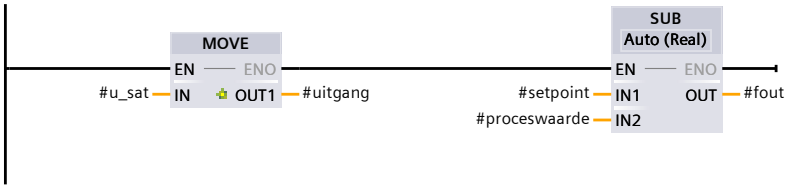
Network 14:



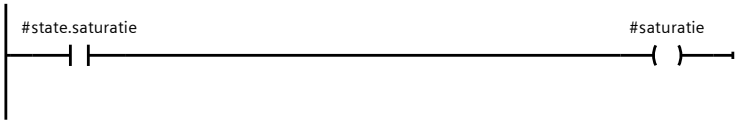
Network 15:



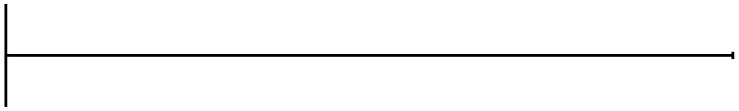
Network 16:



Network 17:



Network 18:



PLC_1 [CPU 1215C AC/DC/Rly] / Program blocks

pid_inst_M2 [DB14]

pid_inst_M2 Properties					
General					
Name	pid_inst_M2	**Number**	14	**Type**	DB
Language	DB	**Numbering**	Automatic		
Information					
Title		**Author**		**Comment**	
Family		**Version**	0.1	**User-defined ID**	

Name	Data type	Start value	Retain
Input			
setpoint	Real	0.0	False
proceswaarde	Real	0.0	False
enable	Bool	false	False
reset	Bool	false	False
Output			
uitgang	Real	0.0	False
fout	Real	0.0	False
saturatie	Bool	false	False
InOut			
state	"UDT_PIDState"		False
Static			
e_huidig	Real	0.0	False
P_term	Real	0.0	False
I_term	Real	0.0	False
D_term	Real	0.0	False
u_raw	Real	0.0	False
u_sat	Real	0.0	False
track_err	Real	0.0	False
alfa_D	Real	0.0	False
Kd	Real	0.0	False
in_deadband	Bool	false	False
reset_prev	Bool	false	False
abs(e_huidig)	Real	0.0	False
abs(state.deadb)	Real	0.0	False

PLC_1 [CPU 1215C AC/DC/Rly] / Program blocks

FB_M2_AngleComp_DB_1 [DB15]

FB_M2_AngleComp_DB_1 Properties					
General					
Name	FB_M2_Angle-Comp_DB_1	**Number**	15	**Type**	DB
Language	DB	**Numbering**	Automatic		
Information					
Title		**Author**		**Comment**	
Family		**Version**	0.1	**User-defined ID**	

Name	Data type	Start value	Retain
Input			
inclinatie_pitch_deg	Real	0.0	False
inclinatie_roll_deg	Real	0.0	False
hoek_M2_deg	Real	0.0	False
radar_dist_m	Real	0.0	False
laser_dist_m	Real	0.0	False
enable	Bool	false	False
reset	Bool	false	False
doel_hoek_deg	Real	0.0	False
L_arm_m	Real	0.0	False
out_min_deg	Real	0.0	False
out_max_deg	Real	0.0	False
Output			
M2_setpoint_deg	Real	0.0	False
FF_bijdrage_deg	Real	0.0	False
FB_bijdrage_deg	Real	0.0	False
fout_deg	Real	0.0	False
lat_err_mm	Real	0.0	False
op_positie	Bool	false	False
InOut			
pid_state	"UDT_PIDState"		False
Static			
roll_rad	Real	0.0	False
pitch_rad	Real	0.0	False
lat_err_m	Real	0.0	False
FF_lat_deg	Real	0.0	False
FF_pitch_deg	Real	0.0	False
FF_totaal_deg	Real	0.0	False
SP_totaal	Real	0.0	False
radar_eff	Real	0.0	False
pid_inst	"FB_PID_M0"		False
div	Real	0.0	False
atan	Real	0.0	False
sub	Real	0.0	False
pid_inst_M2	"FB_PID_M0"		False

PLC_1 [CPU 1215C AC/DC/Rly]

Technology objects

This folder is empty.

PLC_1 [CPU 1215C AC/DC/Rly] / PLC tags / Default tag table [31]

PLC tags

PLC tags				
	Name	Data type	Address	Retain

PLC_1 [CPU 1215C AC/DC/Rly] / PLC tags / Default tag table [31]

User constants

User constants			
	Name	Data type	Value

PLC_1 [CPU 1215C AC/DC/Rly] / PLC data types

UDT_SensorData

UDT_SensorData Properties					
General					
Name	UDT_SensorData	**Number**	1	**Type**	UDT
Language		**Numbering**			
Information					
Title		**Author**		**Comment**	
Family		**Version**		**User-defined ID**	

Name	Data type	Default value
dist_laser_M0	Real	0.0
inclinatie_M0	Real	0.0
inclinatie_M0_rad	Real	0.0
inclinatie_roll_deg	Real	0.0
pos_M1_mm	Real	0.0
snelheid_M1_mm_s	Real	0.0
hoek_M2_deg	Real	0.0
hoek_M2_rad	Real	0.0
radar_dist_M2	Real	0.0
eind_pos_x	Real	0.0
eind_pos_y	Real	0.0
eind_pos_z	Real	0.0
timestamp_ms	DWord	16#0

PLC_1 [CPU 1215C AC/DC/Rly] / PLC data types

UDT_MotorCmd

UDT_MotorCmd Properties					
General					
Name	UDT_MotorCmd	**Number**	2	**Type**	UDT
Language		**Numbering**			
Information					
Title		**Author**		**Comment**	
Family		**Version**		**User-defined ID**	

Name	Data type	Default value
M0_enable	Bool	false
M0_snelheid_pct	Real	0.0
M0_snelheid_m_s	Real	0.0
M1_enable	Bool	false
M1_pos_setpoint_mm	Real	0.0
M1_pos_FF_mm	Real	0.0
M1_pos_FB_mm	Real	0.0
M1_koppel_pct	Real	0.0
M2_enable	Bool	false
M2_hoek_setpoint_deg	Real	0.0
M2_hoek_FF_deg	Real	0.0
M2_hoek_FB_deg	Real	0.0

PLC_1 [CPU 1215C AC/DC/Rly] / PLC data types

UDT_PIDState

UDT_PIDState Properties					
General					
Name	UDT_PIDState	**Number**	3	**Type**	UDT
Language		**Numbering**			
Information					
Title		**Author**		**Comment**	
Family		**Version**		**User-defined ID**	

Name	Data type	Default value
Kp	Real	0.0
Ti_s	Real	0.0
Td_s	Real	0.0
Tt_s	Real	0.0
dt_s	Real	0.0
out_min	Real	0.0
out_max	Real	0.0
deadband	Real	0.0
integraal	Real	0.0
e_prev	Real	0.0
d_gefilterd	Real	0.0
alfa_D	Real	0.0
setpoint	Real	0.0
proceswaarde	Real	0.0
fout	Real	0.0
uitgang_raw	Real	0.0
uitgang	Real	0.0
saturatie	Bool	false

PLC_1 [CPU 1215C AC/DC/Rly] / PLC data types

UDT_ResearchData

UDT_ResearchData Properties					
General					
Name	UDT_ResearchData	**Number**	4	**Type**	UDT
Language		**Numbering**			
Information					
Title		**Author**		**Comment**	
Family		**Version**		**User-defined ID**	

Name	Data type	Default value
n_samples	DWord	16#0
err_M0_huidig	Real	0.0
sum_sq_err_M0	Real	0.0
mse_M0	Real	0.0
rmse_M0	Real	0.0
err_M1_huidig	Real	0.0
sum_sq_err_M1	Real	0.0
mse_M1	Real	0.0
rmse_M1	Real	0.0
err_M2_huidig	Real	0.0
sum_sq_err_M2	Real	0.0
mse_M2	Real	0.0
rmse_M2	Real	0.0
err_3D_mm	Real	0.0
sum_sq_err_3D	Real	0.0
mse_3D	Real	0.0
rmse_3D	Real	0.0
dominante_motor	Int	0
dominantie_ratio_12	Real	0.0
dominantie_ratio_01	Real	0.0
reset	Bool	false

PLC_1 [CPU 1215C AC/DC/Rly] / PLC data types

UDT_StateMachine

UDT_StateMachine Properties					
General					
Name	UDT_StateMachine	**Number**	5	**Type**	UDT
Language		**Numbering**			
Information					
Title		**Author**		**Comment**	
Family		**Version**		**User-defined ID**	

Name	Data type	Default value
toestand	Int	0
toestand_prev	Int	0
toestand_timer_ms	DWord	16#0
transitie_verzoek	Int	0
fout_code	Int	0
fout_tekst	String[80]	''

PLC_1 [CPU 1215C AC/DC/Rly] / Watch and force tables

Force table

Name	Address	Display format	Force value

PLC_1 [CPU 1215C AC/DC/Rly]

Traces

Name

PLC_1 [CPU 1215C AC/DC/Rly] / Traces

Measurements

This folder is empty.

PLC_1 [CPU 1215C AC/DC/Rly] / Traces

Combined measurements

Name

PLC_1 [CPU 1215C AC/DC/Rly] / OPC UA communication

Server interfaces

This folder is empty.

PLC_1 [CPU 1215C AC/DC/Rly]

PLC alarm text lists

This folder is empty.

PLC_1 [CPU 1215C AC/DC/Rly] / Local modules

PLC_1 [CPU 1215C AC/DC/Rly]

PLC_1			
Project information			
Name	PLC_1	**Author**	beirn
Comment		**Slot**	1
Rack	0		
Catalog information			
Short designation	CPU 1215C AC/DC/Rly	**Description**	Work memory 200 KB; 120/240VDC power supply with DI14 x 24VDC SINK/SOURCE, DQ10 x relay and AI2 and AQ2 on board; 6 high-speed counters and 4 pulse outputs on-board; signal board expands on-board I/O; up to 3 communication modules for serial communication; up to 8 signal modules for I/O expansion; PROFINET IO controller, 2 ports, I-device, transport protocol TCP/IP, secure Open User Communication, S7 communication, Web server, OPC UA: Server DA
Article number	6ES7 215-1BG40-0XB0	**Firmware version**	V4.6
	False		

Connection resources

	Station resources - Reserved - Maximum	Station resources - Reserved - Configured	Station resources - Dynamic - Configured	Module resources - PLC_1 [CPU 1215C AC/DC/Rly] - Configured
Maximum number of resources:		34	34	68
	Maximum	Configured	Configured	Configured
PG communication:	4	-	-	-
HMI communication:	12	0	0	0
S7 communication:	8	0	0	0
Open user communication:	8	0	0	0
Web communication:	2	-	-	-
OPC UA client/server communication:	0	-	-	-
Other communication:	-	-	0	0
Total resources used:		0	0	0
Available resources:		34	34	68

Overview of addresses\Overview of addresses\Overview of addresses			
Inputs	True	**Outputs**	True
Address gaps	False	**Slot**	True

Type	Addr. from	Addr. to	Module	PIP	Device name	Device number	Size	Master / IO system	Rack	Slot
I	0	1	DI 14/DQ 10_1	Automatic update	PLC_1 [CPU 1215C AC/DC/Rly]	-	2 Bytes	-	0	1 1
O	0	1	DI 14/DQ 10_1	Automatic update	PLC_1 [CPU 1215C AC/DC/Rly]	-	2 Bytes	-	0	1 1
I	64	67	AI 2/AQ 2_1	Automatic update	PLC_1 [CPU 1215C AC/DC/Rly]	-	4 Bytes	-	0	1 2
O	64	67	AI 2/AQ 2_1	Automatic update	PLC_1 [CPU 1215C AC/DC/Rly]	-	4 Bytes	-	0	1 2
I	1000	1003	HSC_1	Automatic update	PLC_1 [CPU 1215C AC/DC/Rly]	-	4 Bytes	-	0	1 16
I	1004	1007	HSC_2	Automatic update	PLC_1 [CPU 1215C AC/DC/Rly]	-	4 Bytes	-	0	1 17
I	1008	1011	HSC_3	Automatic update	PLC_1 [CPU 1215C AC/DC/Rly]	-	4 Bytes	-	0	1 18
I	1012	1015	HSC_4	Automatic update	PLC_1 [CPU 1215C AC/DC/Rly]	-	4 Bytes	-	0	1 19
I	1016	1019	HSC_5	Automatic update	PLC_1 [CPU 1215C AC/DC/Rly]	-	4 Bytes	-	0	1 20
I	1020	1023	HSC_6	Automatic update	PLC_1 [CPU 1215C AC/DC/Rly]	-	4 Bytes	-	0	1 21
O	1000	1001	Pulse_1	Automatic update	PLC_1 [CPU 1215C AC/DC/Rly]	-	2 Bytes	-	0	1 32
O	1002	1003	Pulse_2	Automatic update	PLC_1 [CPU 1215C AC/DC/Rly]	-	2 Bytes	-	0	1 33
O	1004	1005	Pulse_3	Automatic update	PLC_1 [CPU 1215C AC/DC/Rly]	-	2 Bytes	-	0	1 34
O	1006	1007	Pulse_4	Automatic update	PLC_1 [CPU 1215C AC/DC/Rly]	-	2 Bytes	-	0	1 35